

Homochiral ideas: symmetry centers

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Related links

- ICQ16 conference slides (2025)
<https://softqc.wordpress.com/presentations/>
- Homochiral inflation paper (2025)
<https://arxiv.org/abs/2502.15608>
- Github (2026)
<https://github.com/marimperorclerc/homochiral>

Objectives

In the whole infinite spectre tiling:

- Find the location of global 2-fold and 3-fold symmetry centers

Around a global 2-fold or 3-fold symmetry center:

- Build an homochiral inflation construction

Objective

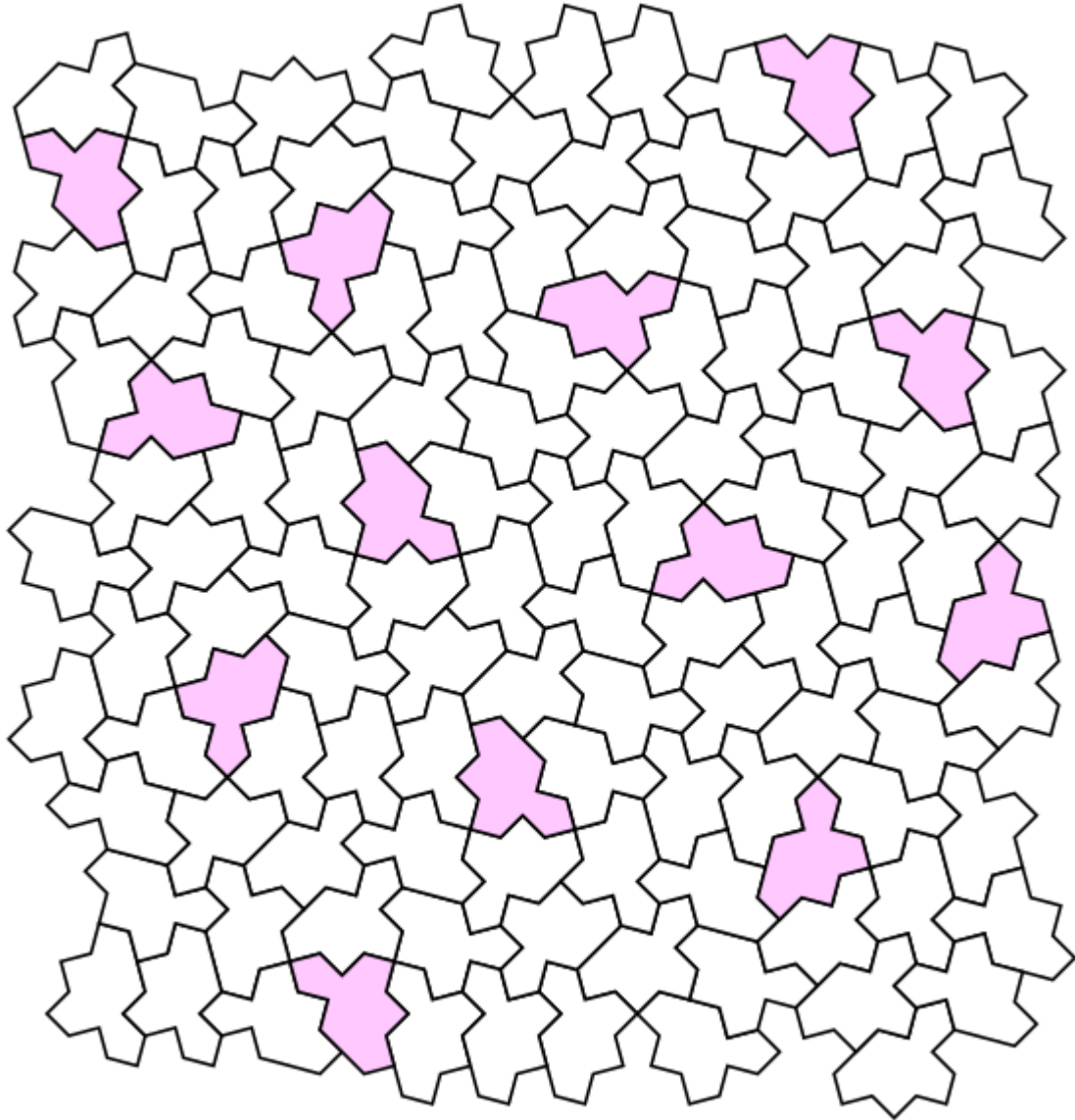
- Construction a sequence of patches by homochiral inflation.
- In the sequence, all patches have the same rotationnal symmetry around the center.

Difference between global and local symmetry

We distinguish between:

- a global symmetry center where rotation operation around the center applies to the whole infinite tiling
- And a local symmetry center, where rotation operation around the center applies locally to a finite subset of tiles

Spectre tiling (left chirality): ODD and EVEN monotiles



- ODD (pink):
M1,M3,M5,M7,M9,M11
- EVEN (no color):
M2,M4,M6,M8,M10,M12

Six orientations for ODD and EVEN monotiles

$M_1 = M_{O_1}$

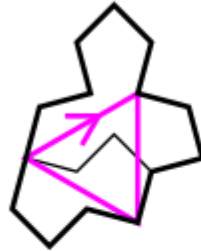
$M_3 = M_{O_2}$

$M_5 = M_{O_3}$

$M_7 = M_{O_4}$

$M_9 = M_{O_5}$

$M_{11} = M_{O_6}$



$M_2 = M_{E_1}$

$M_4 = M_{E_2}$

$M_6 = M_{E_3}$

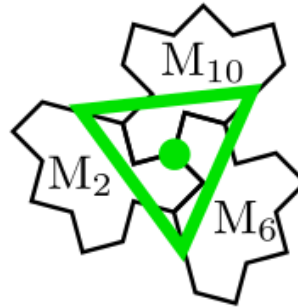
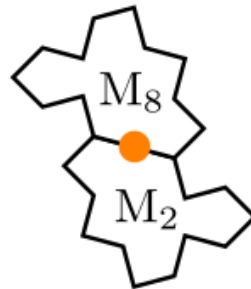
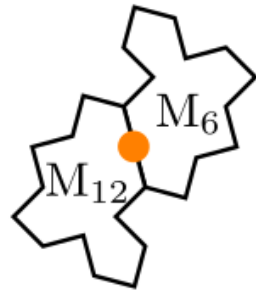
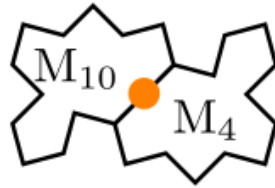
$M_8 = M_{E_4}$

$M_{10} = M_{E_5}$

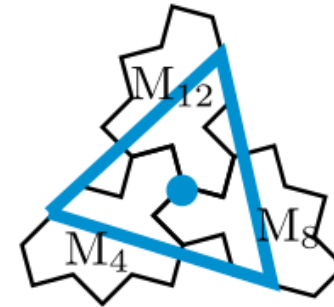
$M_{12} = M_{E_6}$

Symmetry centers are located on the contour of adjacent EVEN monotiles

2-fold



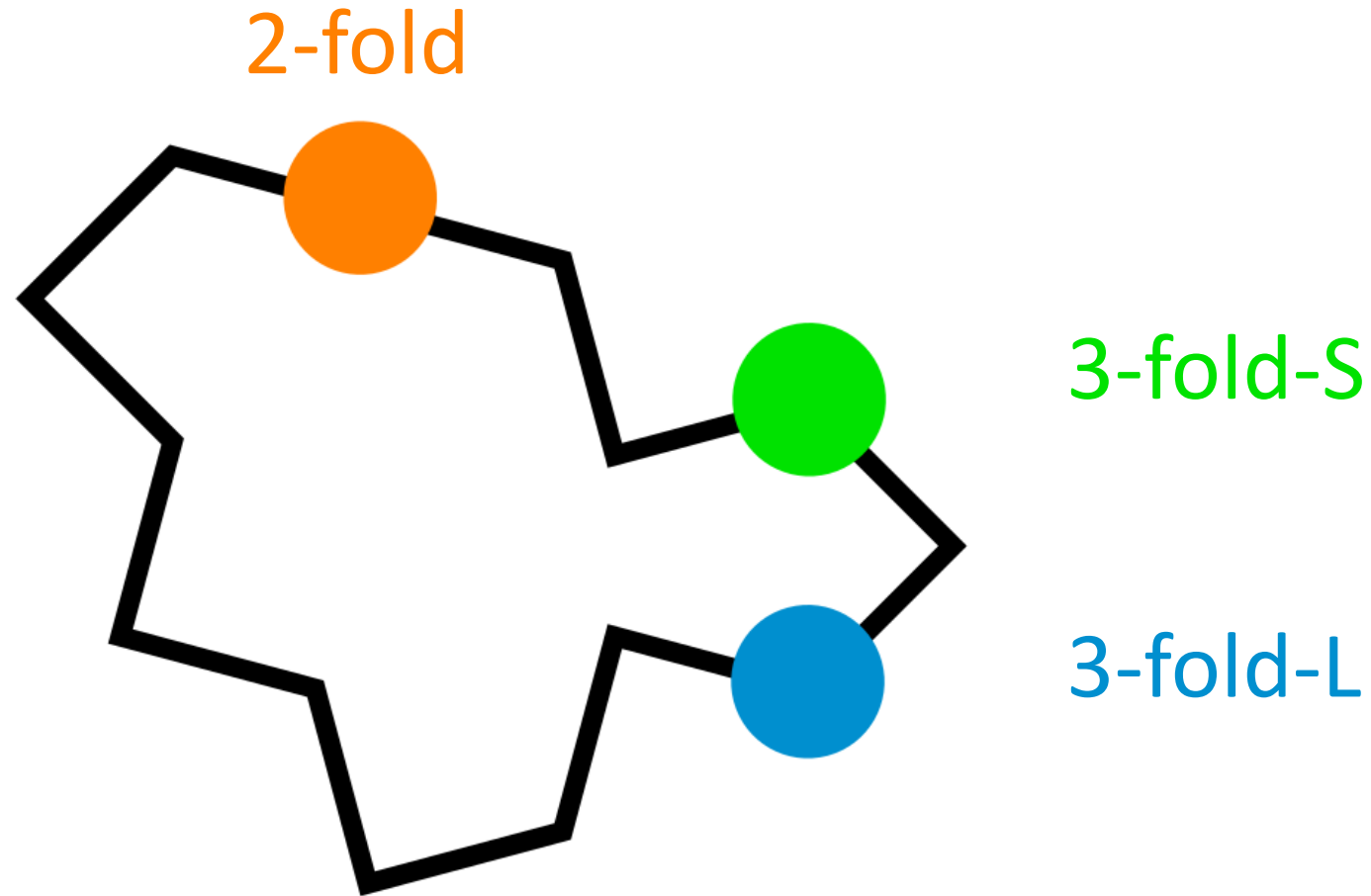
3-fold-S



3-fold-L

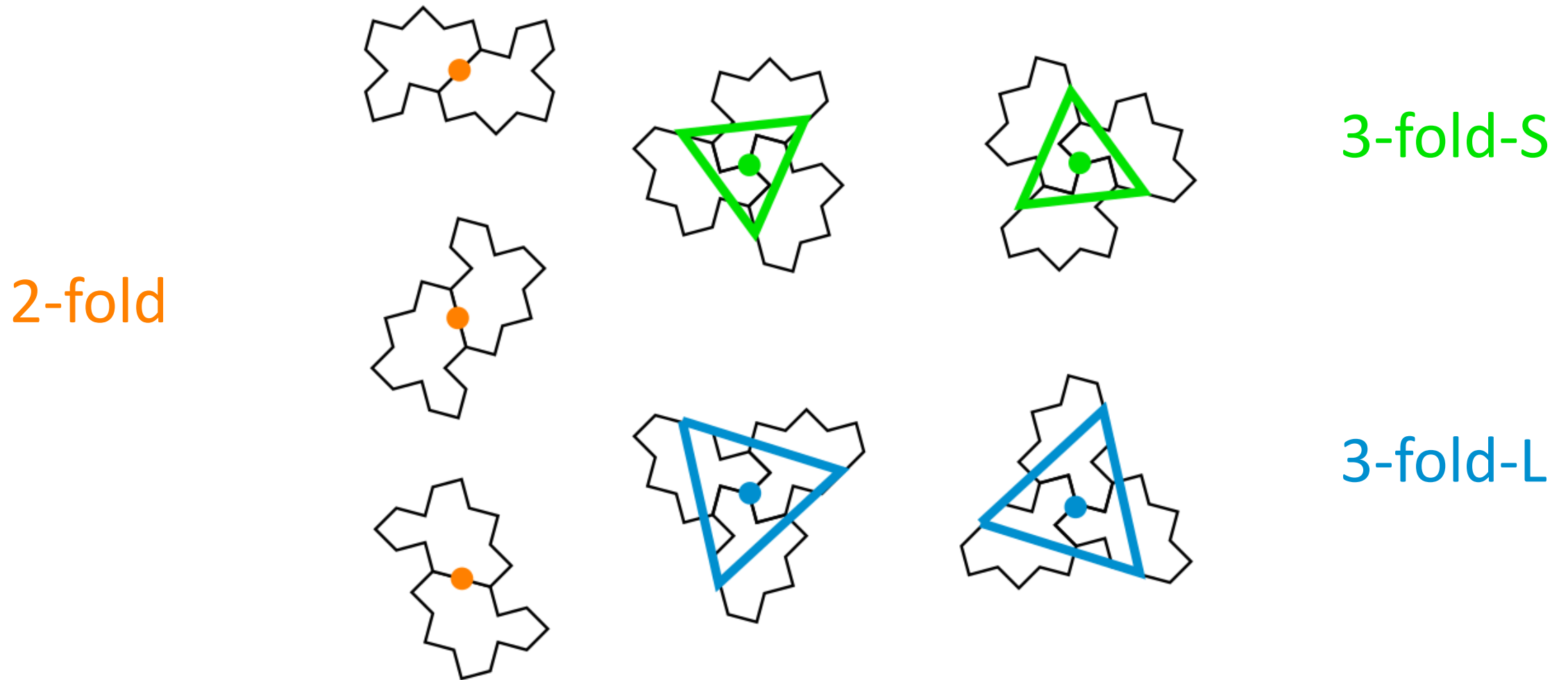
- EVEN monotiles are M2, M4, M6, M8, M10 and M12

Location on the contour of one EVEN monotile



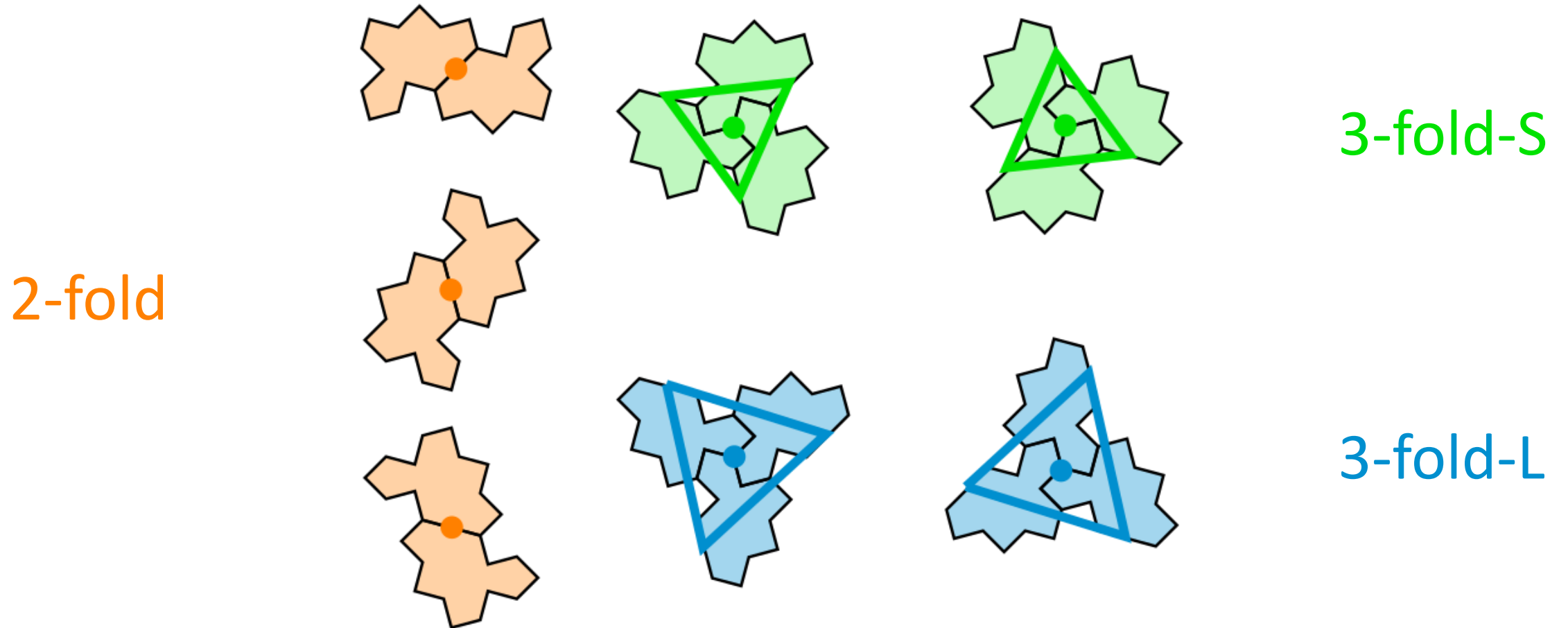
- This figure shows the origin of the two different, S and L, 3-fold centers
- It is linked with the chirality of the spectre tile (here with left handedness)

configuration of monotiles at a symmetry center



- One configuration for 2-fold (orange color) and two configuration for 3-fold (green and blue). Figures show all possible orientations in the tiling.

configuration of monotiles at a symmetry center



- Color is used when rotation around symmetry center is applied locally.

Two different global symmetry centers are infinitely far away from each other in the infinite tiling

- a global symmetry center is placed on purpose at the origin of the infinite tiling
- Then all other types of global symmetry centers are at infinite distance from it
- example: a 2-fold global symmetry at the origin
- Argument: if two global rotation centers were at a finite distance from each other, the tiling would be periodic.
- Two rotation centers at finite distance generate a periodic lattice of symmetry centers.

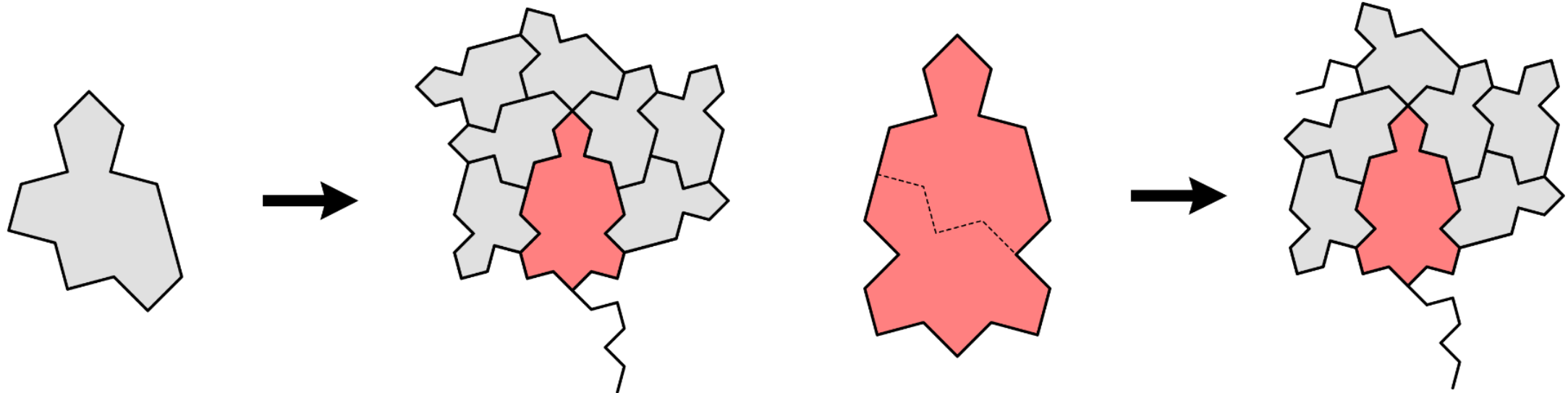
H7/H8 by Simon Tatham

<https://www.chiark.greenend.org.uk/~sgtatham/quasiblog/aperiodic-refine/>

<https://arxiv.org/abs/2512.16595>

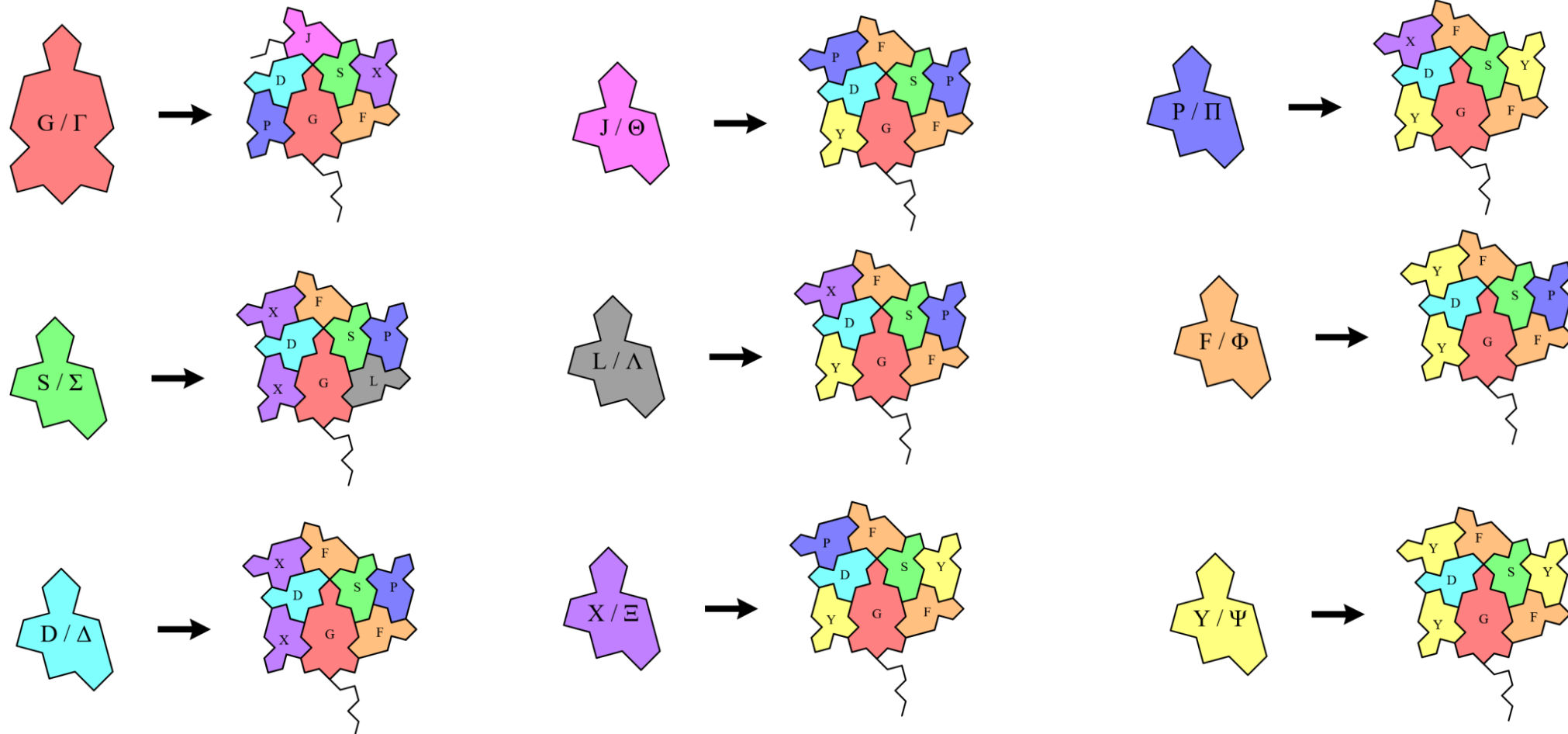
Code in RUST language:

<https://crates.io/crates/substitution-tiling-transducers>

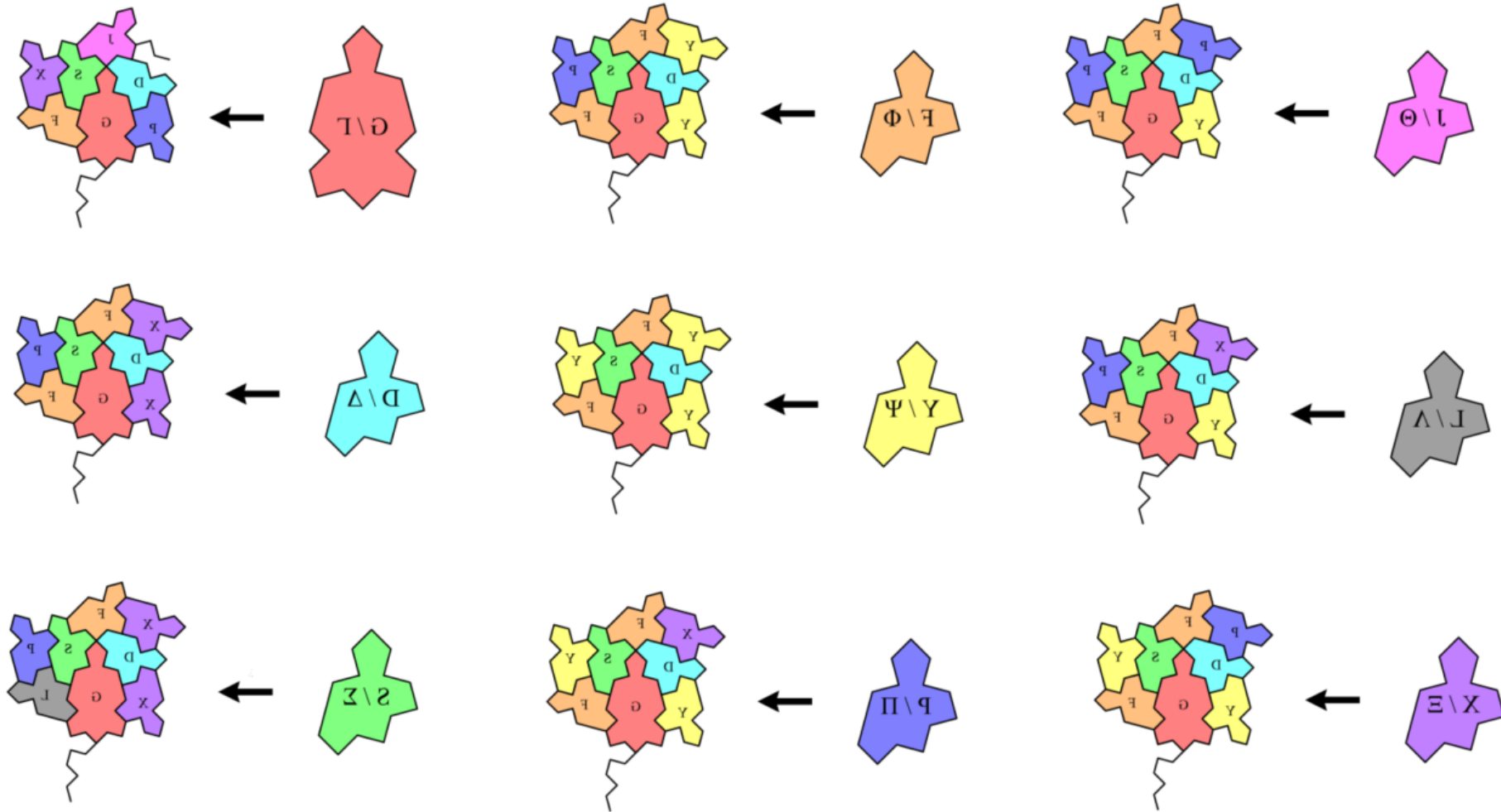


The nine homochiral substitutions by Simon Tathan

<https://www.chiark.greenend.org.uk/~sgtatham/quasiblog/aperiodic-refine/>



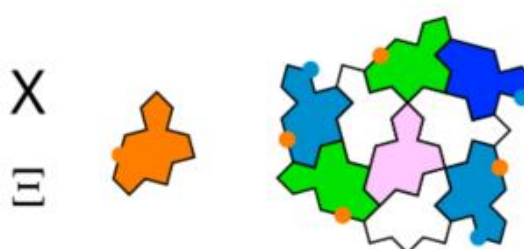
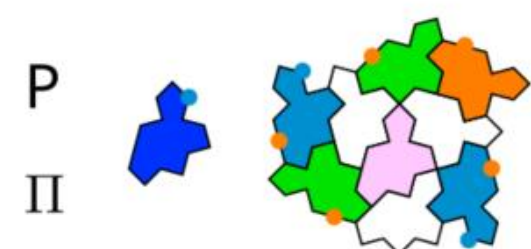
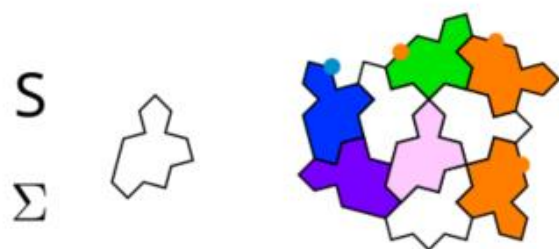
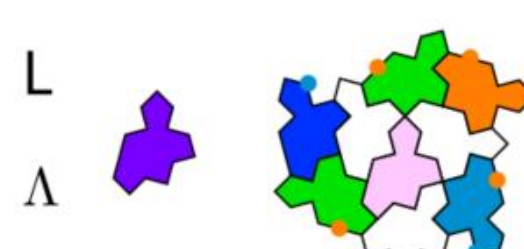
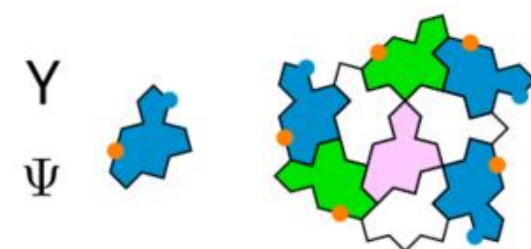
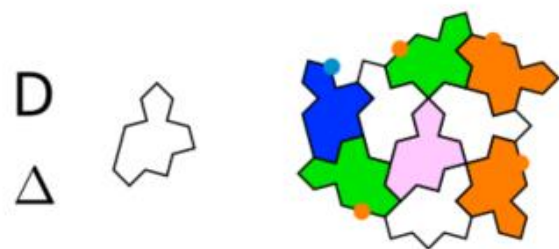
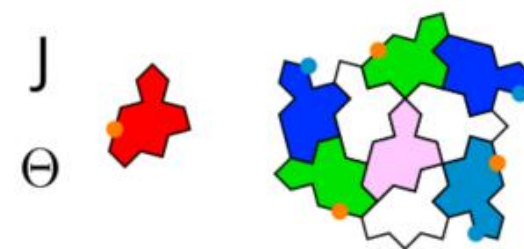
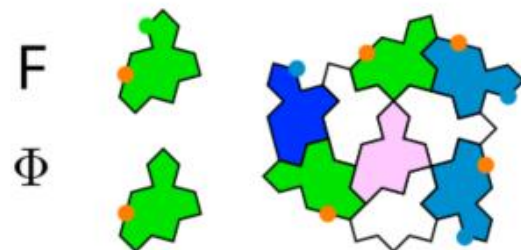
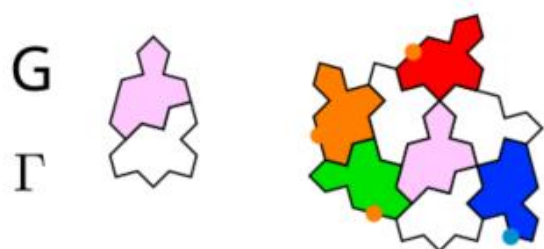
H7/H8 for left chirality



New color code

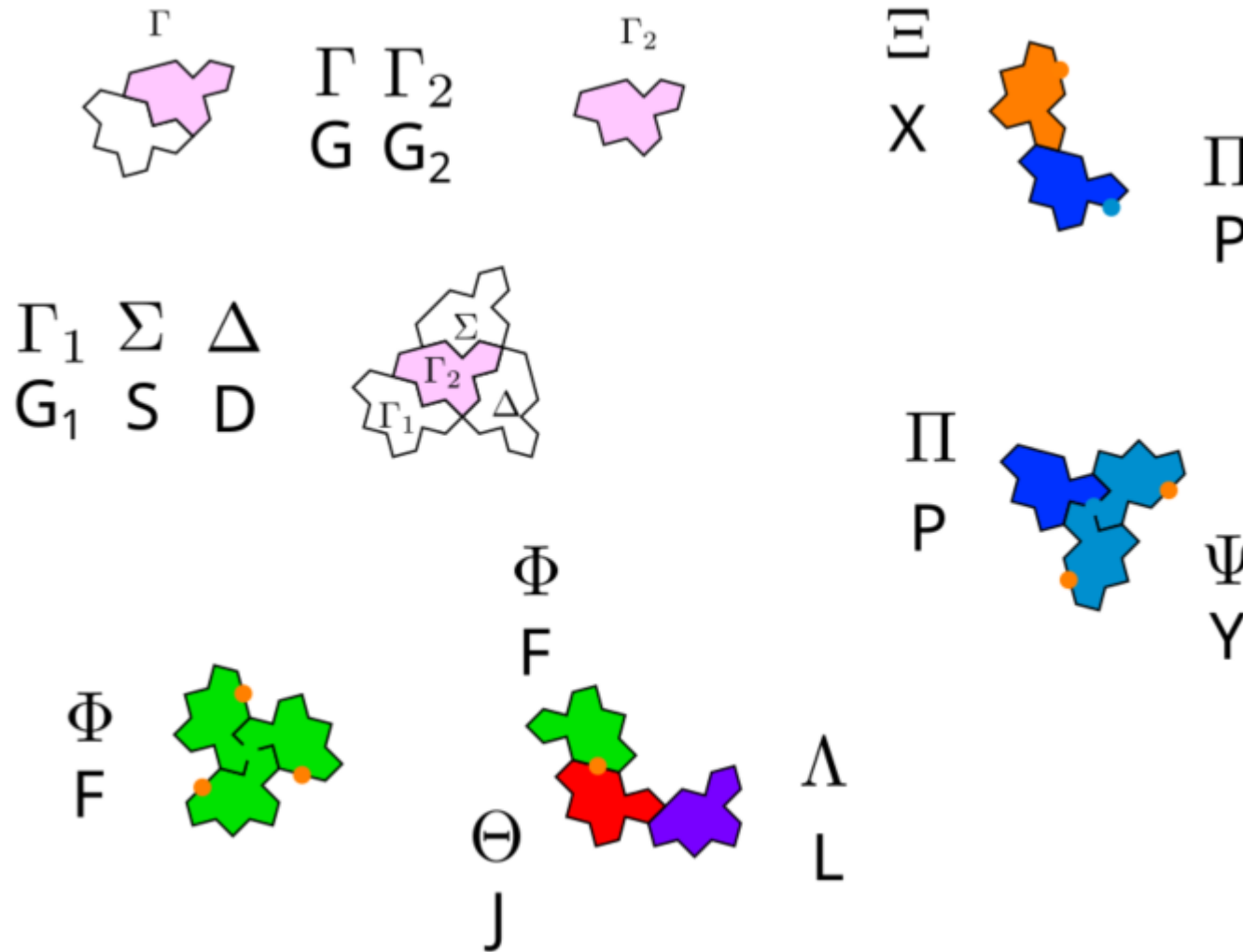
G Γ		F Φ		J Θ	
D Δ		Y Ψ		L Λ	
S Σ		P Π		X Ξ	

H7/H8 with left chirality and new color code



New color code: identify patterns in the tiling

- ODD tiles
Pink color
- Small cluster
 $\backslash$ sigma,
 $\backslash$ delta,
 $\backslash$ gamma1
white color
- $\backslash$ phi: Small green
stars + some
isolated EVEN
tiles

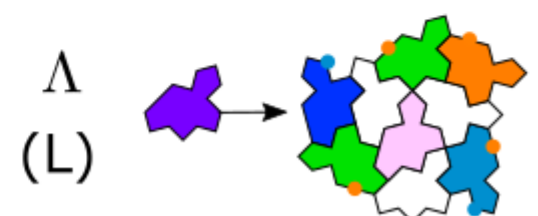
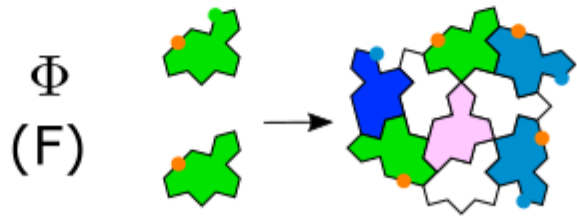
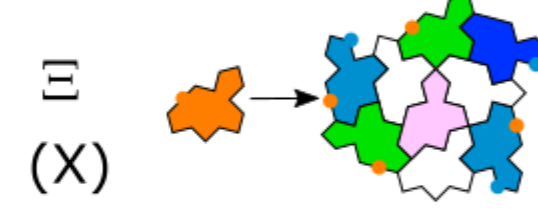
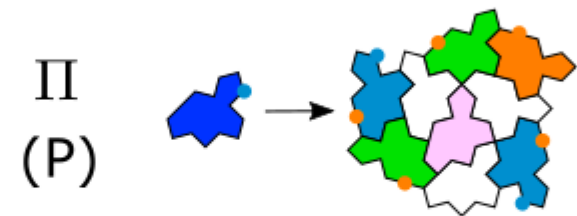
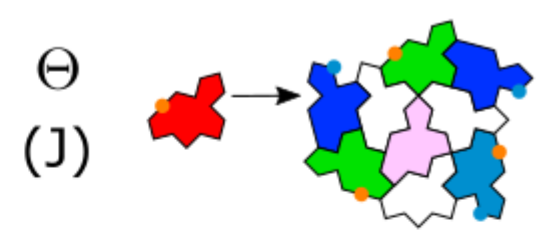
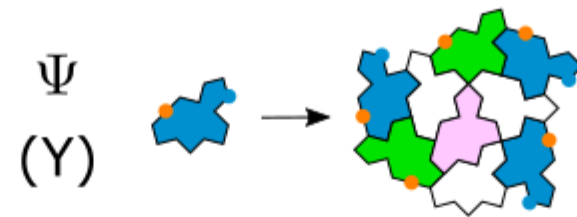
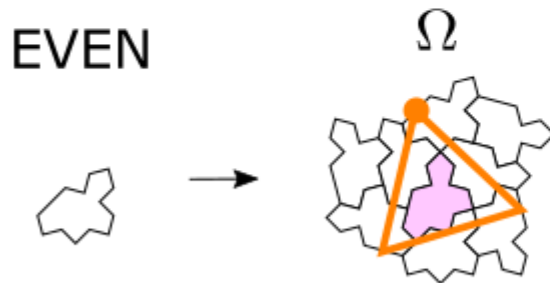
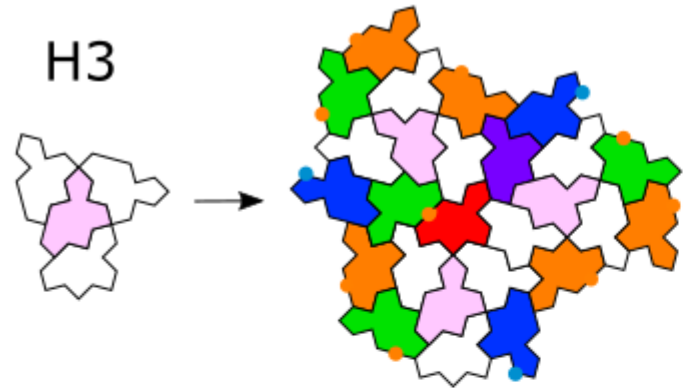


- All $\backslash$ chi tiles are linked to a $\backslash$ pi tile and consequently to a large star.

Blue large stars correspond to all $\backslash$ Pi and $\backslash$ Psi tiles with different combinations

- $\backslash$ isolated green + $\backslash$ theta + $\backslash$ lambda always together

Alternative substitution rules



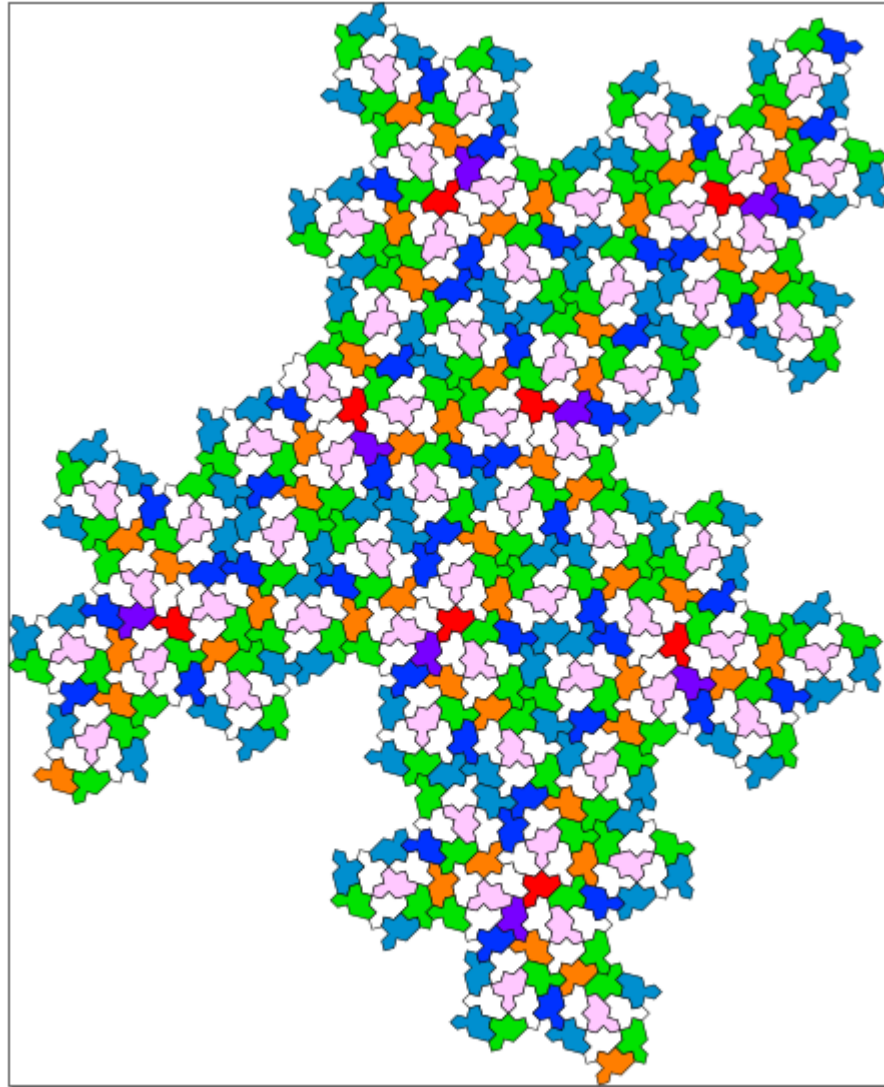
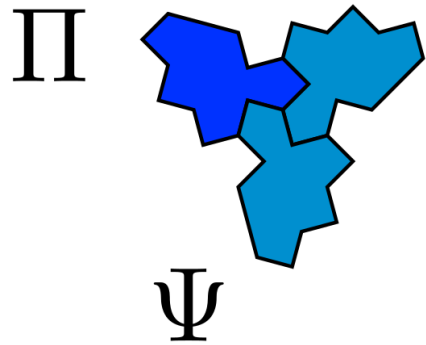
- No Y blue tiles in H3 substitution !

- Add $\gamma + \delta + \sigma == \text{H3 construction for substitution}$

Alternative substitution rules

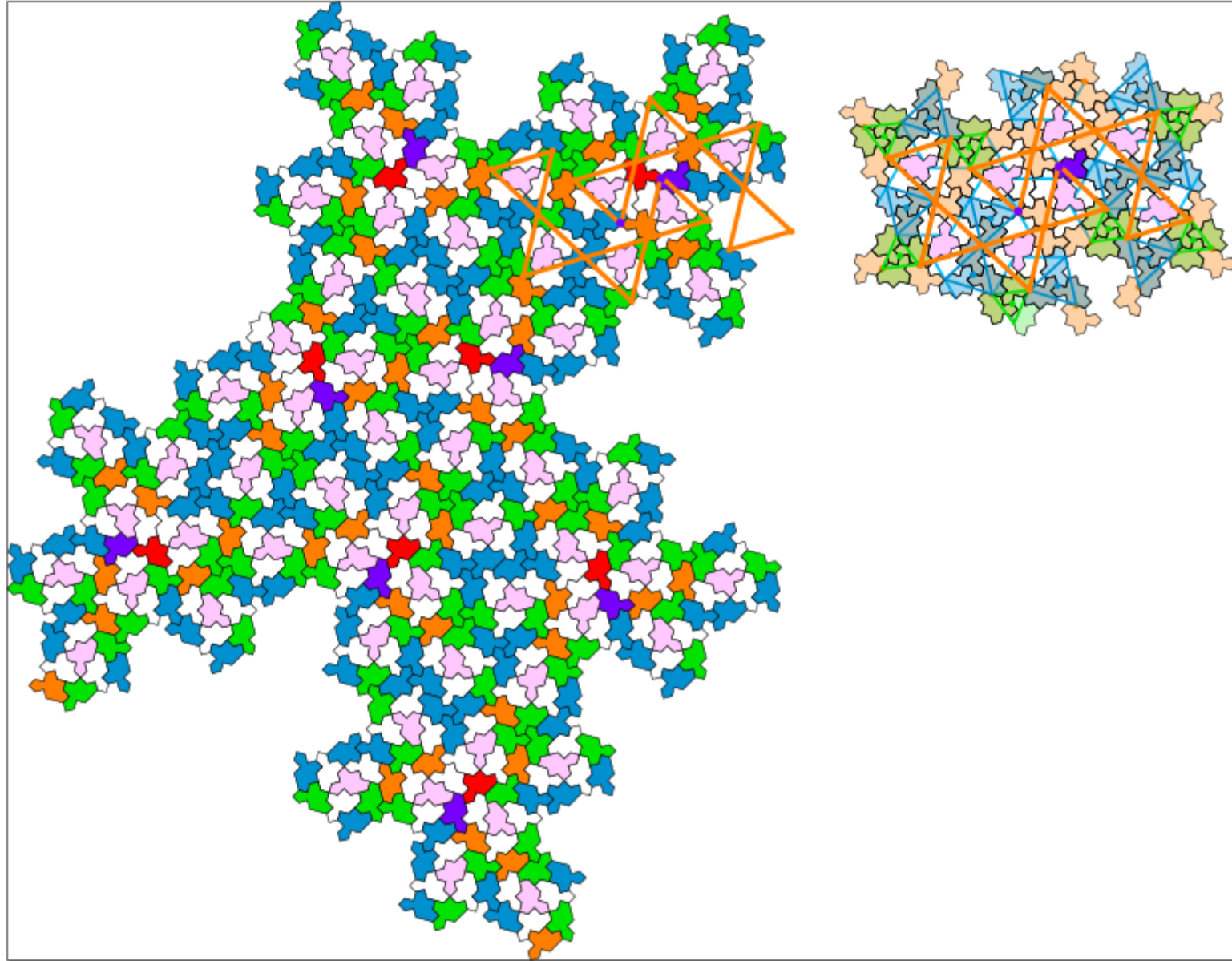
- separate the green tiles with two different colors:
- A dark green when attached to the red tile
- Find the appropriate coloring in all substitutions (green and dark green) ???
(probably dark green when attached to dark blue or orange)

Inflation using python code



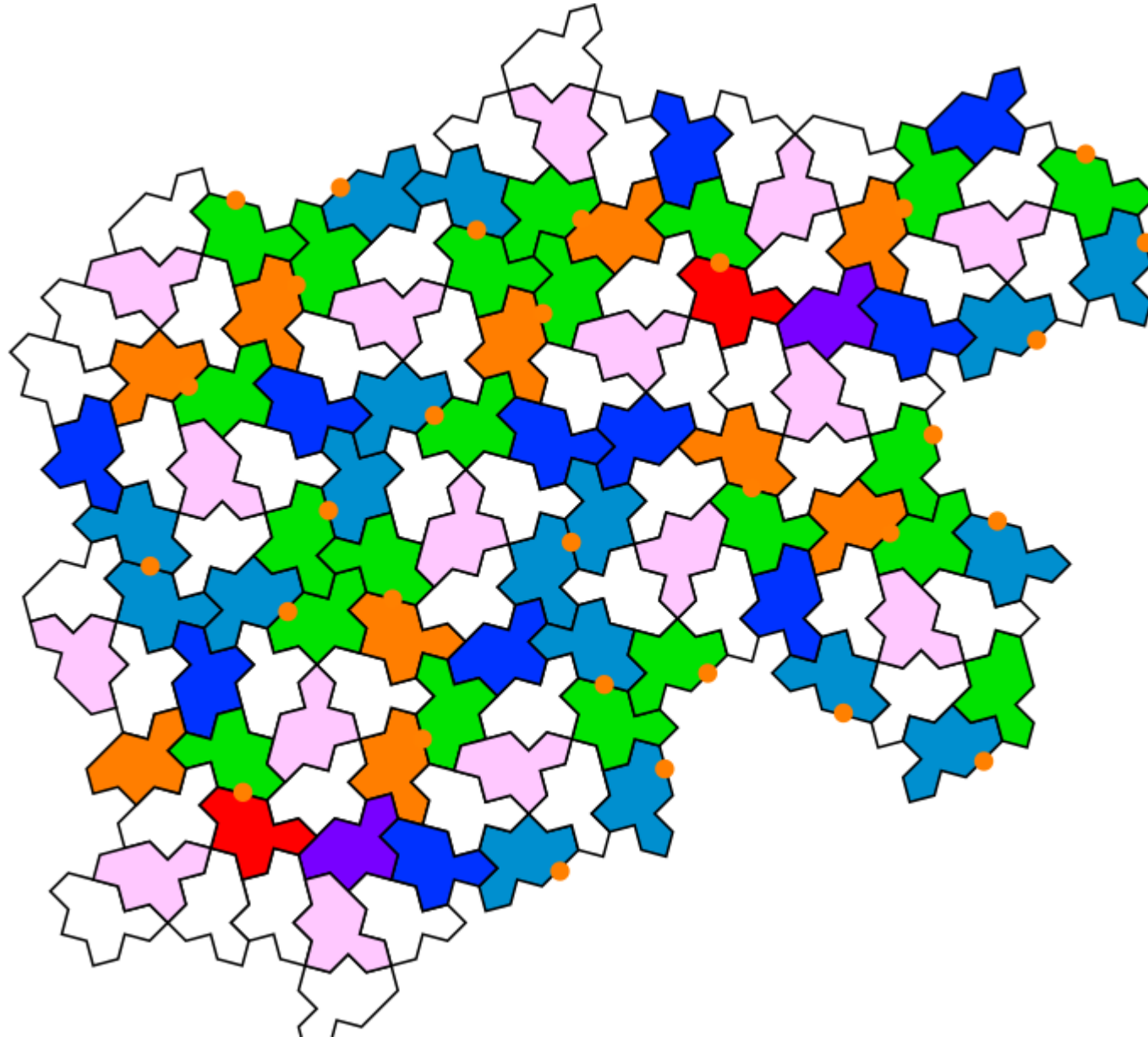
- Blue large stars correspond to all Π and Ψ tiles with different combinations

Inflation using python code

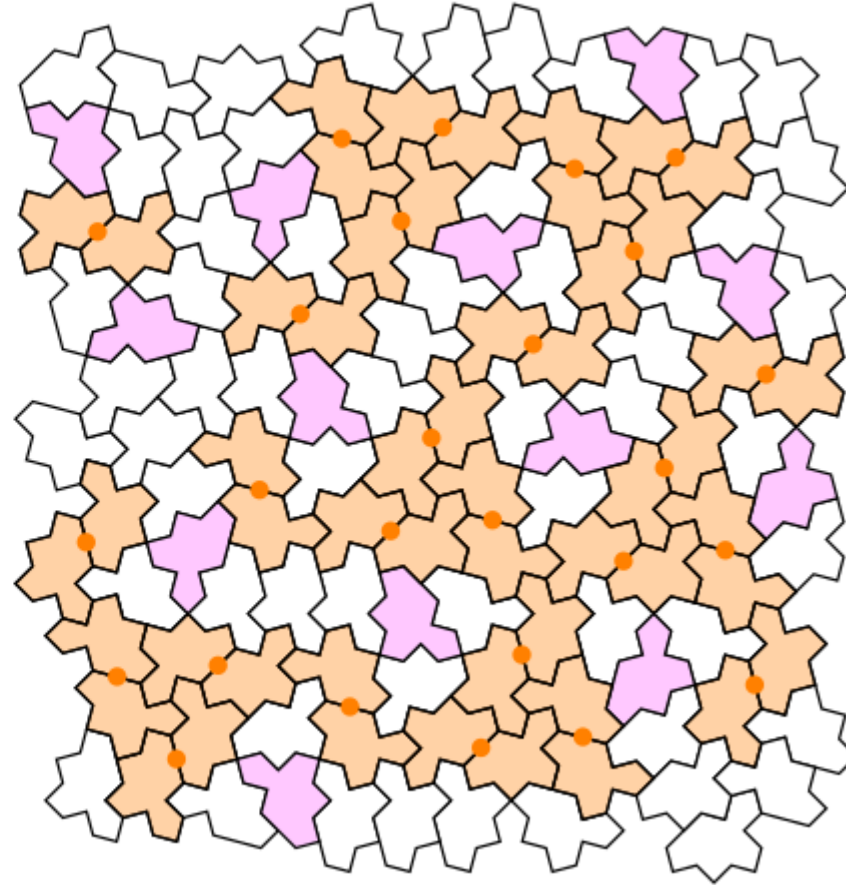
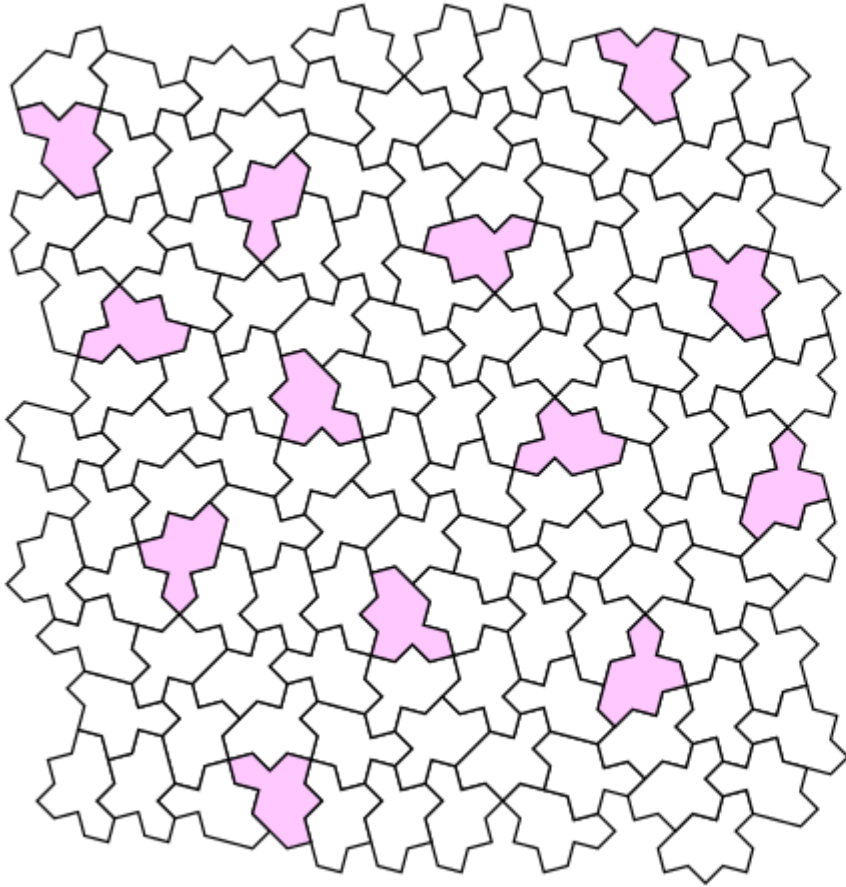


Local two fold symmetry centers

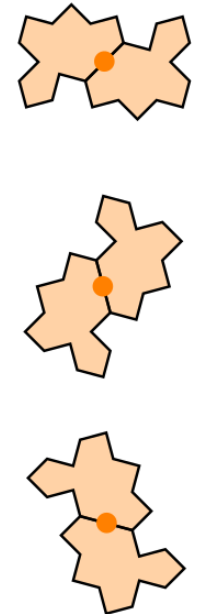
2-fold



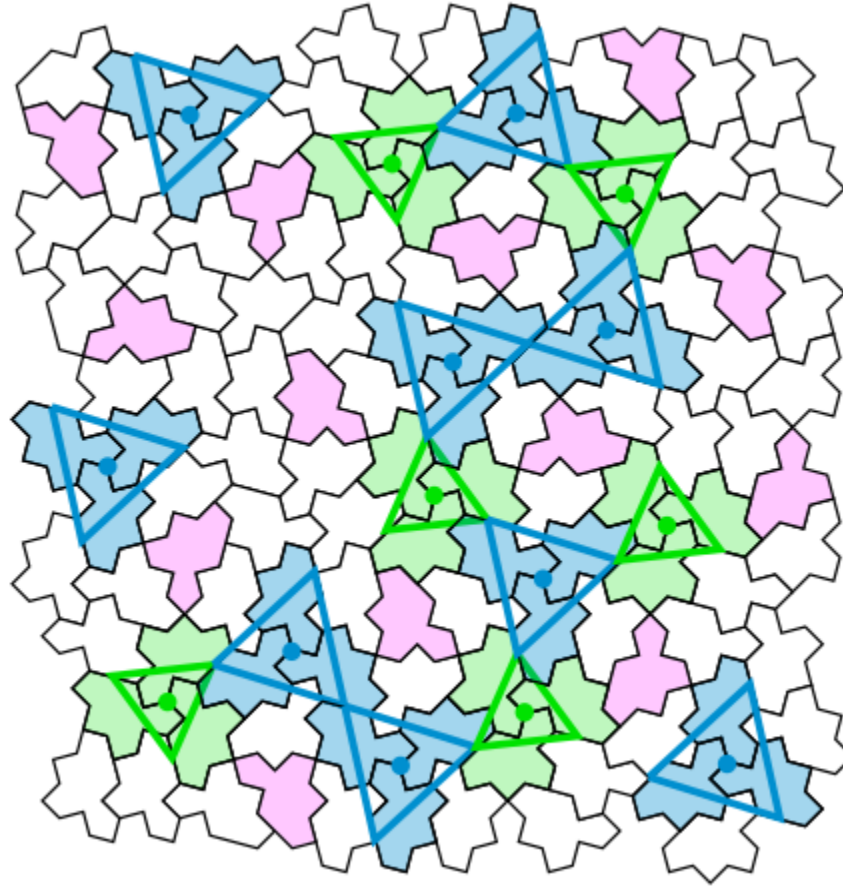
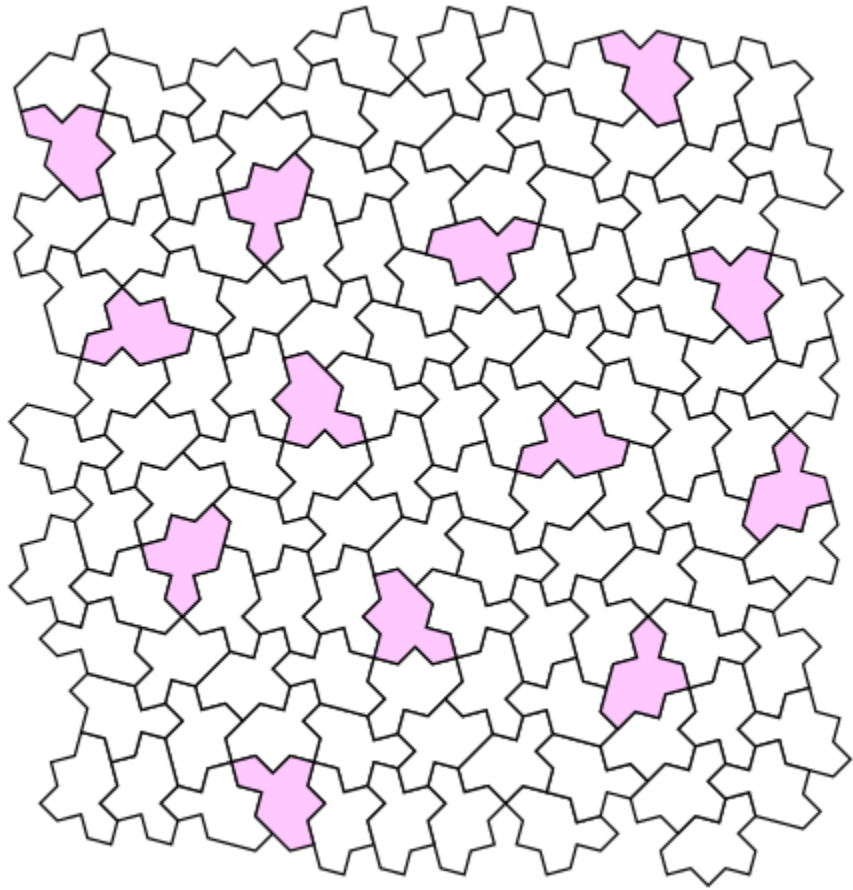
Example patch with all local 2-fold centers (orange color)



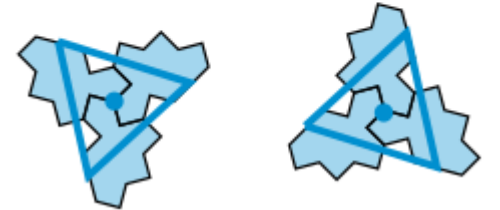
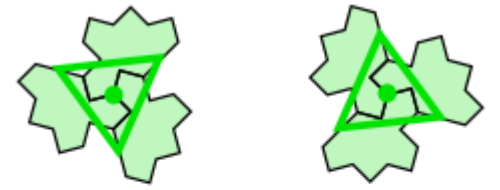
Local 2-fold



Same patch with all local 3-fold centers (green and blue colors)



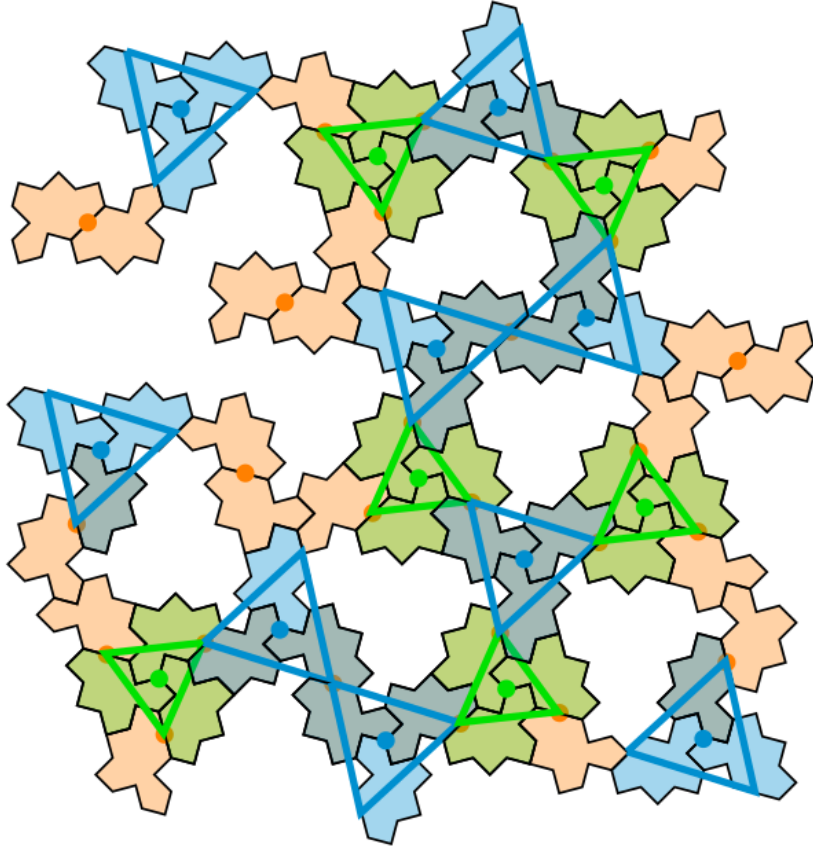
3-fold-S



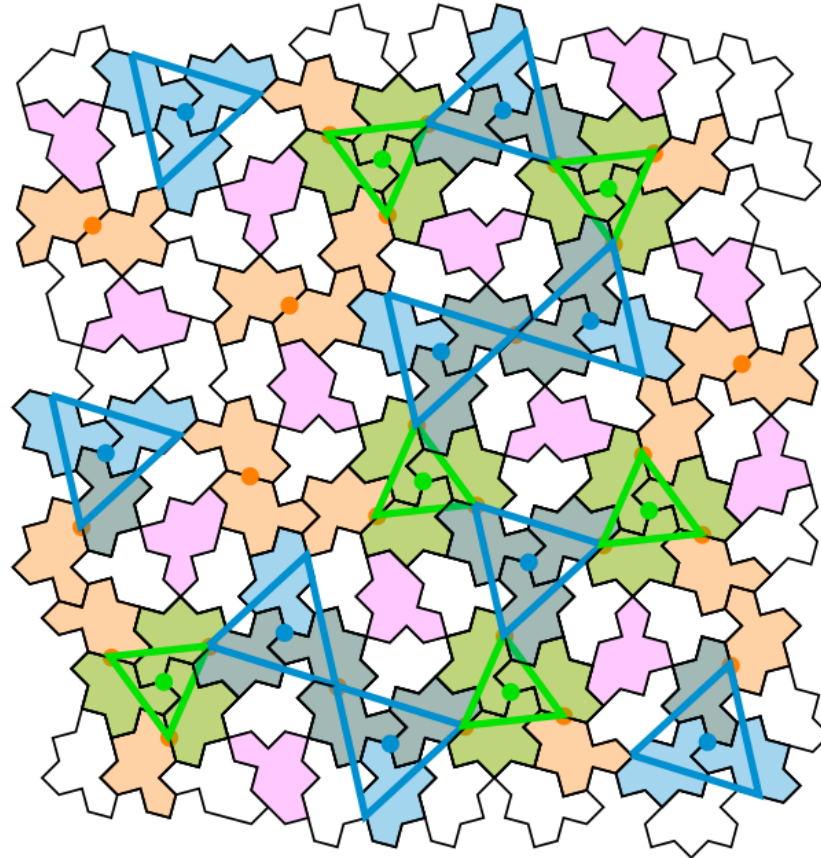
3-fold-L

Superposition of (orange/blue/green) colors

2-fold

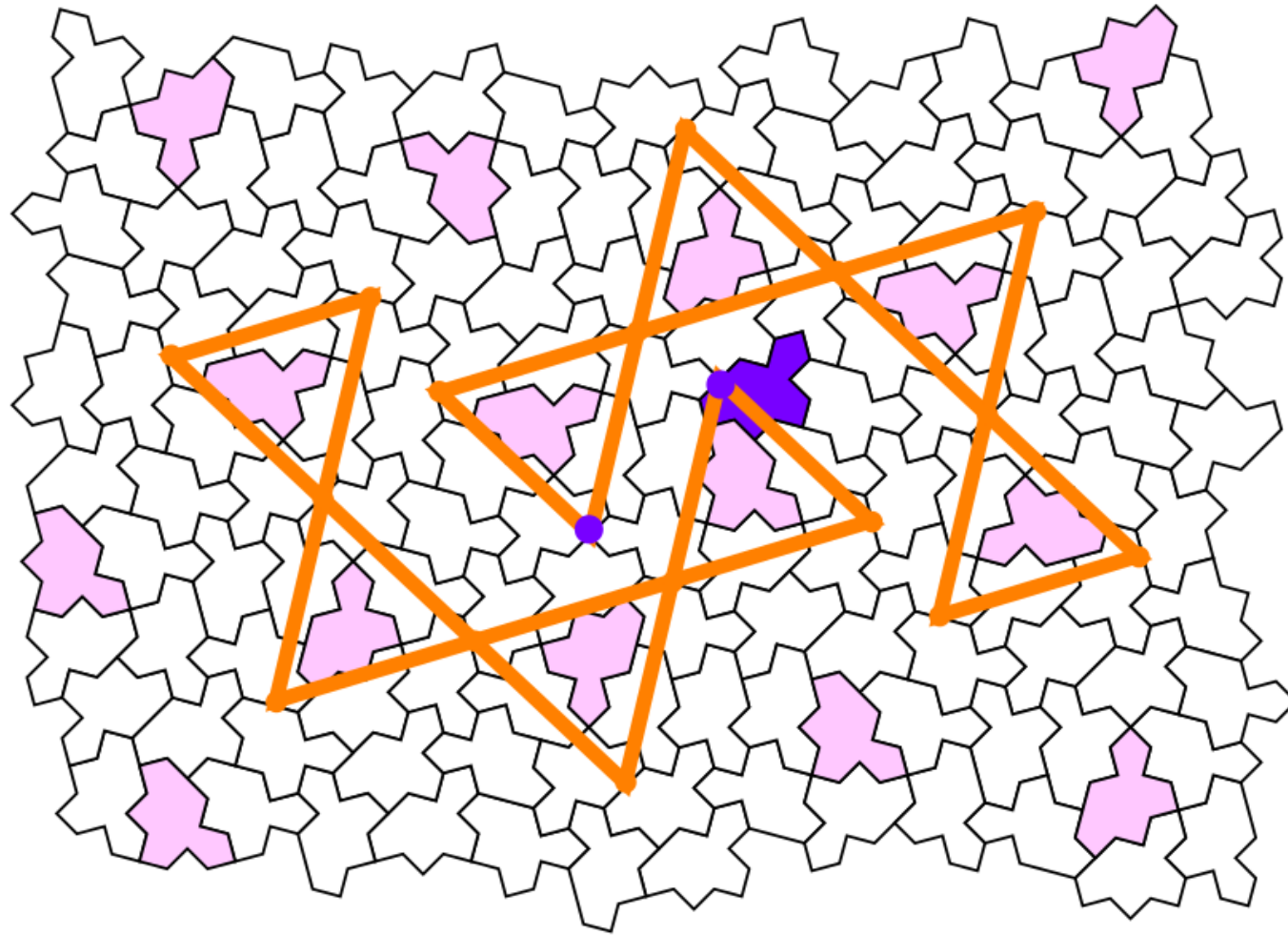


3-fold-S

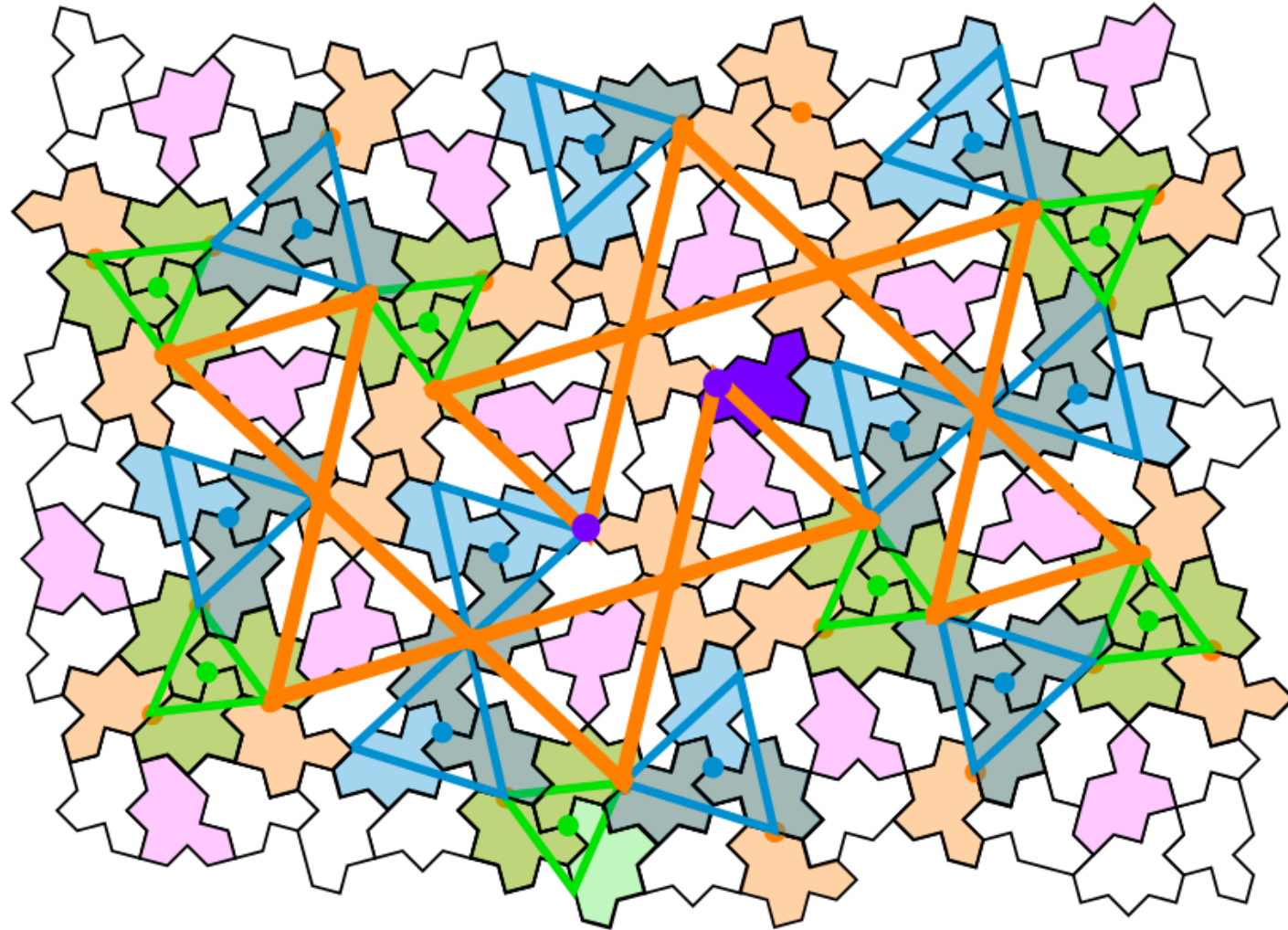


3-fold-L

2-arms

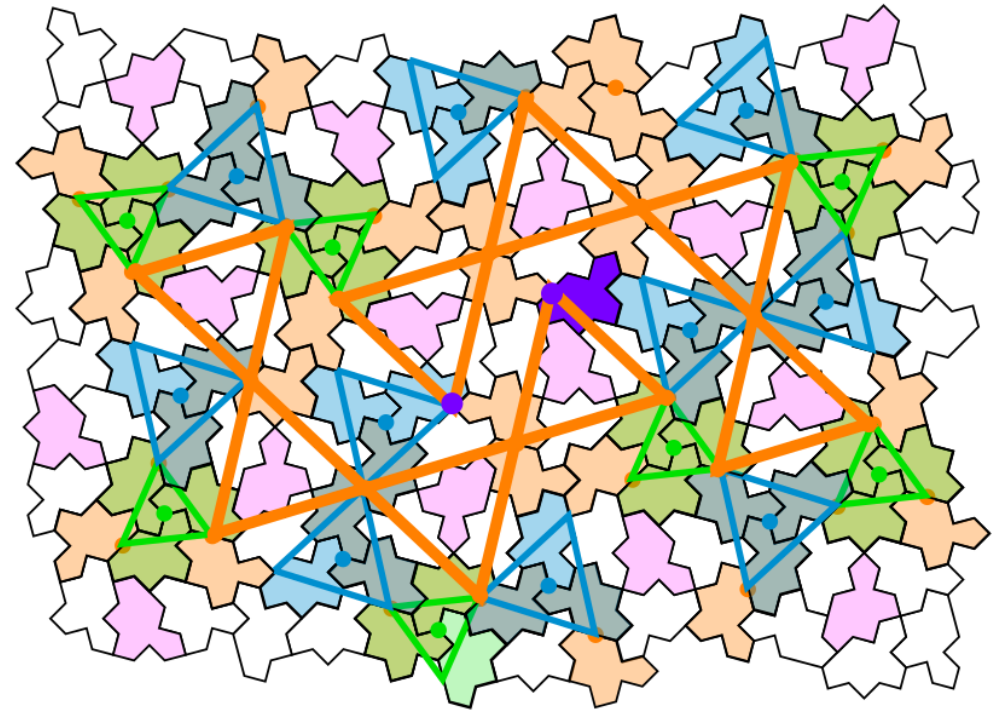
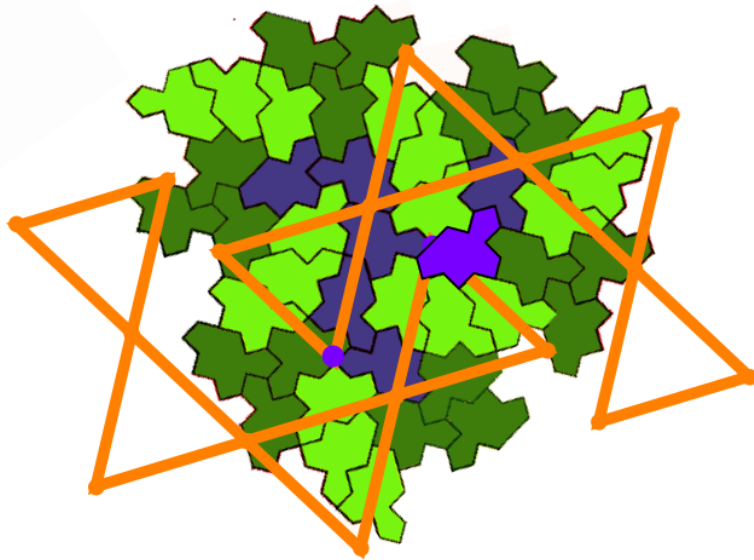
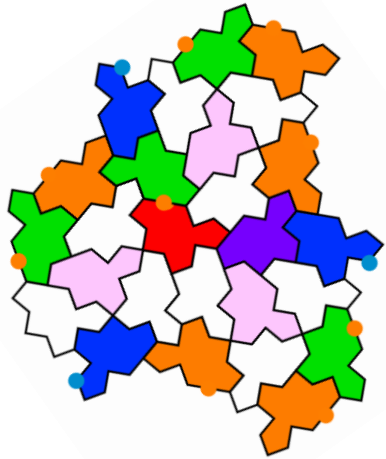


ALL EVEN tiles:
orange/blue/green colors + one violet isolated tile



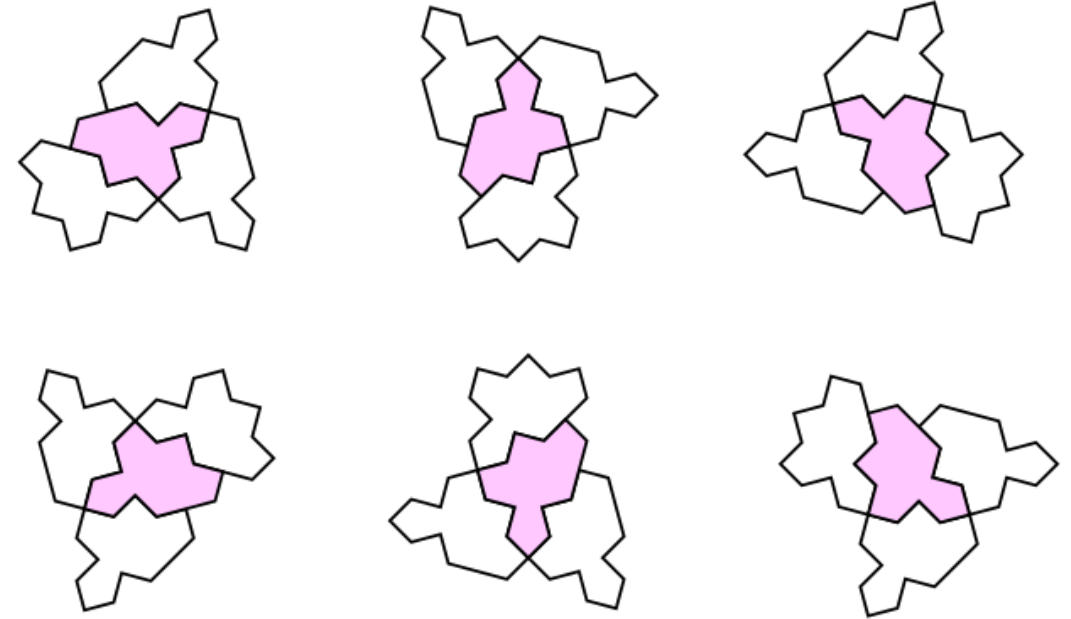
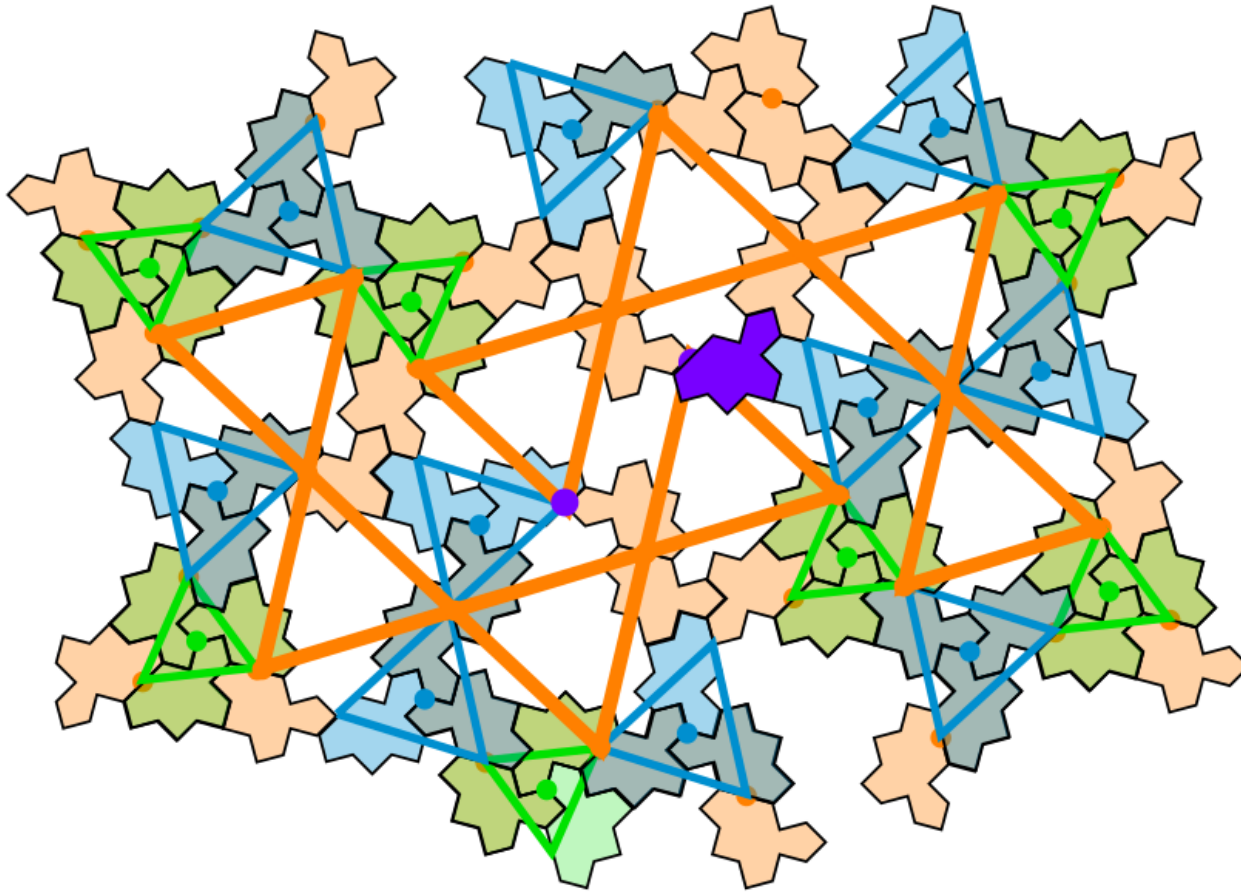
■ 2-arms

ALL EVEN tiles: orange/blue/green colors + one violet isolated tile



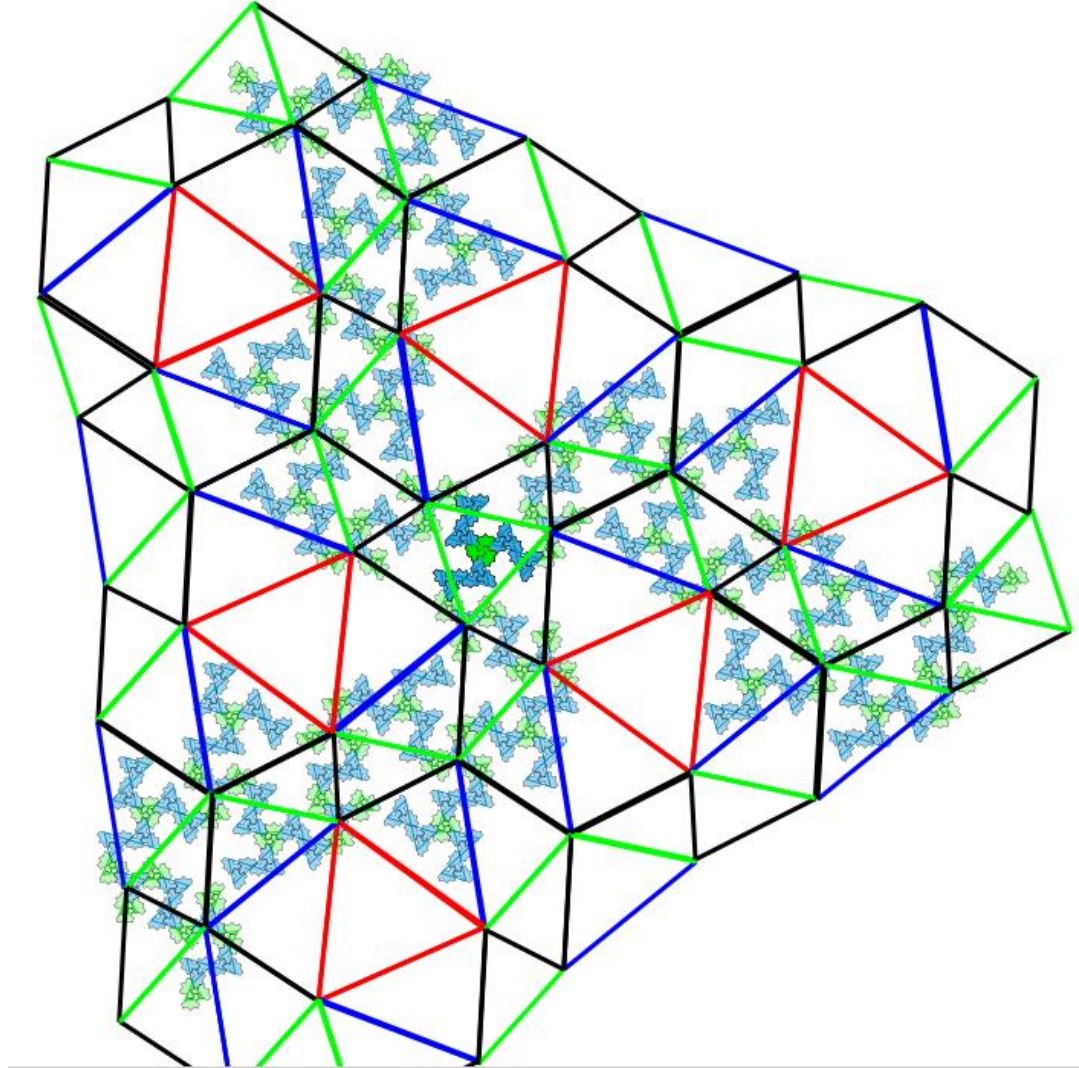
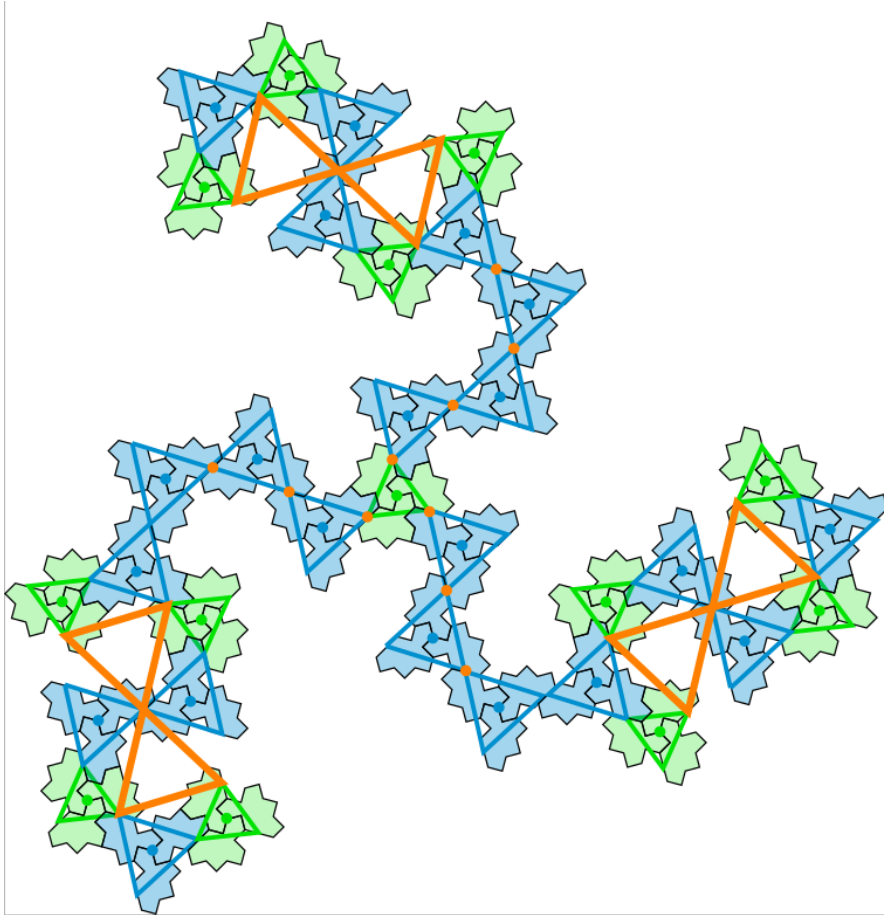
- 2-arms and superposition with chimera pattern

Orange/blue/green + violet: a continuous cluster of ODD tiles around small clusters.



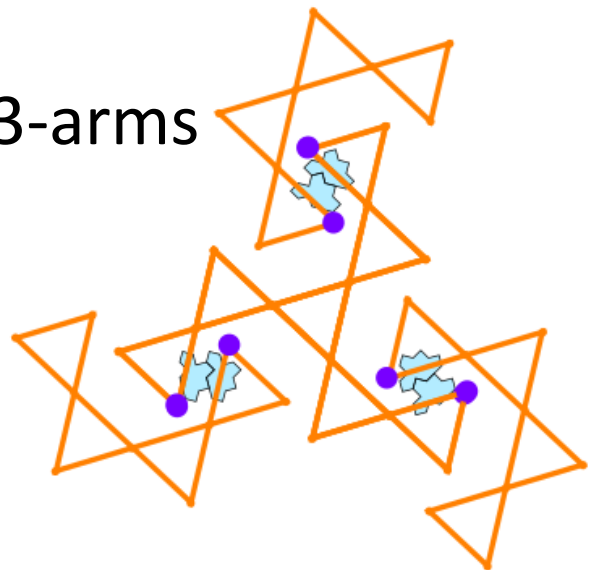
- Small clusters of 4 tiles (one ODD in pink γ_2 + three EVEN γ_1 , Δ and Σ)

To continue ? generate star clusters by applying locally 2-fold and 3-fold symmetry between EVEN monotiles

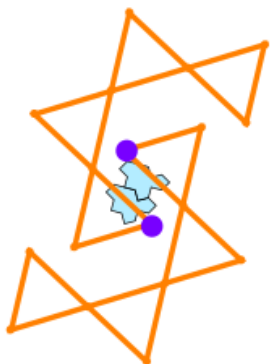


Three patterns in the whole tiling 😊

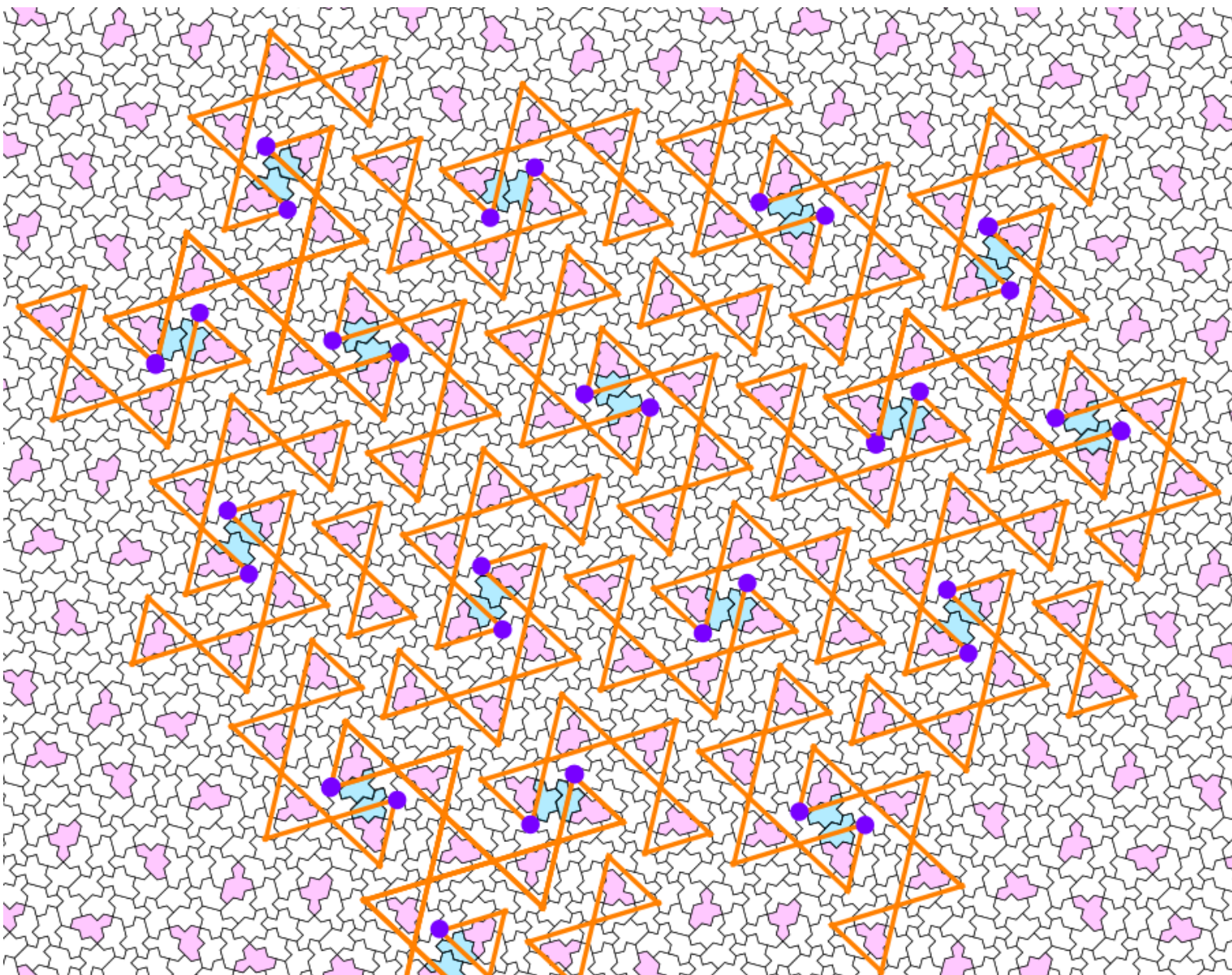
■ 3-arms



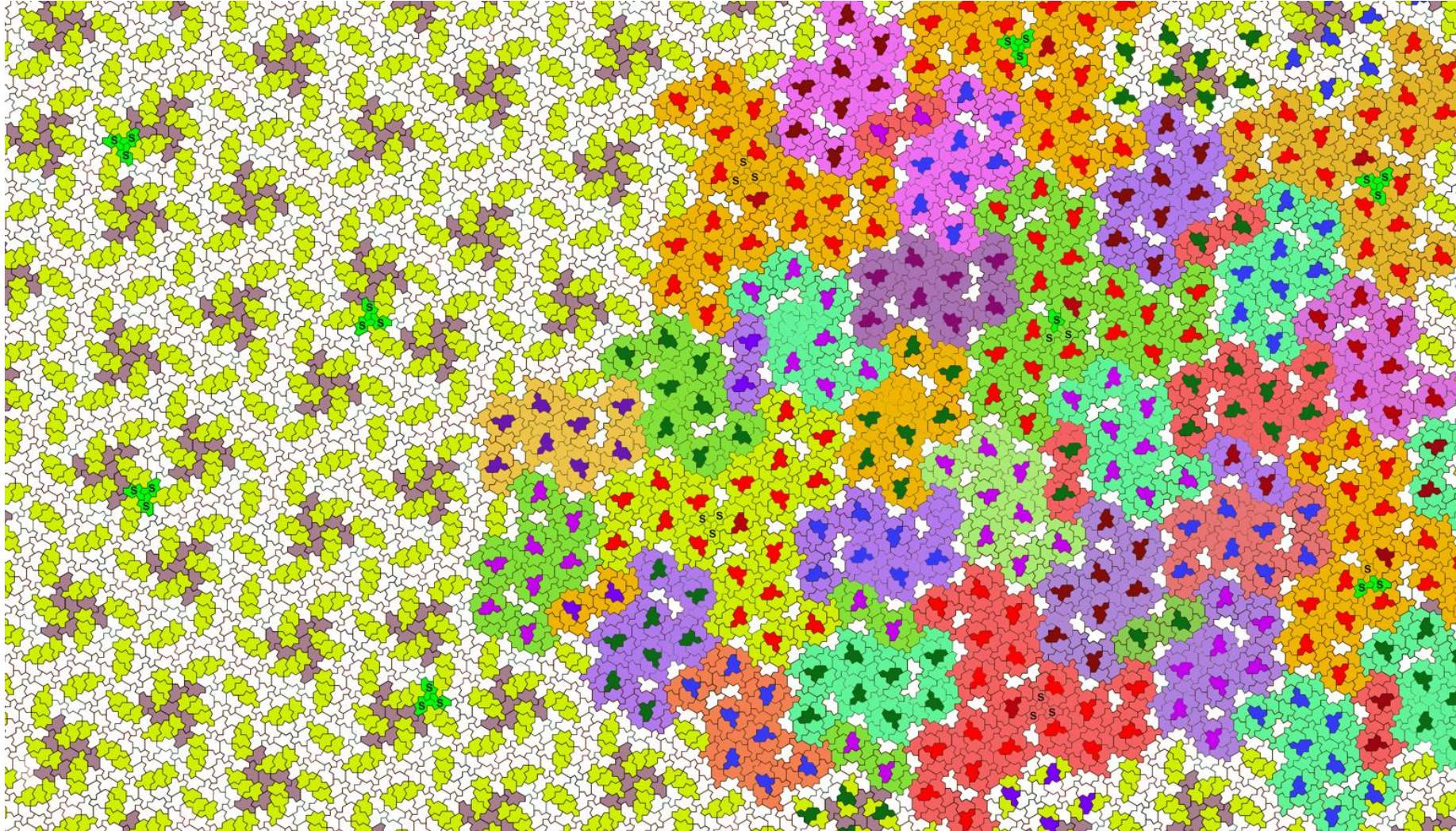
■ 2-arms



■ diabolos

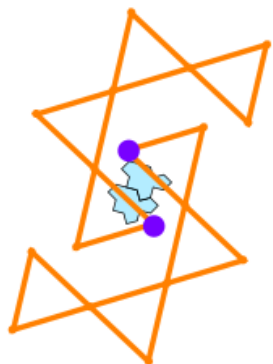
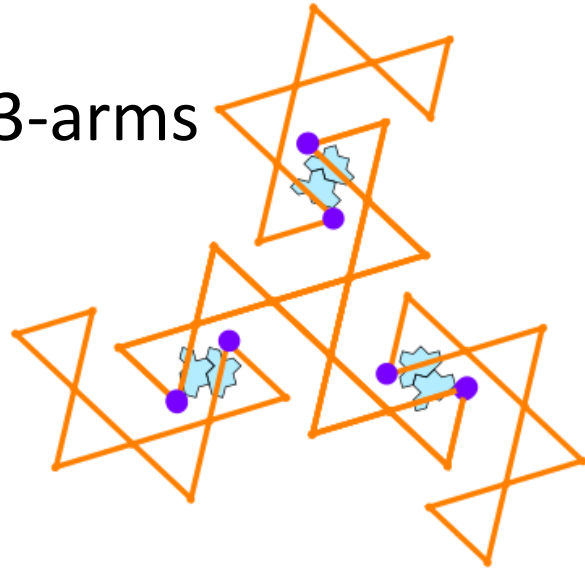


Diabolo/2-arms/3-arms at level 2 by Boris Horvat



Frequency of the three patterns (Boris Horvat)

■ 3-arms



■ 2-arms

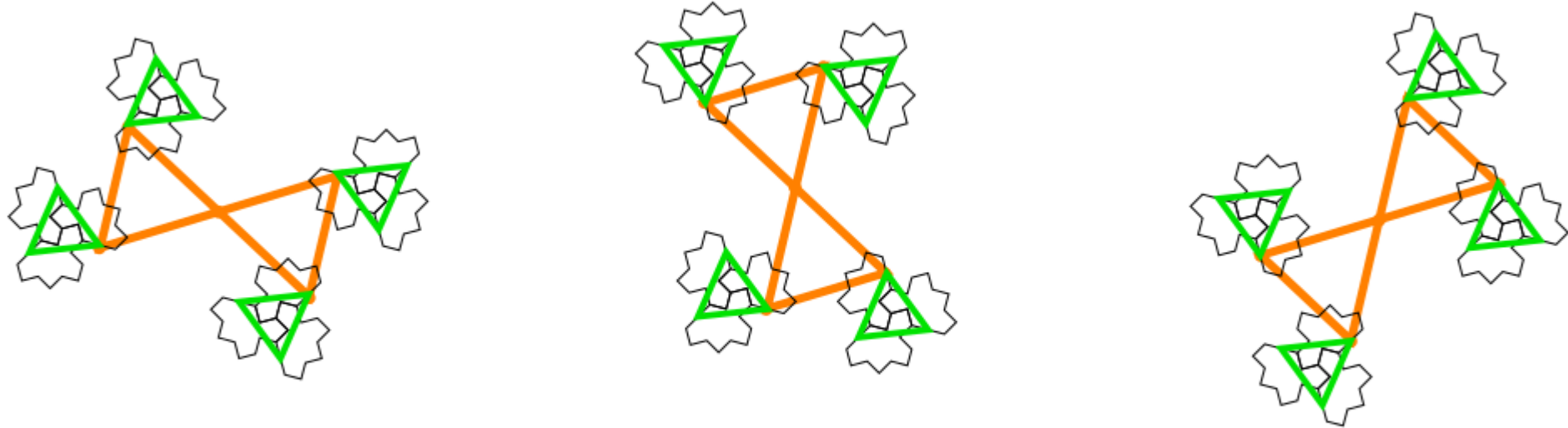
■ diablo



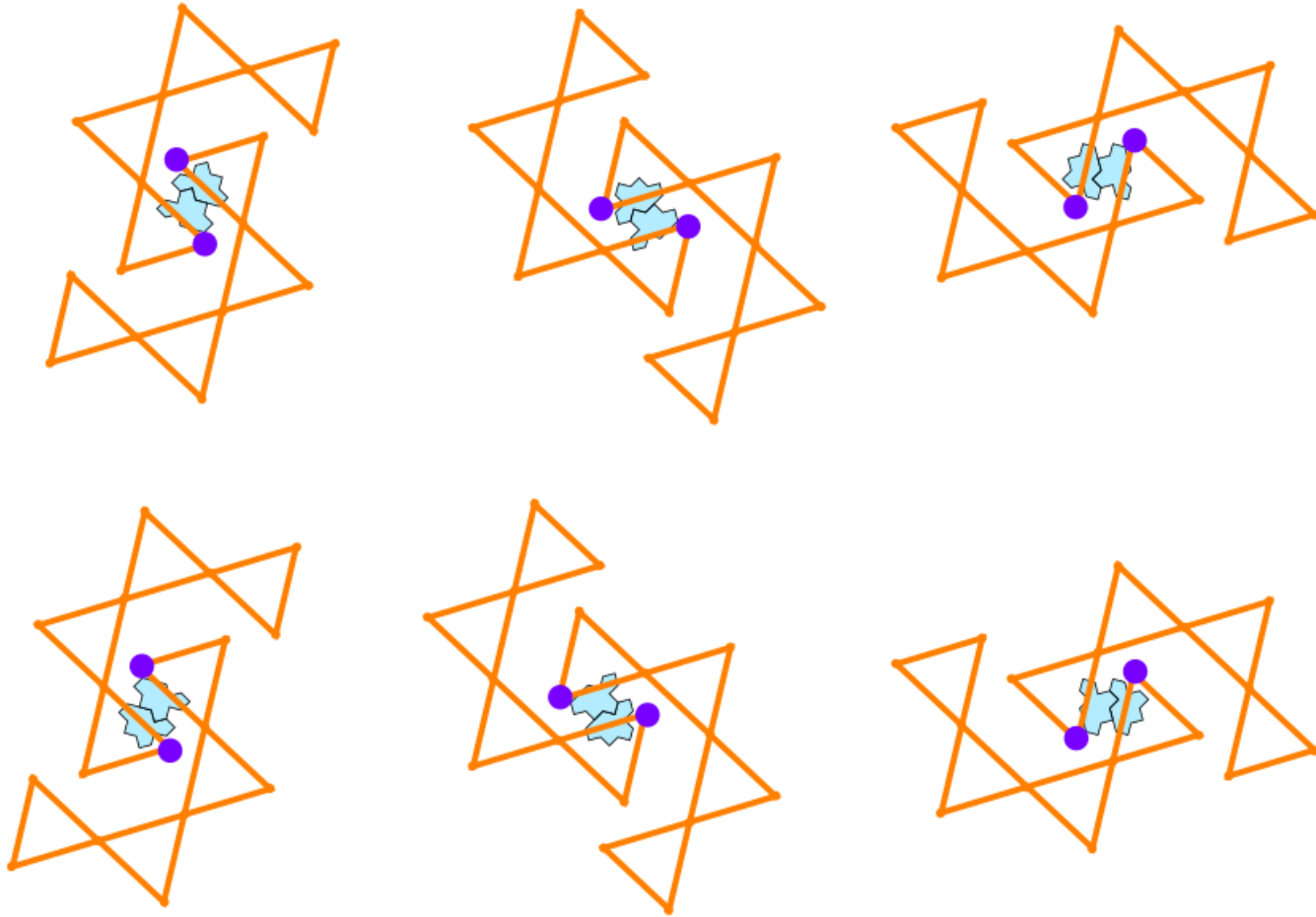
After colouring (manually) the tiling with your objects, I realized that the frequency of diabolos is exactly twice the frequency of 3-way objects. My observations confirm that **every 3-way object (Object-19) has exactly two diabolos attached to two of its arms in a consistent topological orientation**. This allows us to define a new primary super-structure: Object-23 ($1 \times 3\text{-way} + 2 \times \text{Diablo}$), thus partitioning the tiling into two large objects only (Object-23, and Object-8). The 3-way objects (the nodes of the tiling, centred around lone stars) connect to each other via three distinct modes:

- 1) Direct Contact: Arm-to-arm contact between two 3-way objects (Object-19).
- 2) Diablo Link: Connection mediated by a single diablo (Object-2).
- 3) 2-way link: Connection mediated by a 2-way object (Object-8).

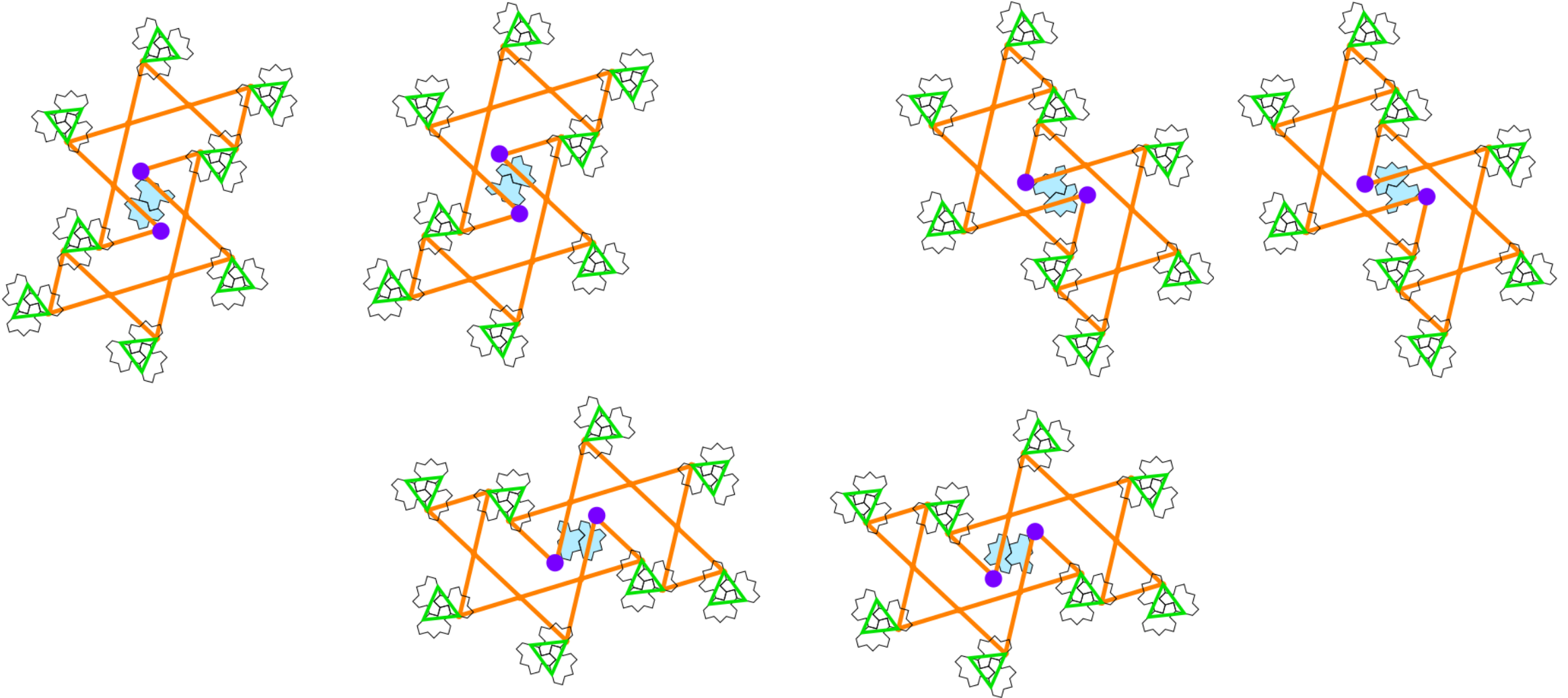
three orientations for diabolos



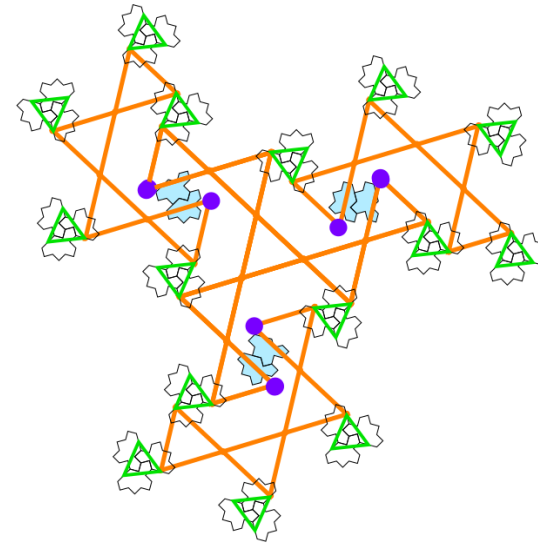
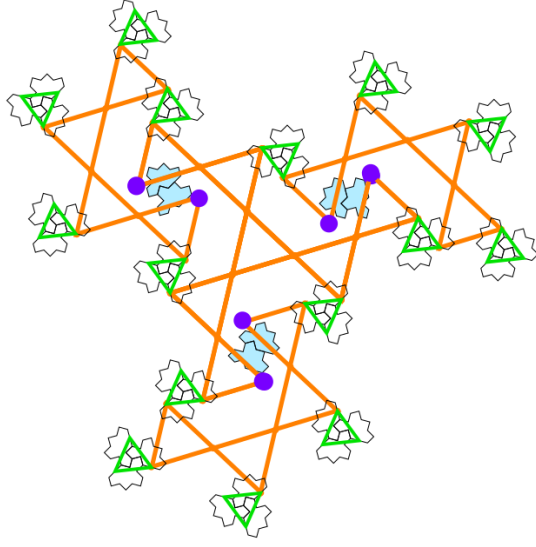
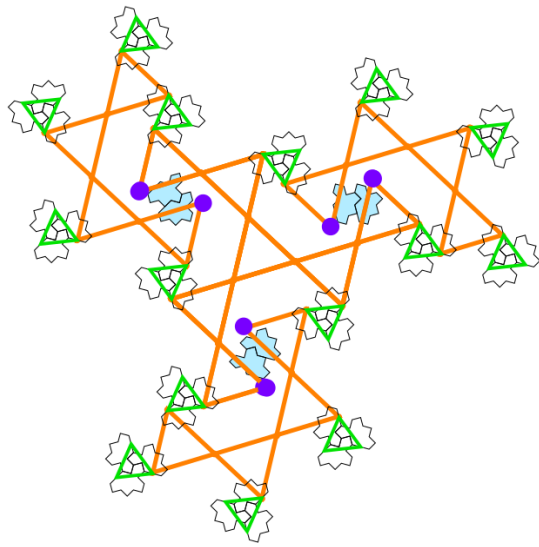
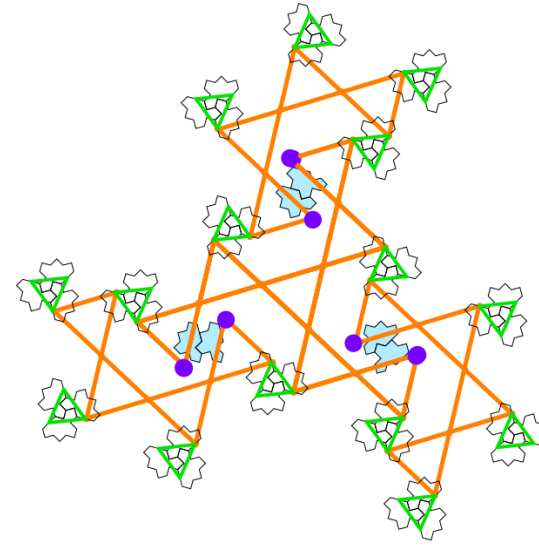
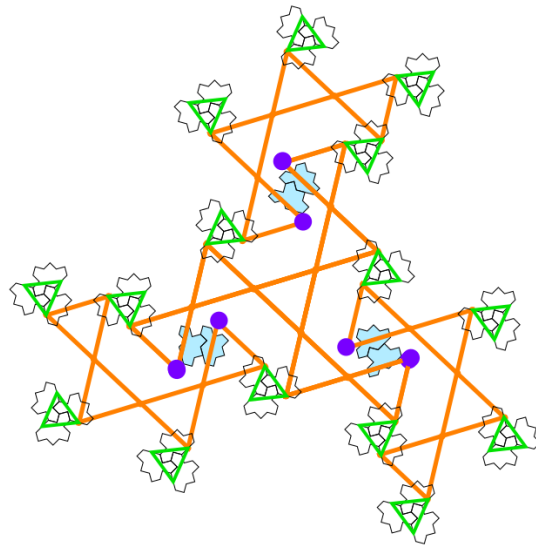
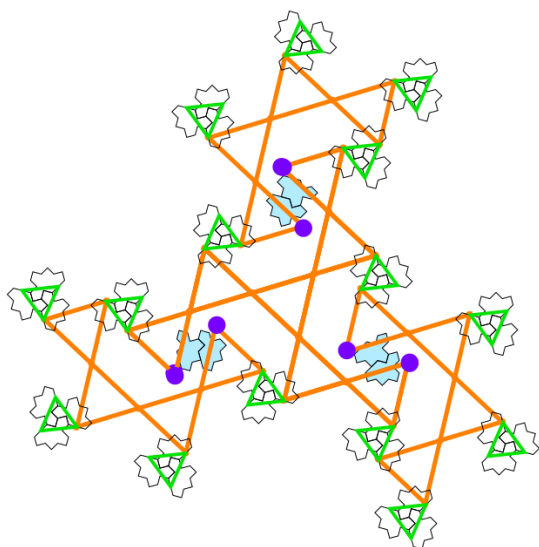
Six orientations of the 2-arms



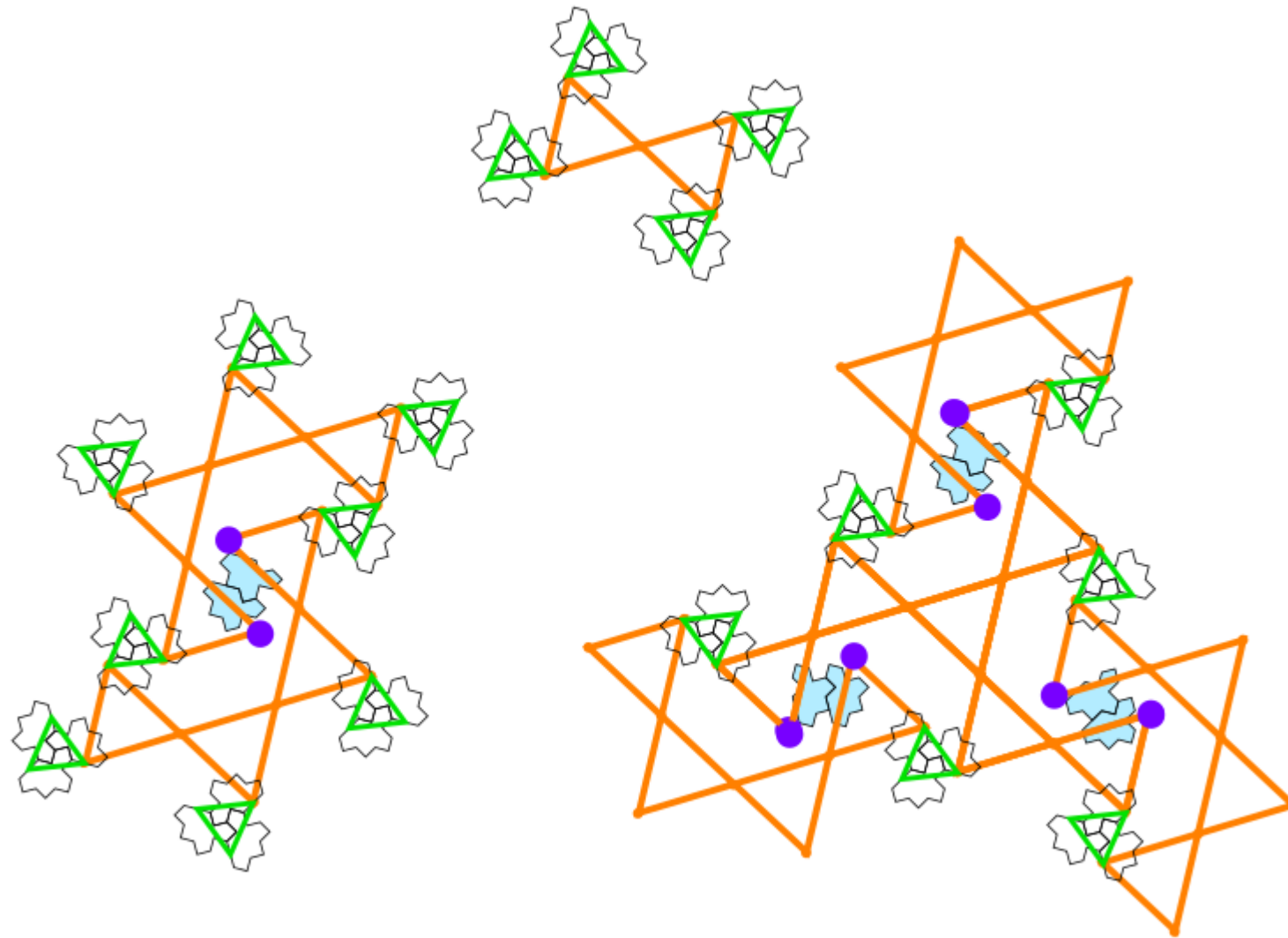
Six orientations of the 2-arms



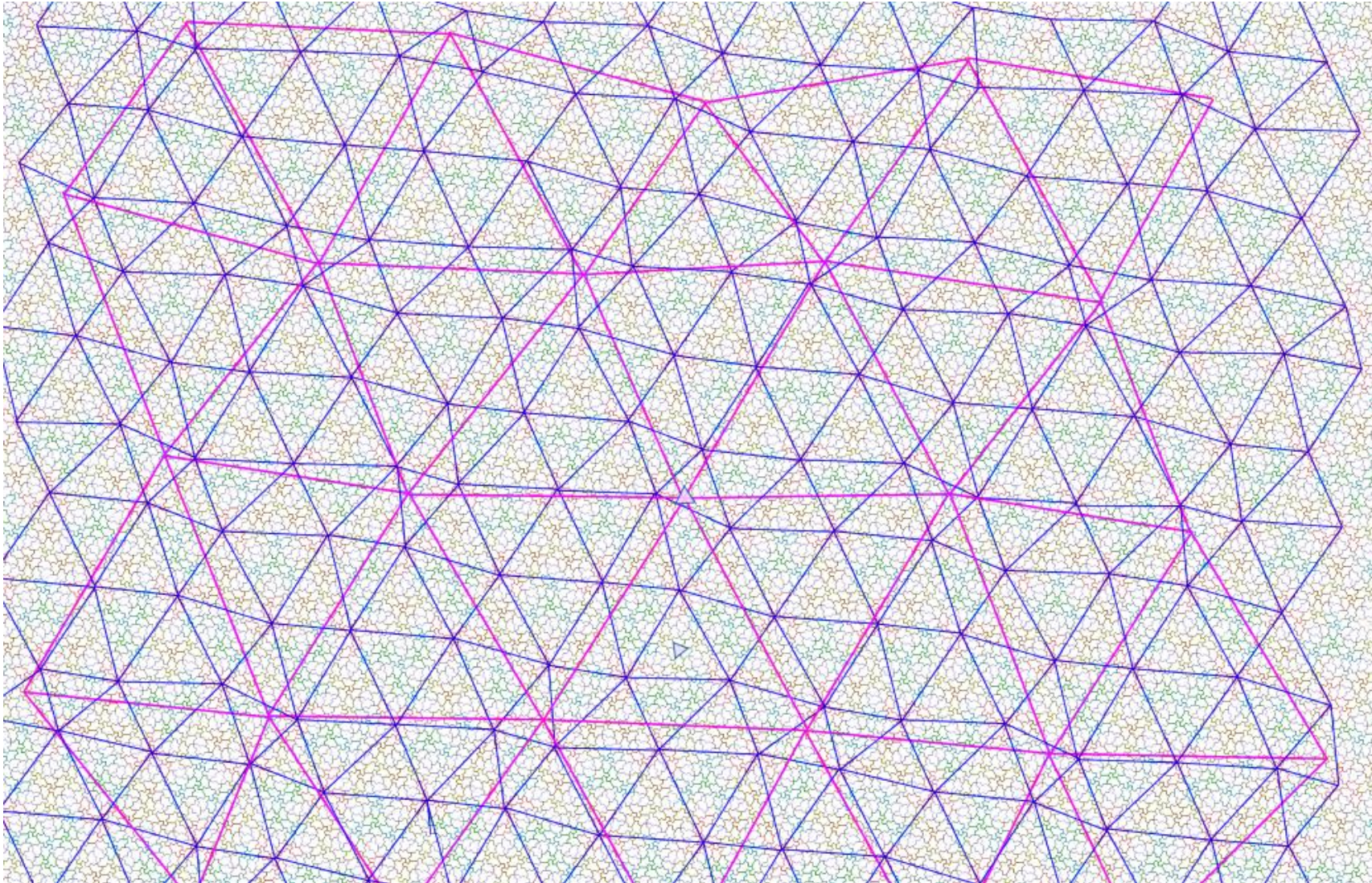
six orientations of the 3-arms



New puzzle

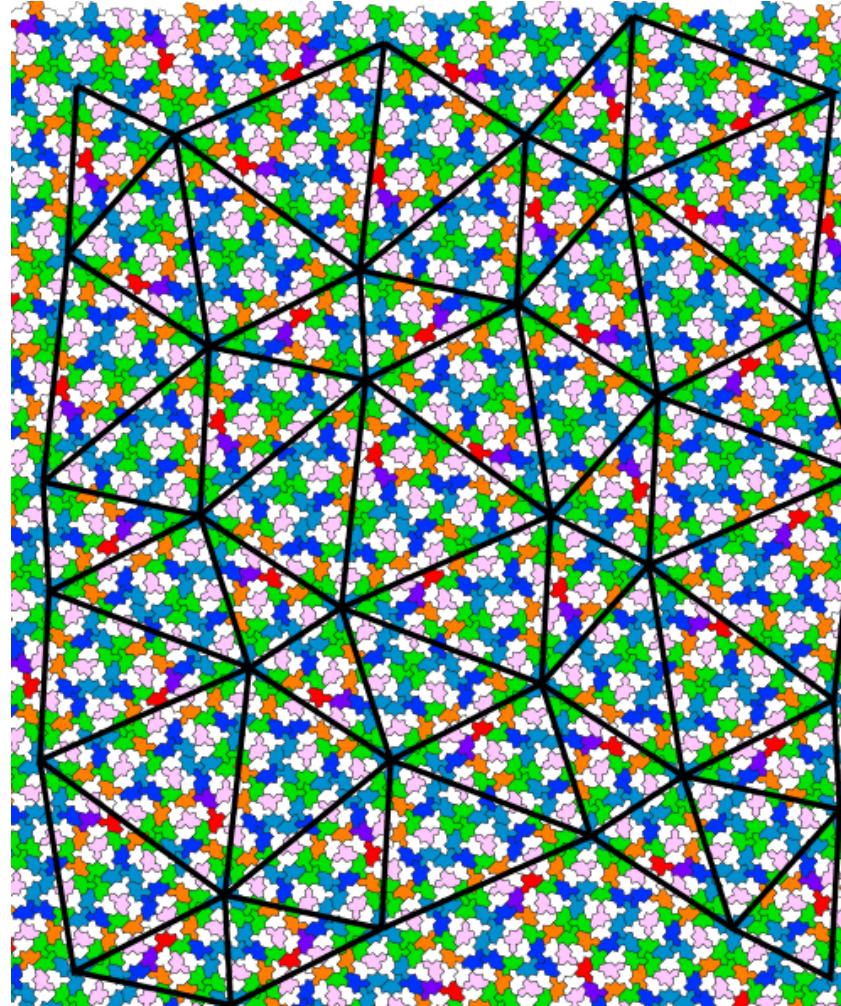
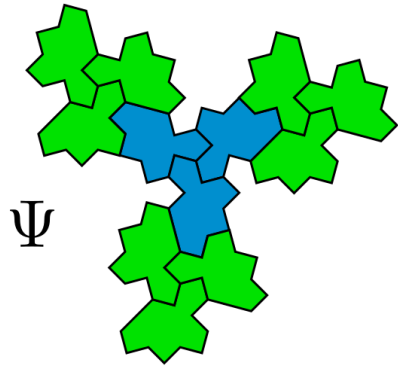


Triangular array by Boris Horvat



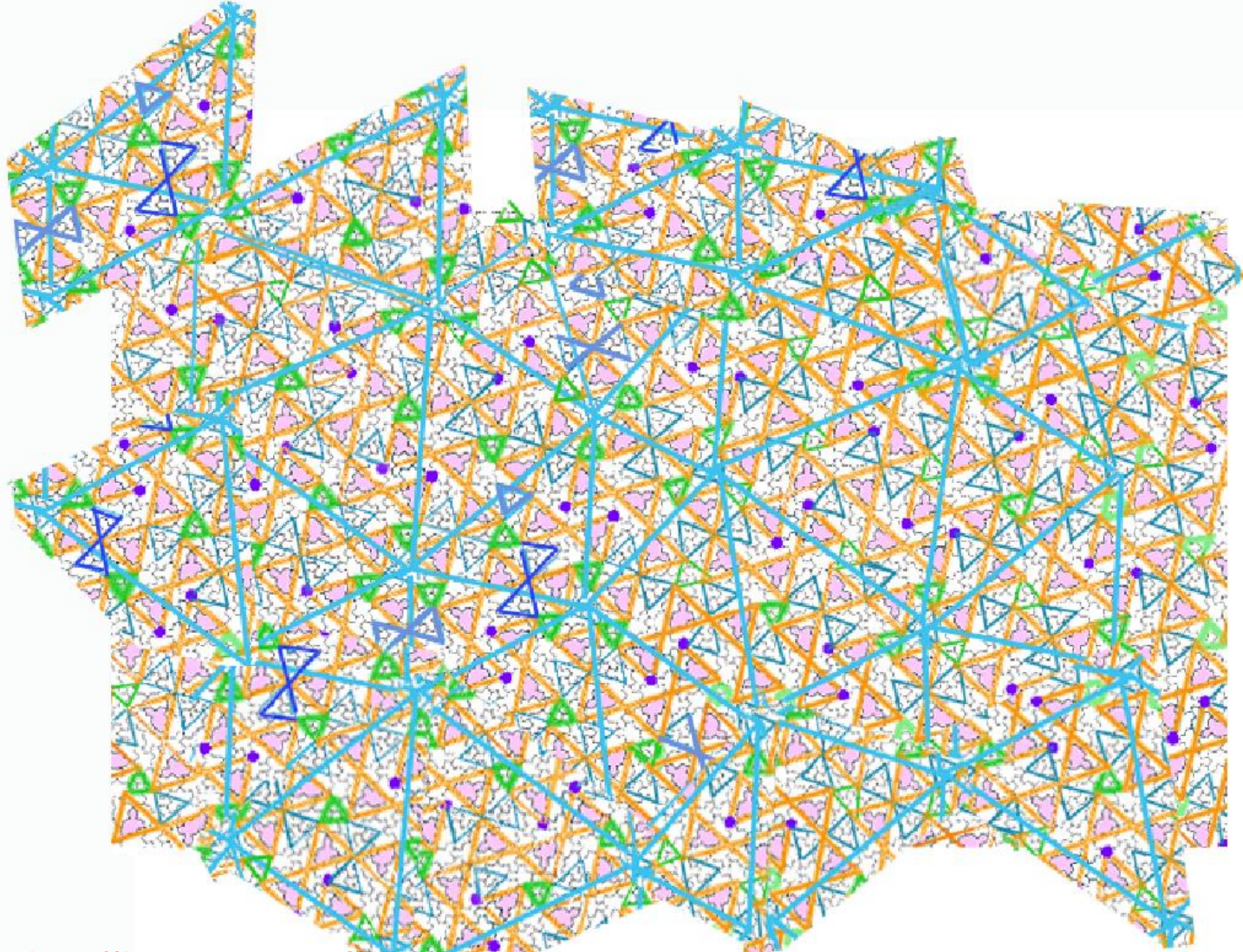
blue color: triangular array between the “lone stars”

Inflation using python code



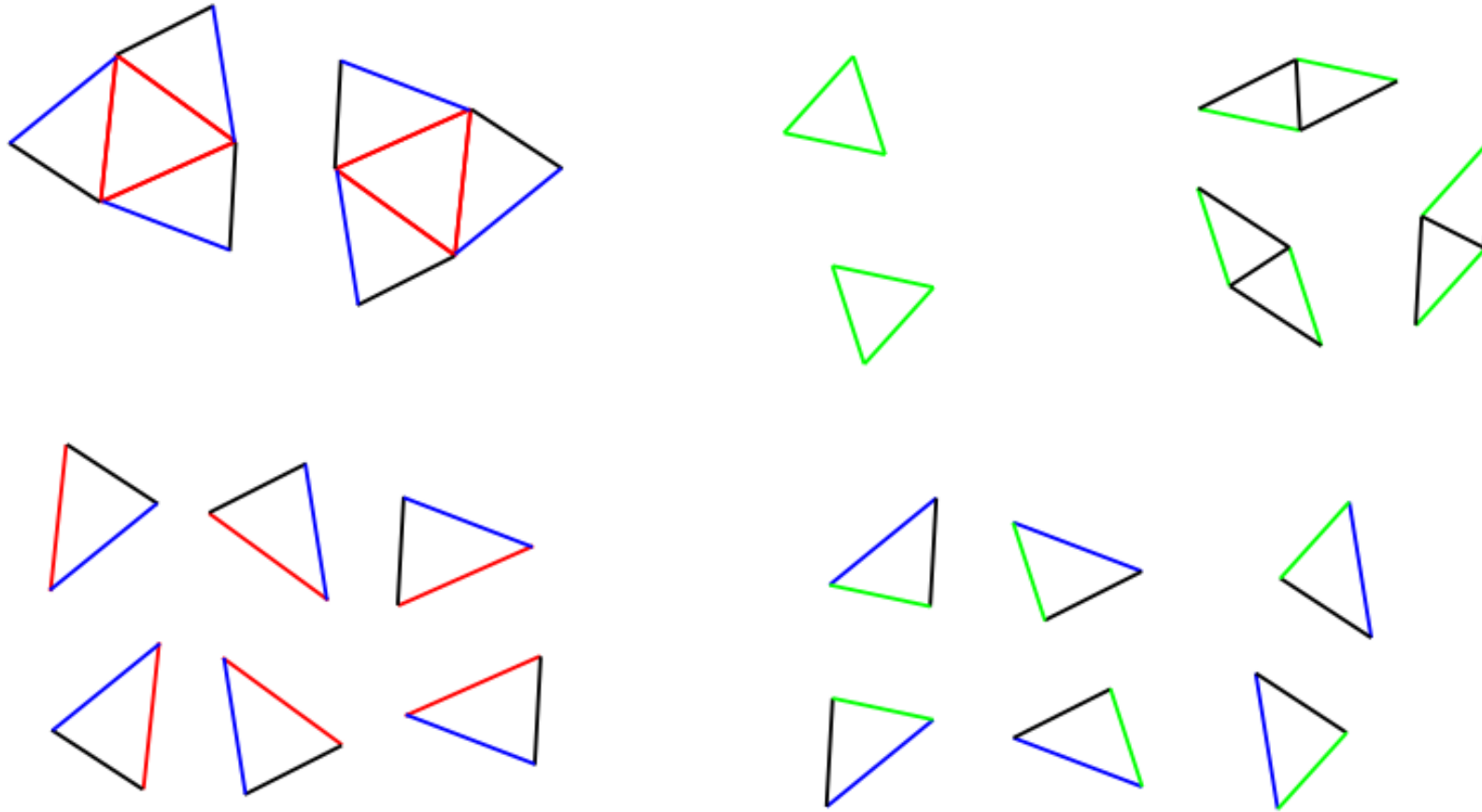
- Triangular array between large blue stars made of three Ψ tiles surrounded by three S2 green stars

Triangular networks by Jean-François



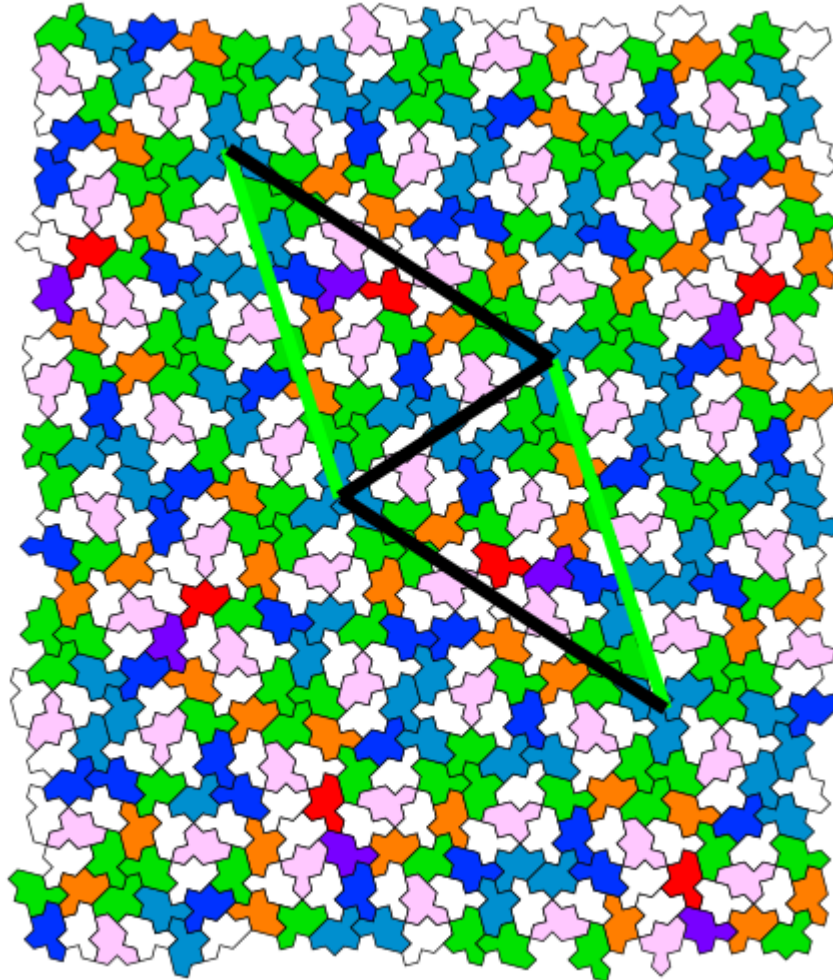
Same network between the “lone stars”

A finite set of triangles ☺

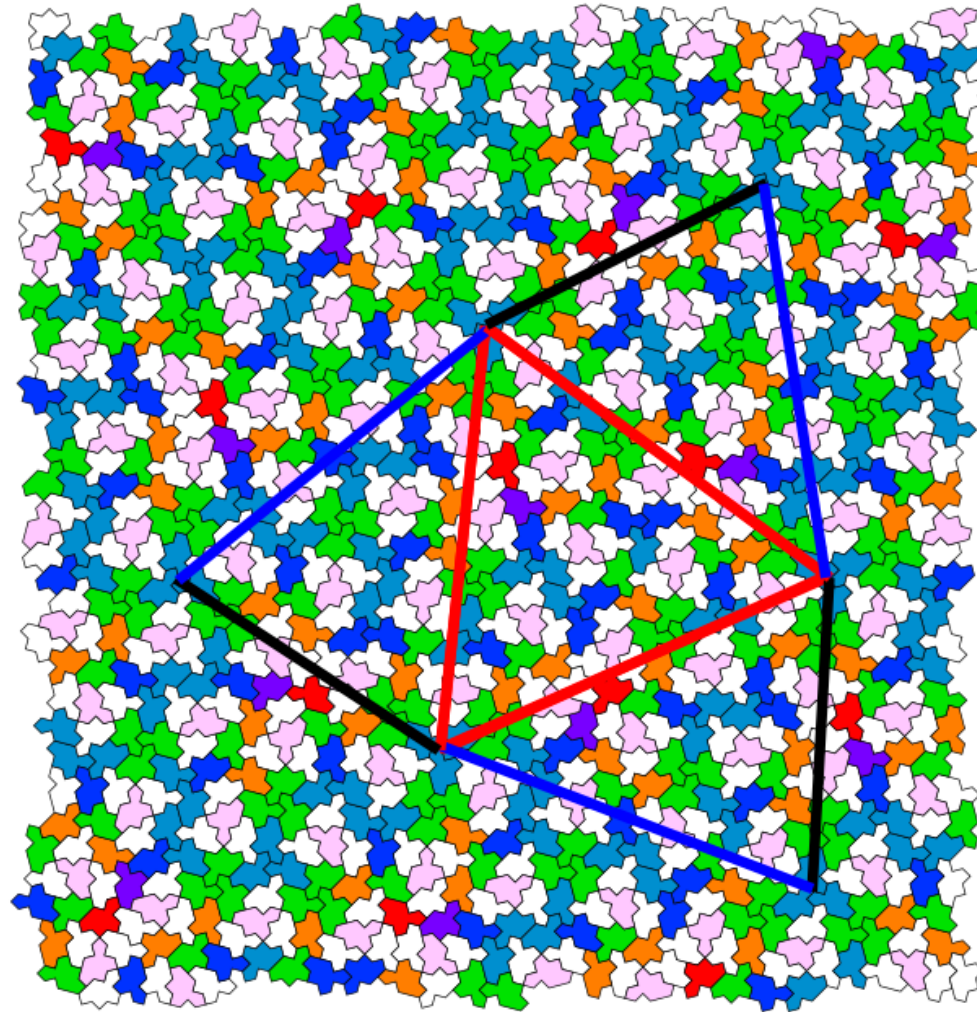


5 different triangles with two regular ones -> 4 edge lengths

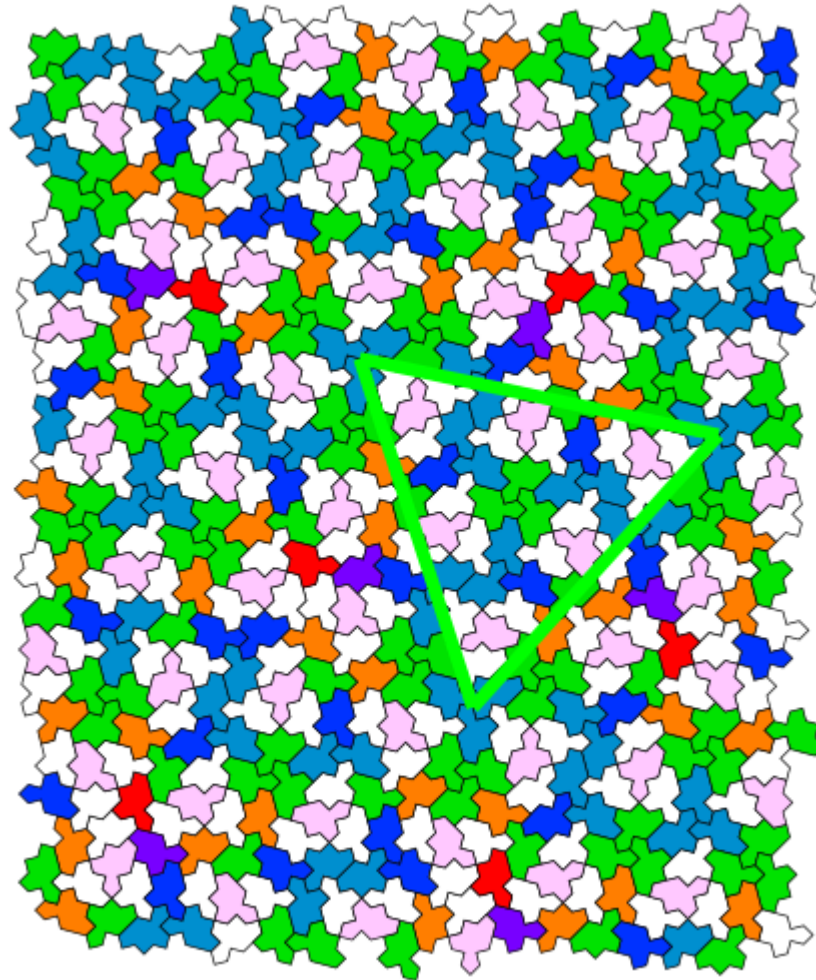
Set of triangles: diablo



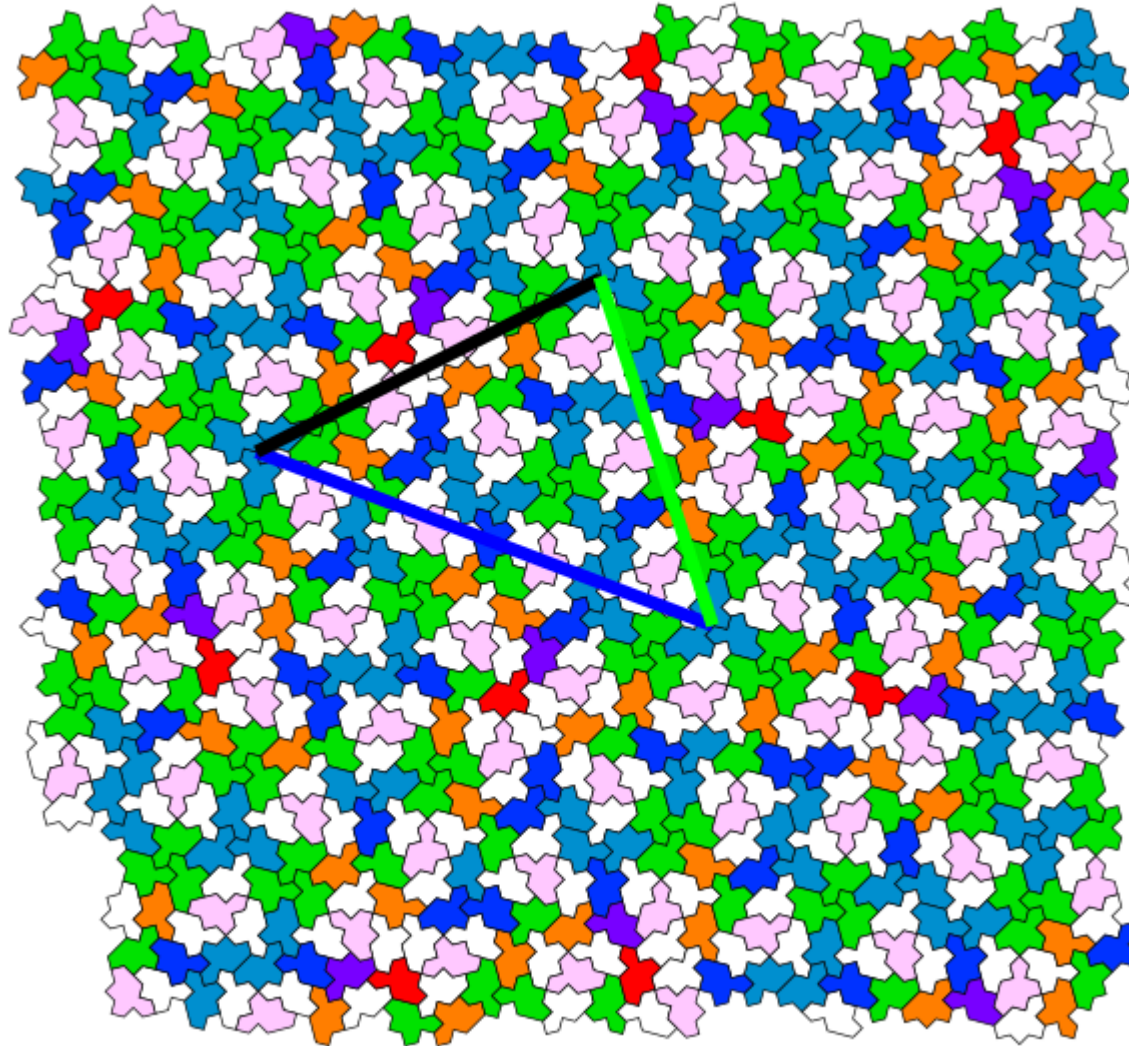
Set of triangles: three arms



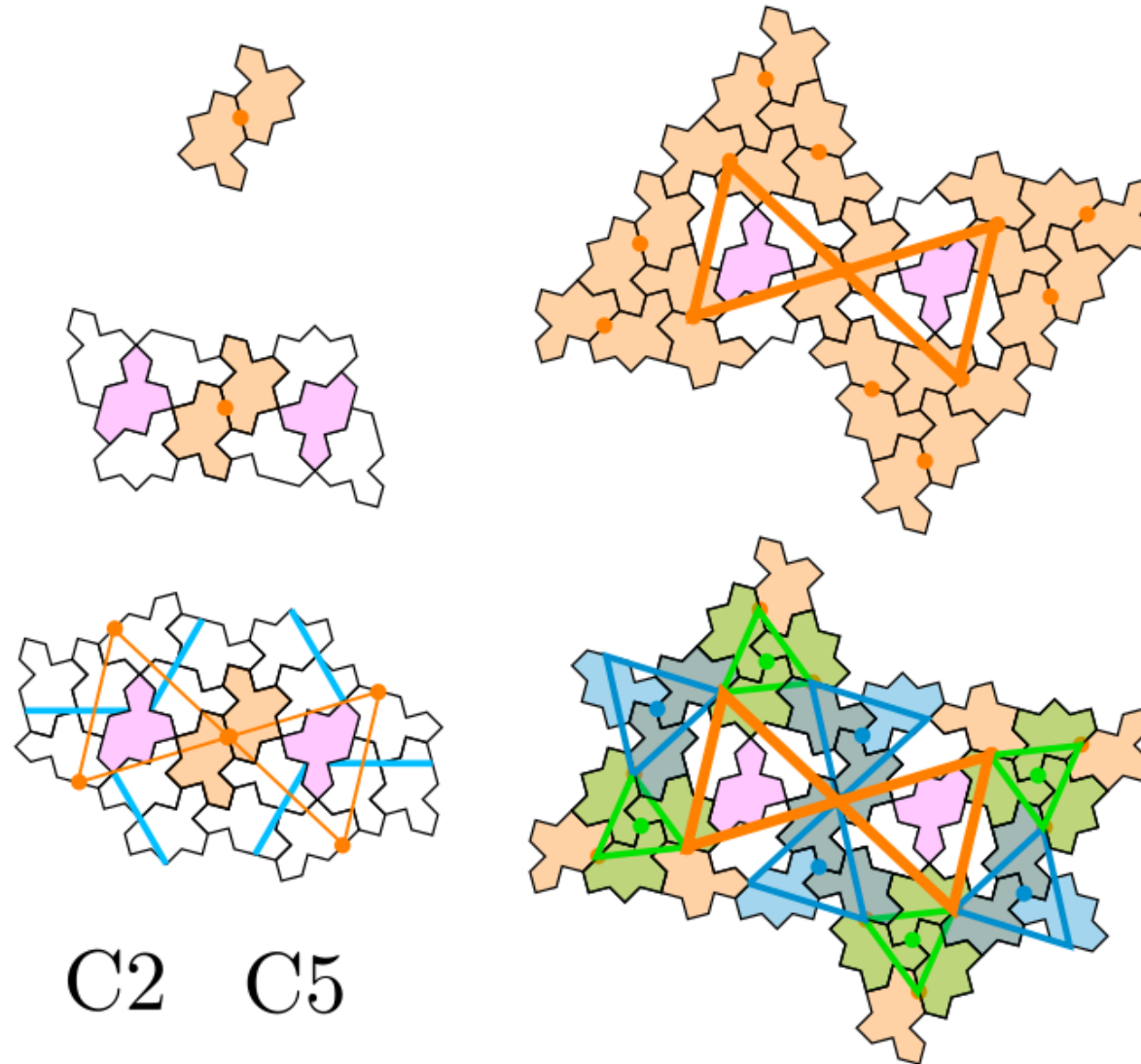
Set of triangles: green triangle



Set of triangles: last triangle

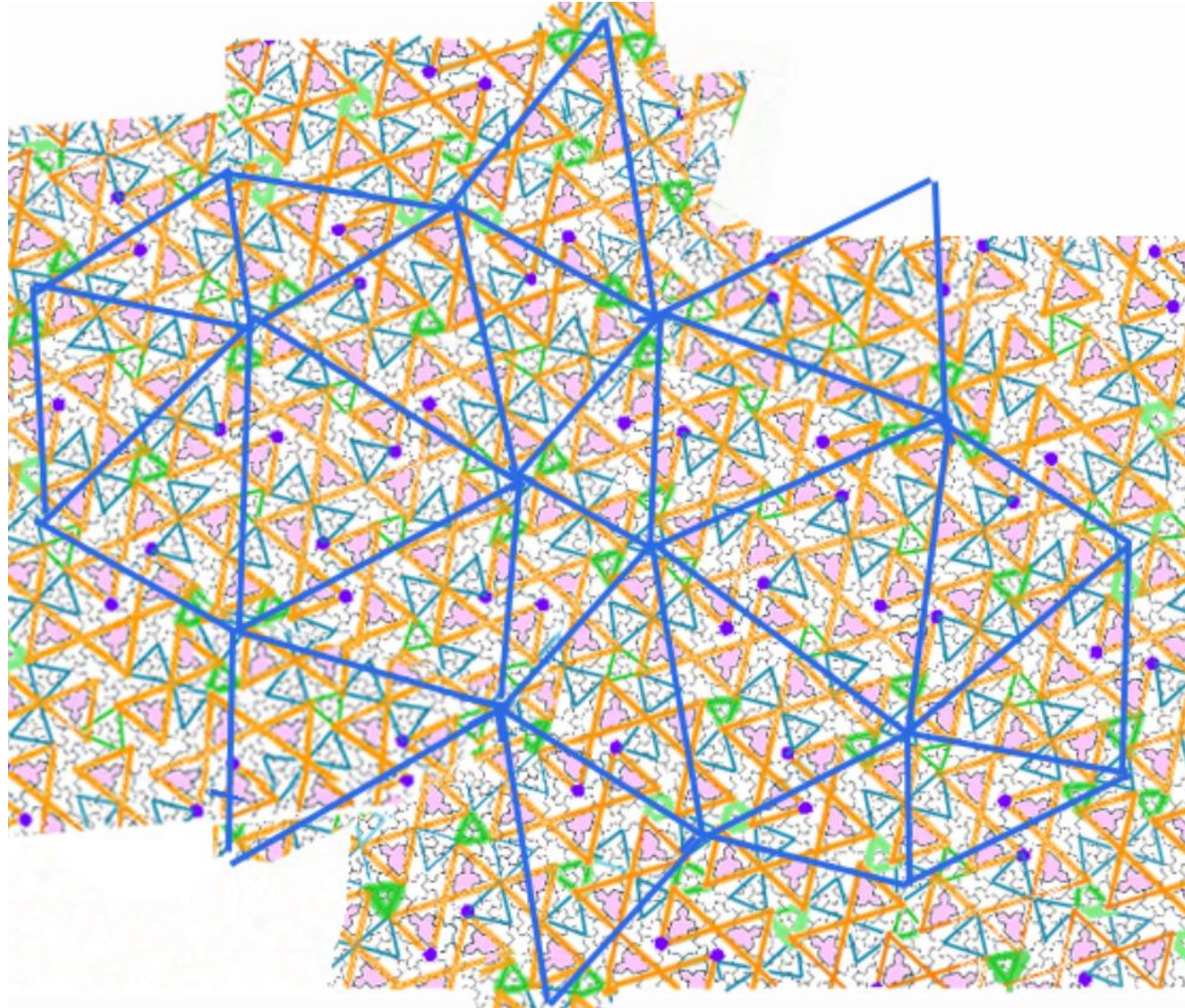


Global 2-fold symmetry: center of a diablo pattern

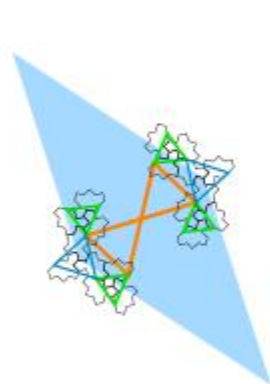
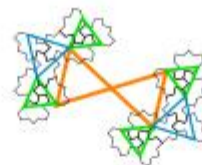


- From local to global 2-fold symmetry ???

Global 2-fold symmetry: center of a diablo pattern



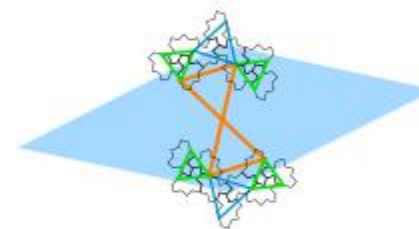
Rhombi for diablo



D1



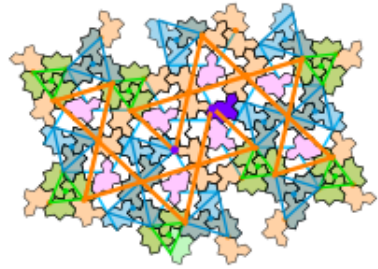
D3



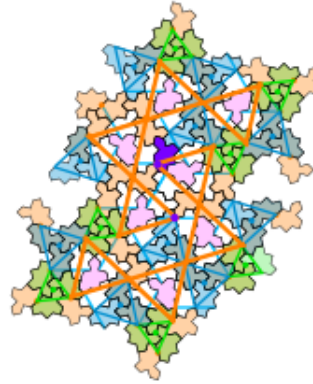
D5

Six ORIENTATIONS OF 2-arms pattern

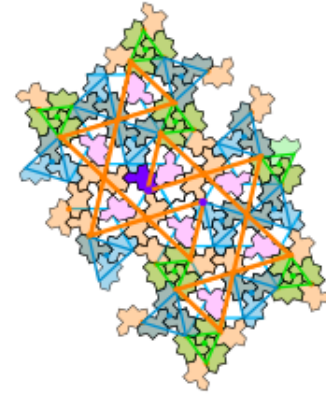
P1



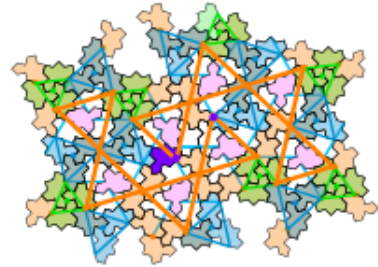
P2



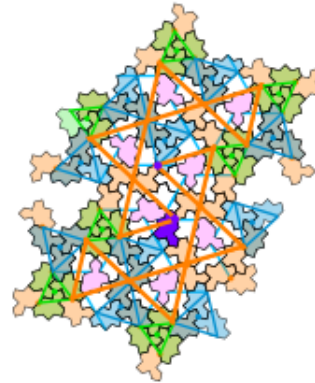
P3



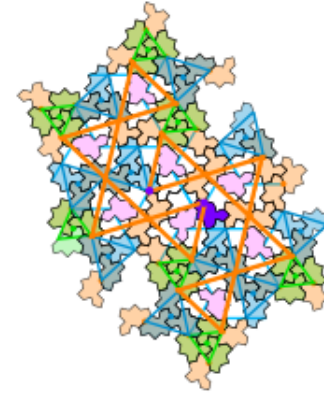
P4



P5



P6



FOUR possible seeds with 3-fold symmetry

- COMBINE Three 2-arms rotated from each other by $2\pi/3$

3-fold-S1



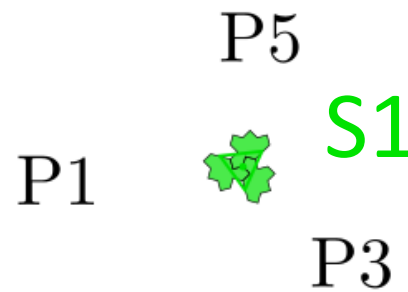
3-fold-S2



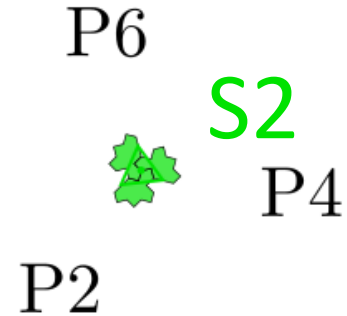
3-fold-L1



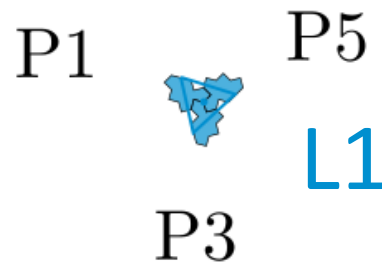
3-fold-L2



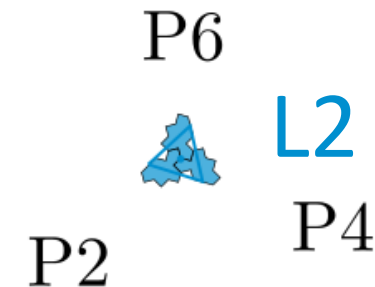
S1



S2



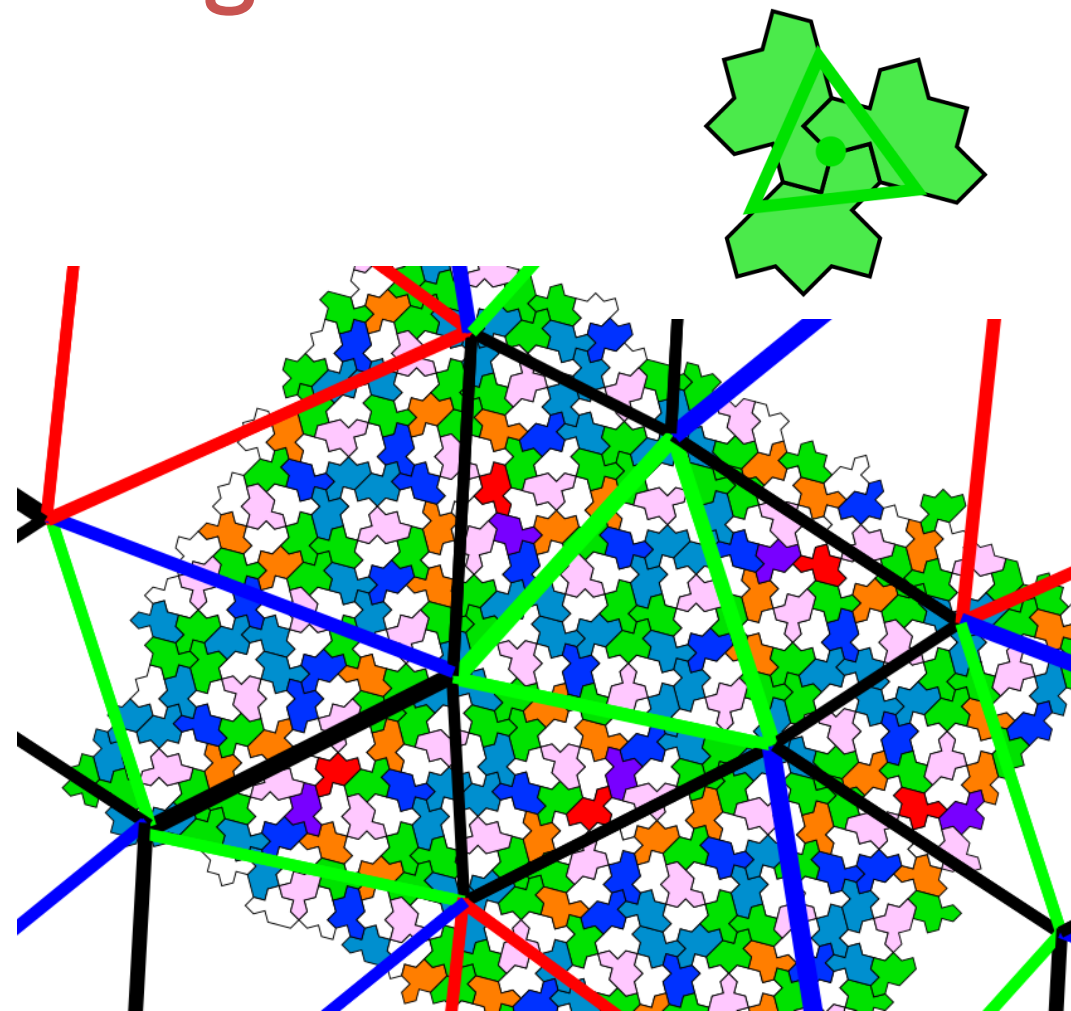
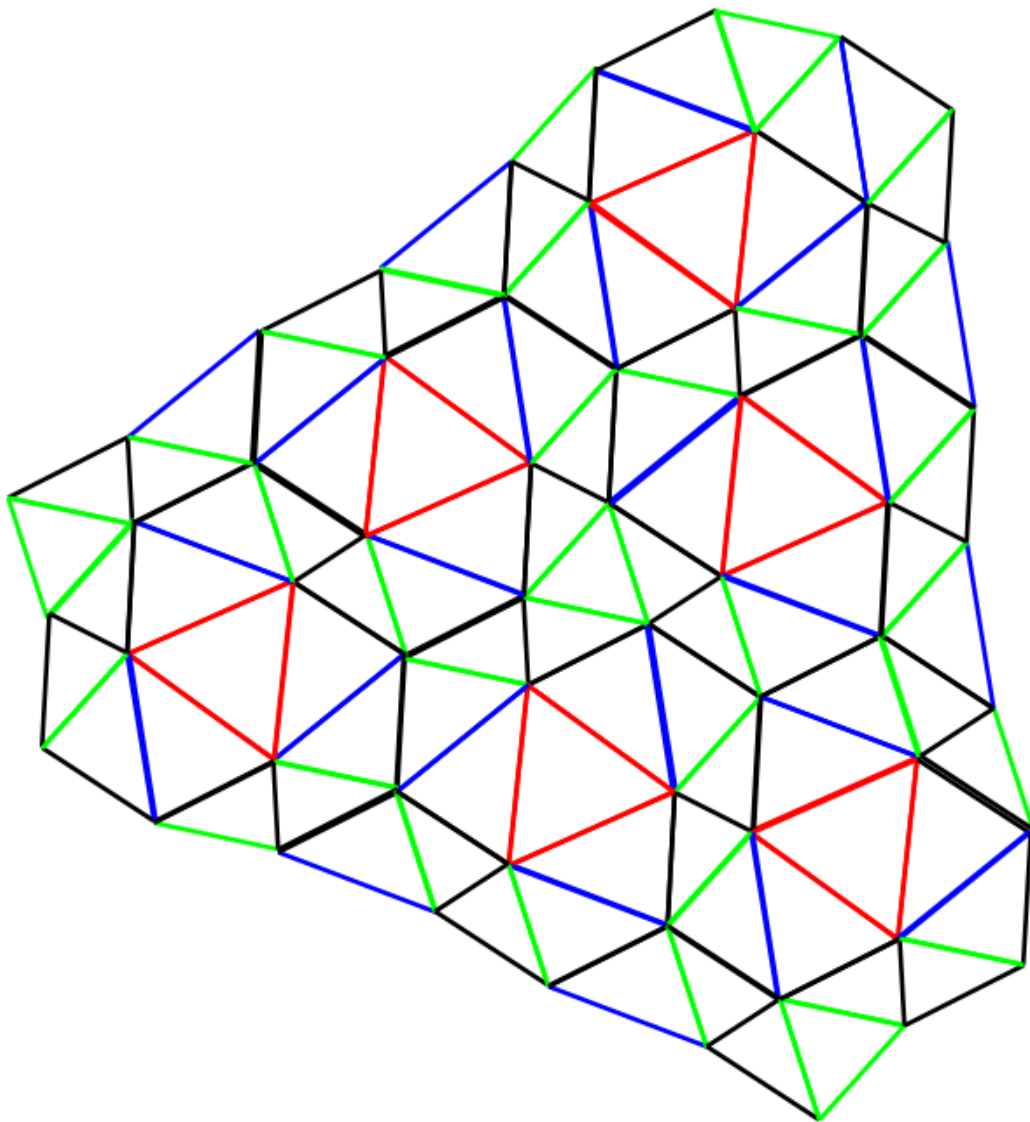
L1



L2

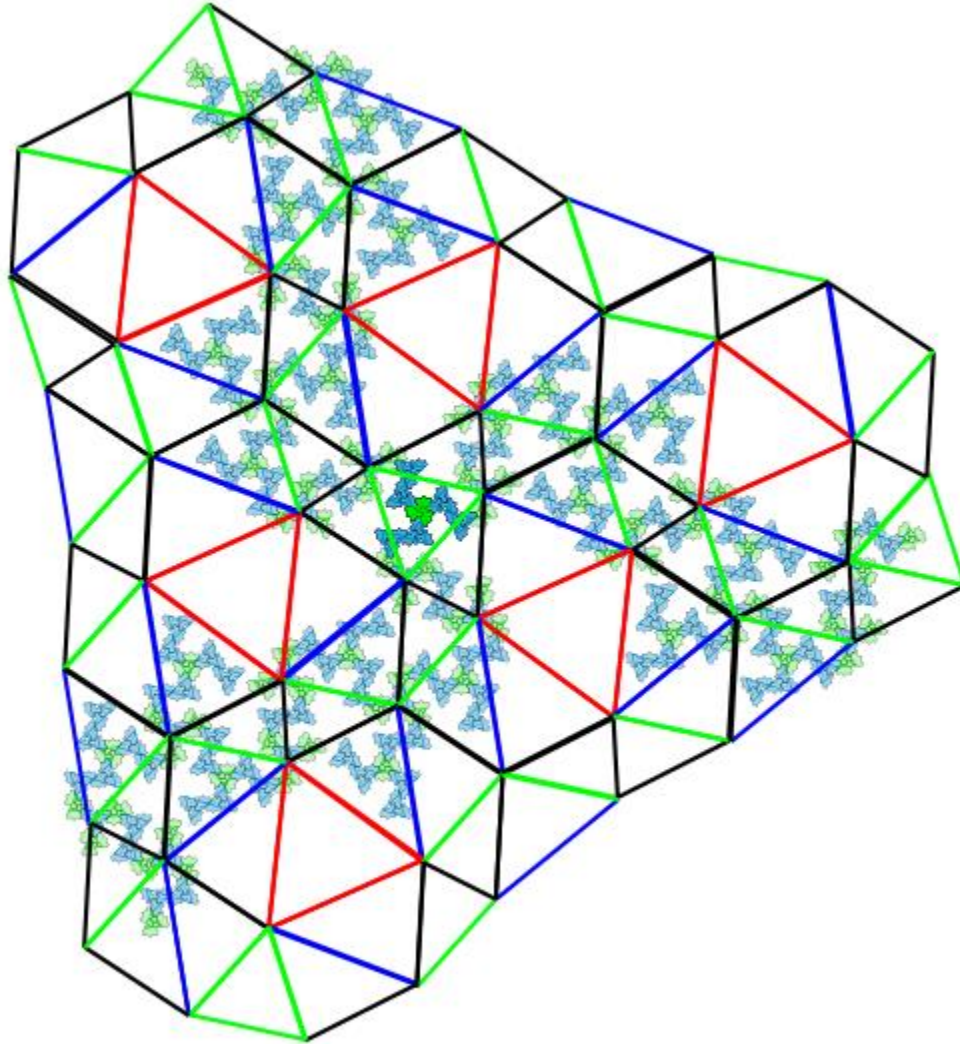
- Question: are all of them present in the infinite tiling ???

Using set of triangles



3-fold-S2

Using set of triangles

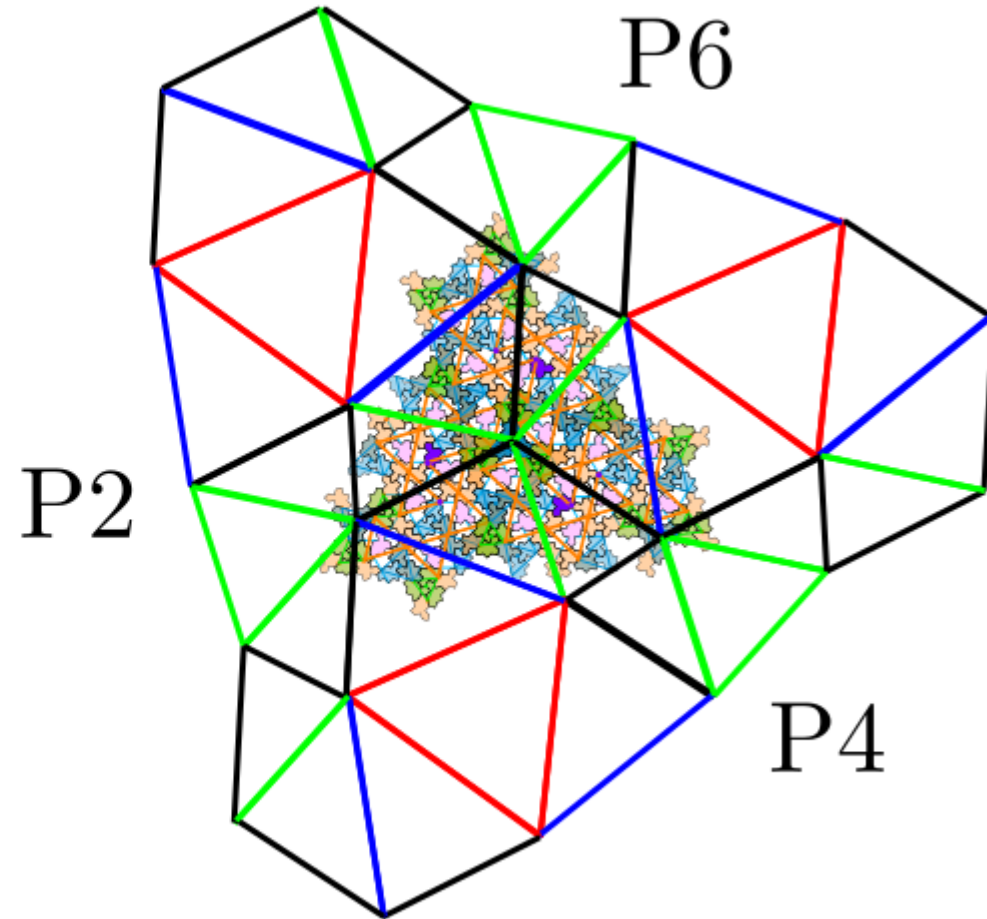


3-fold-S1

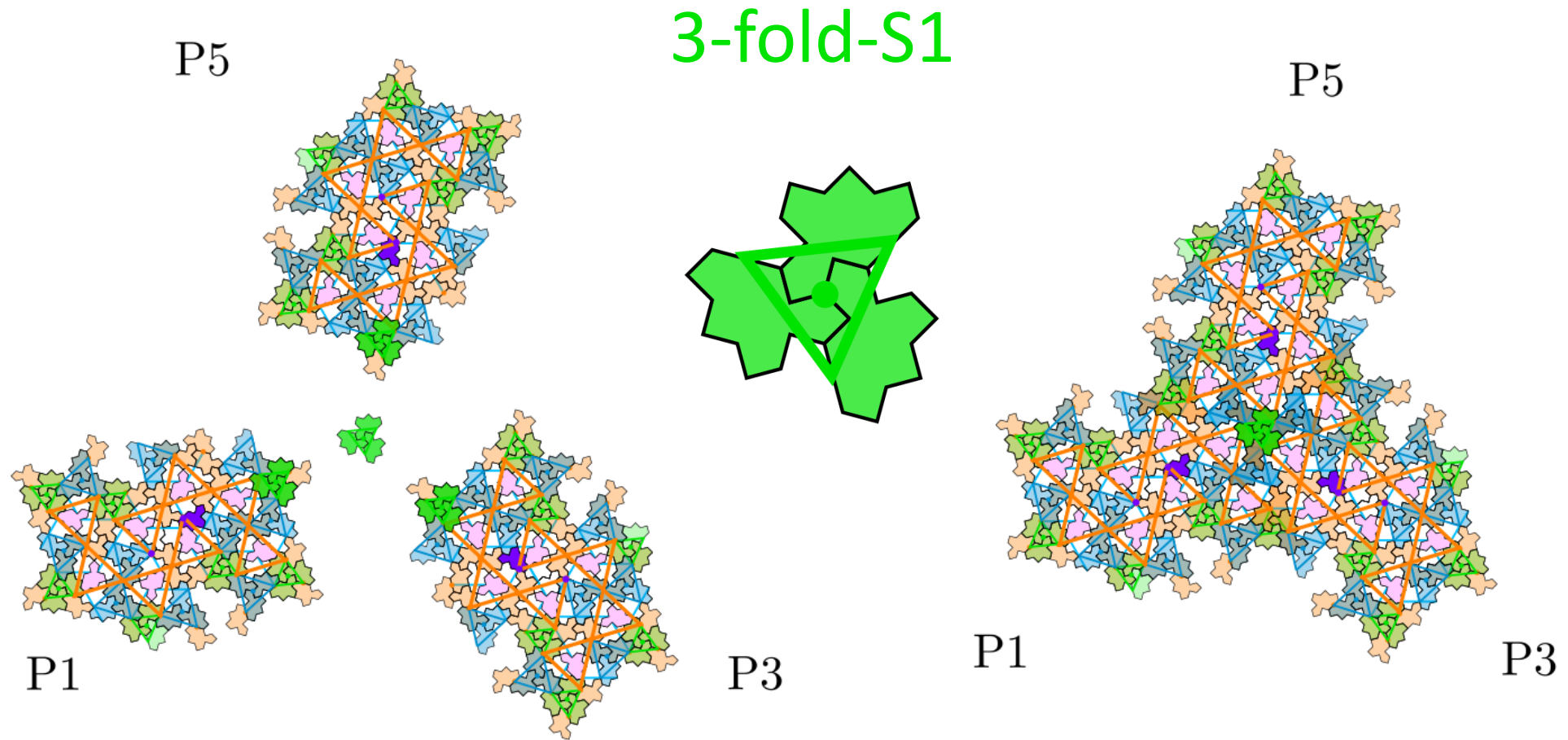
Using set of triangles



3-fold-L2

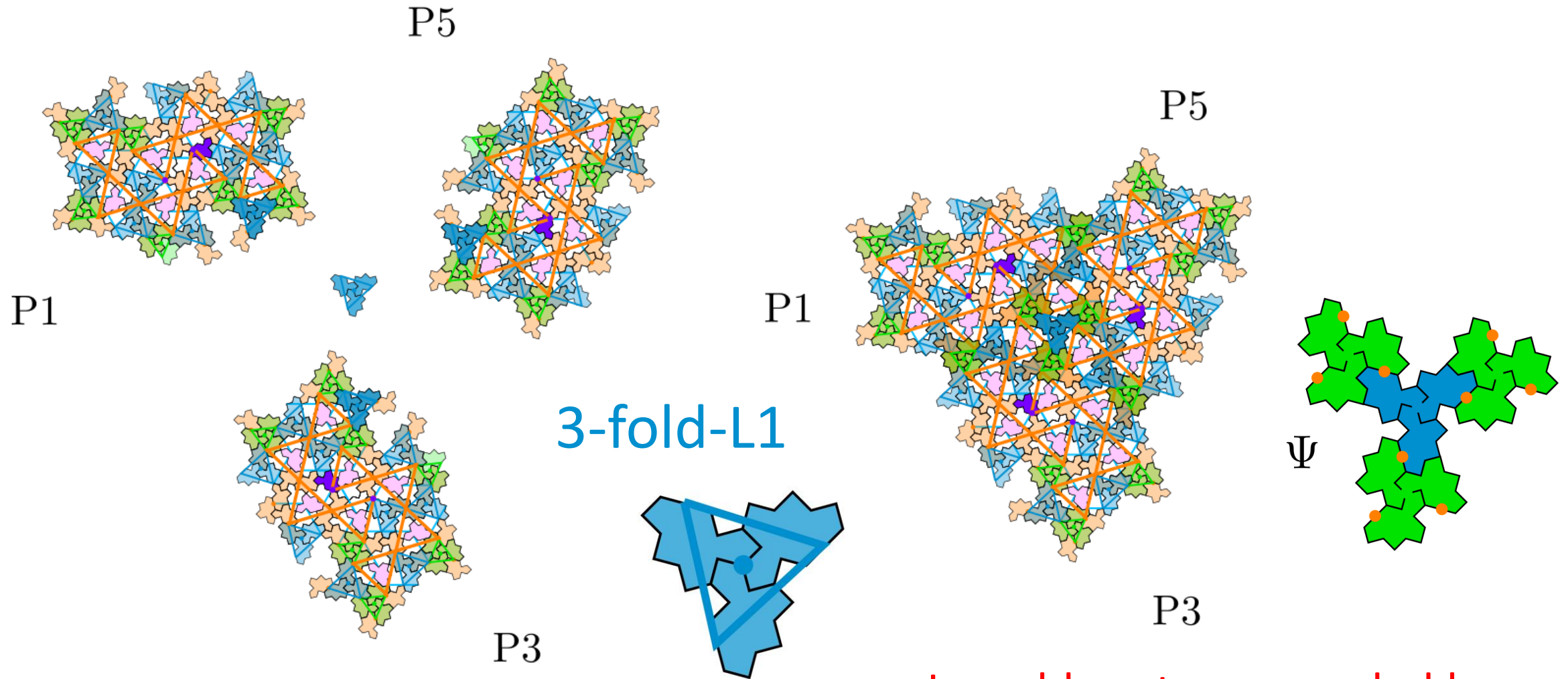


Seed 1 with 3-fold symmetry



- Green star S1 at the center

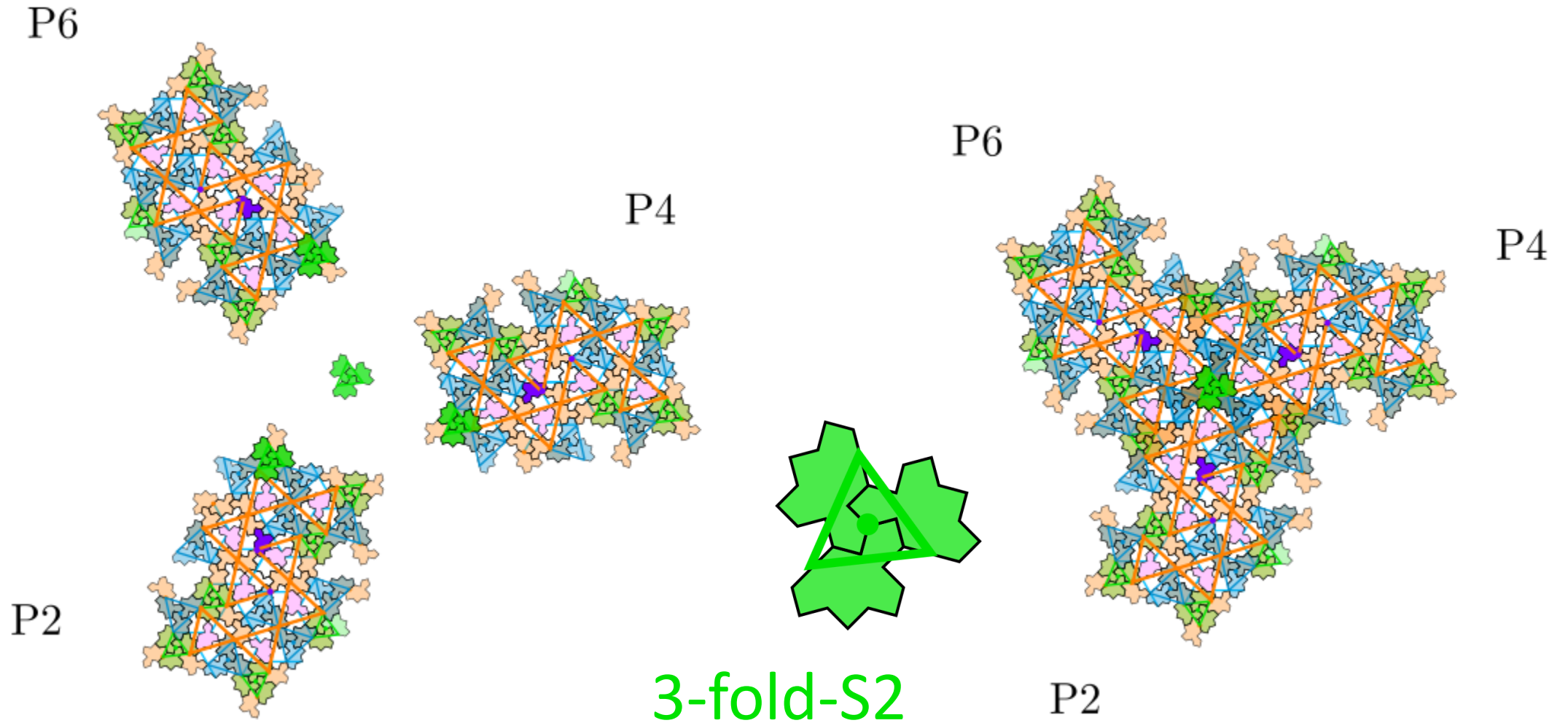
Seed 2 with 3-fold symmetry



- blue star L1 at the center

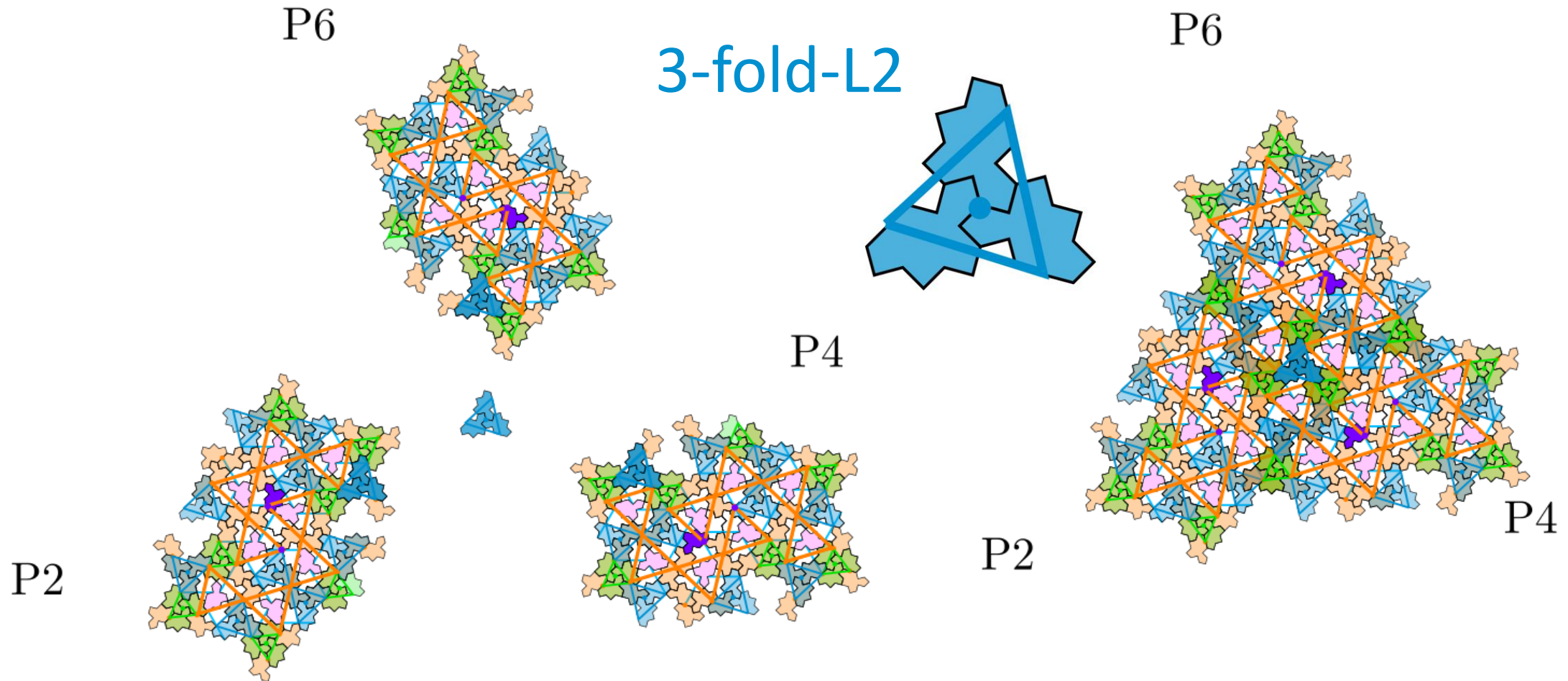
- Lone blue star surrounded by three S2 green stars

Seed 3 with 3-fold symmetry



- Green star S2 at the center

Seed 4 with 3-fold symmetry



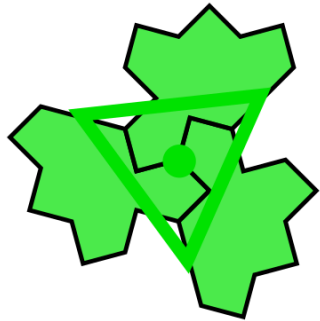
- blue star L2 at the center

Two possible seeds with global 3-fold symmetry

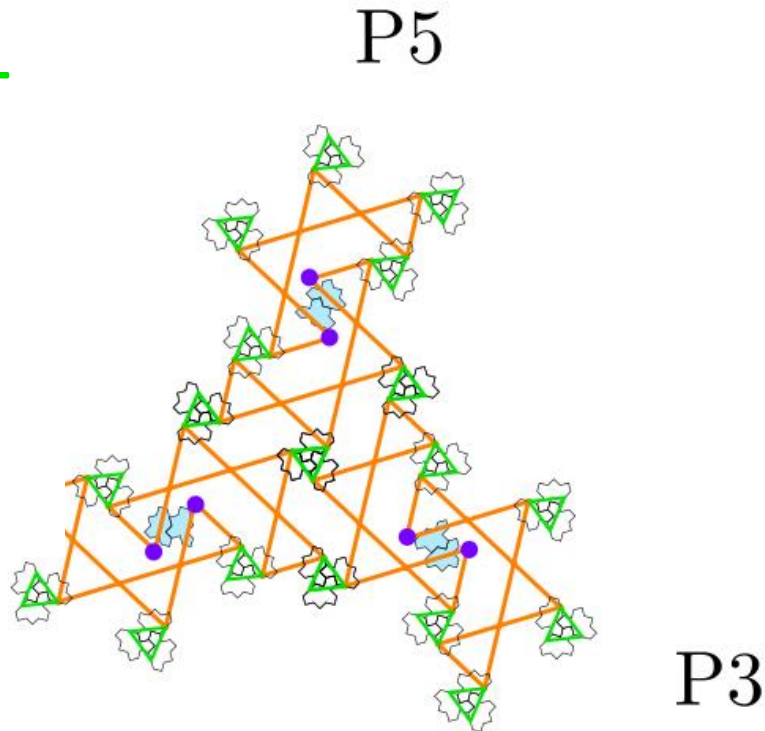
seed1

- Green star at the center

3-fold-S1



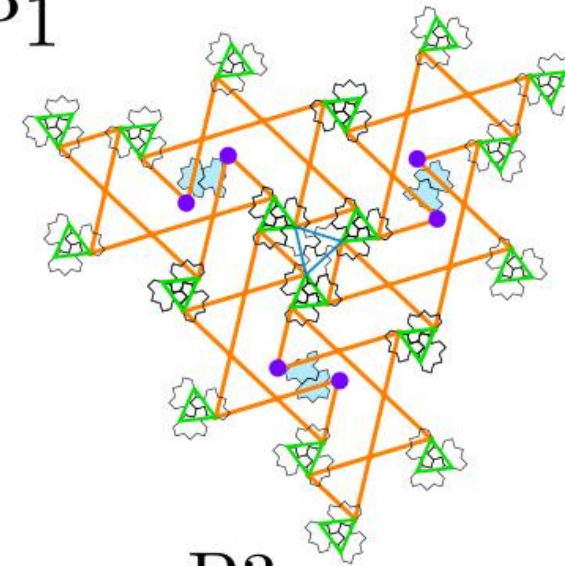
P1



seed2

- blue star at the center

P1



P5

3-fold-L1



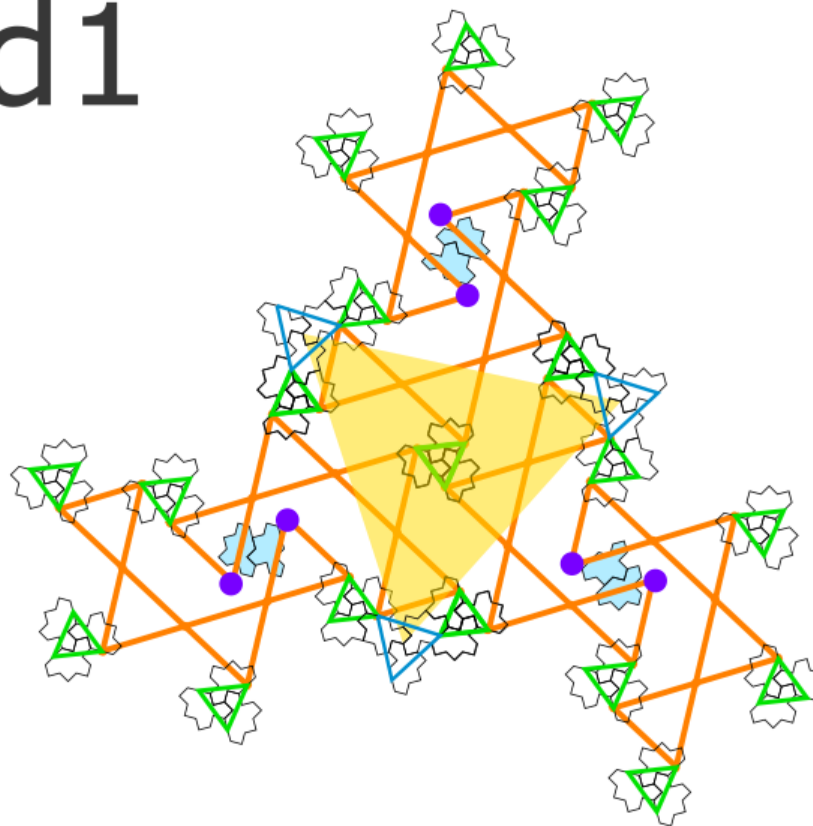
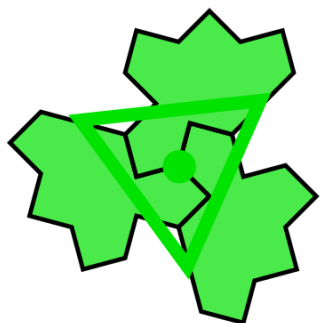
- Three 2-arms rotated from each other by $2\pi/3$

Seed 1

P5

seed1

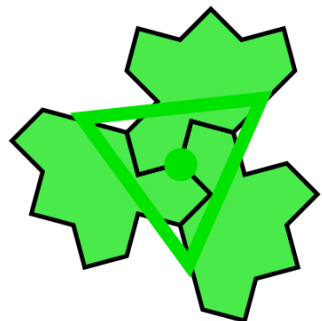
3-fold-S1



P1

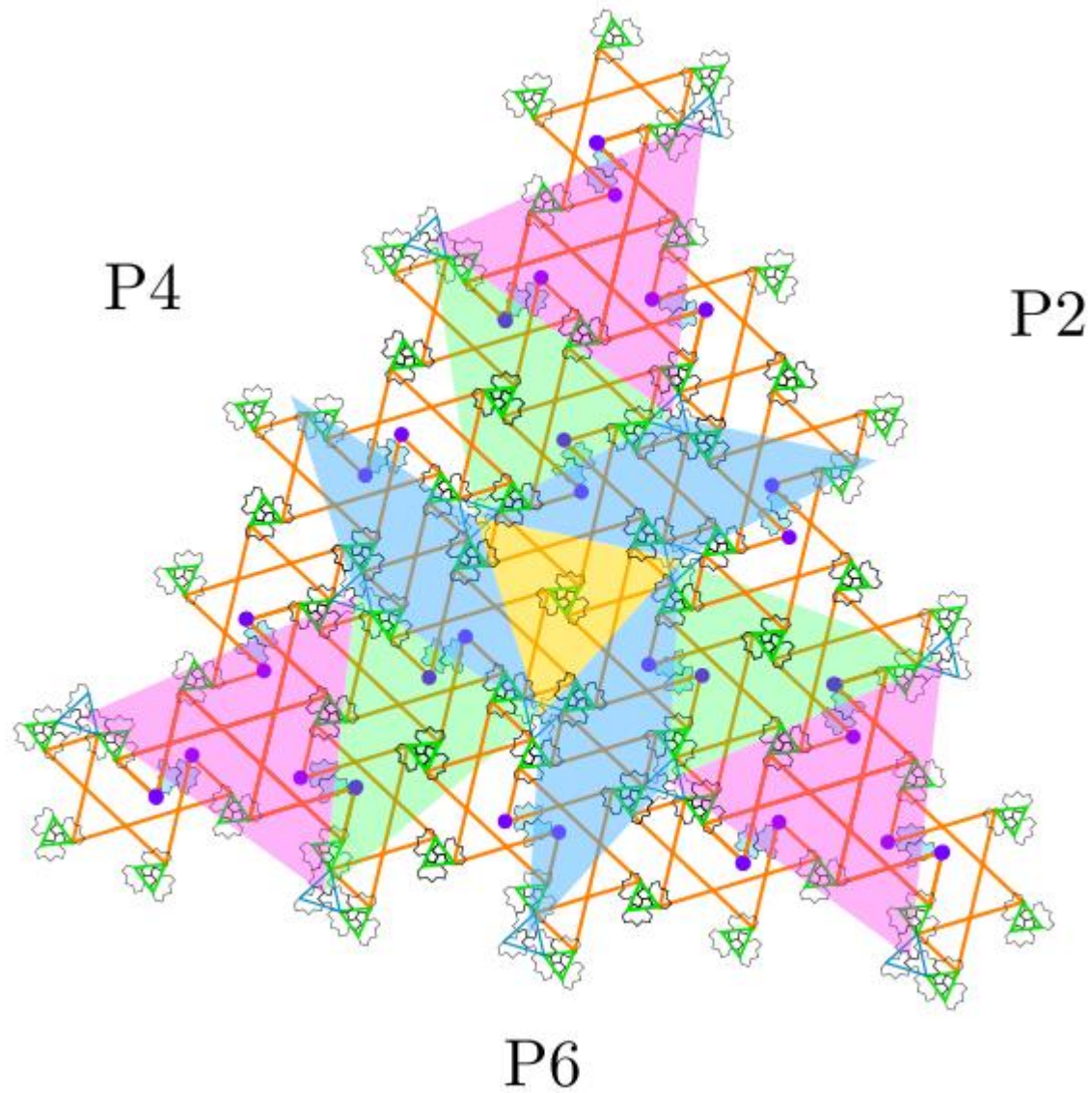
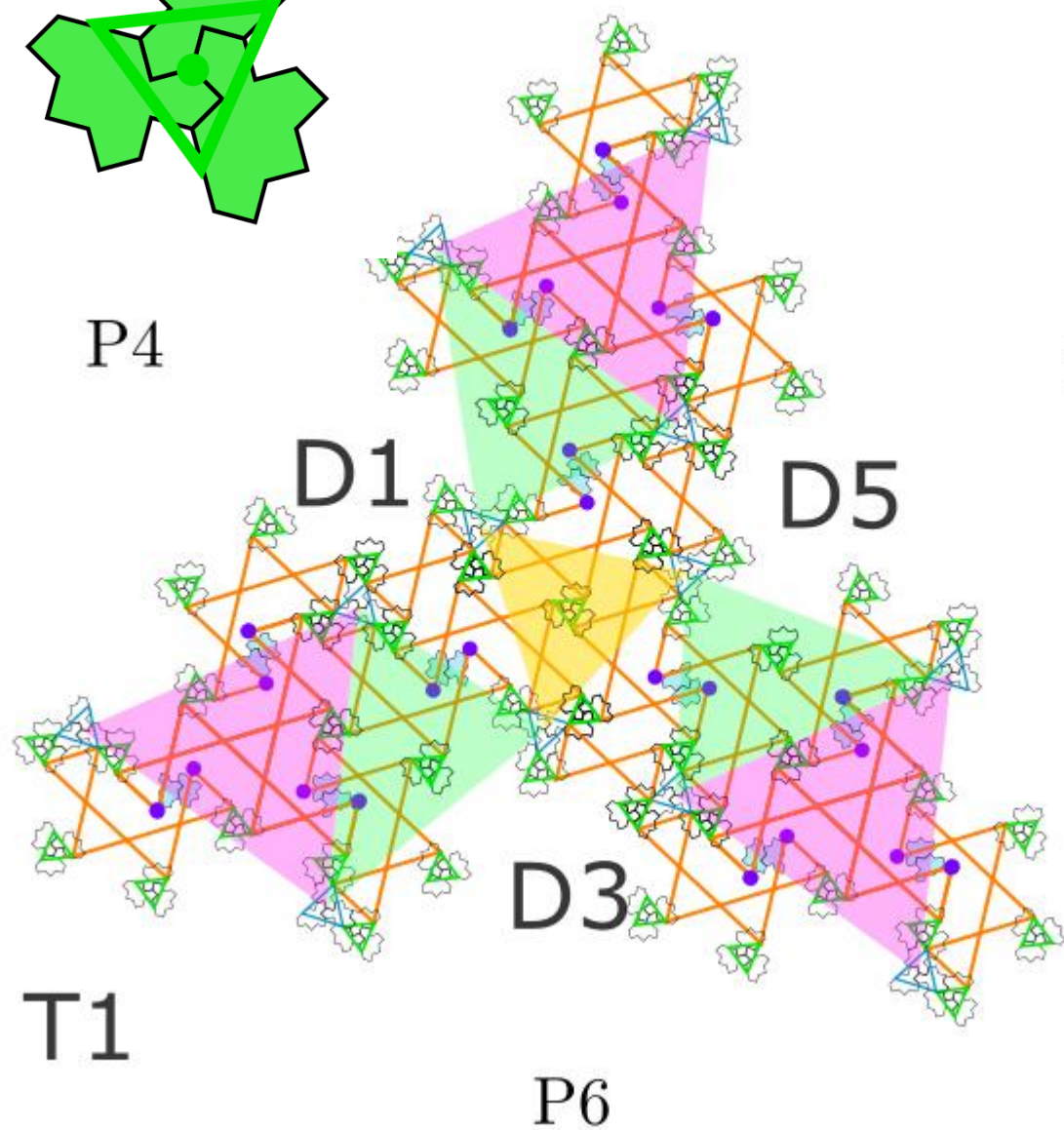
P3

3-fold-S1

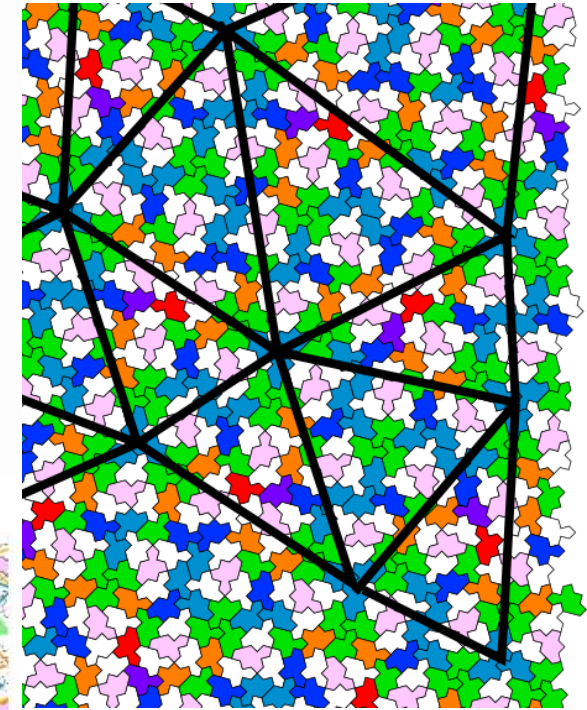
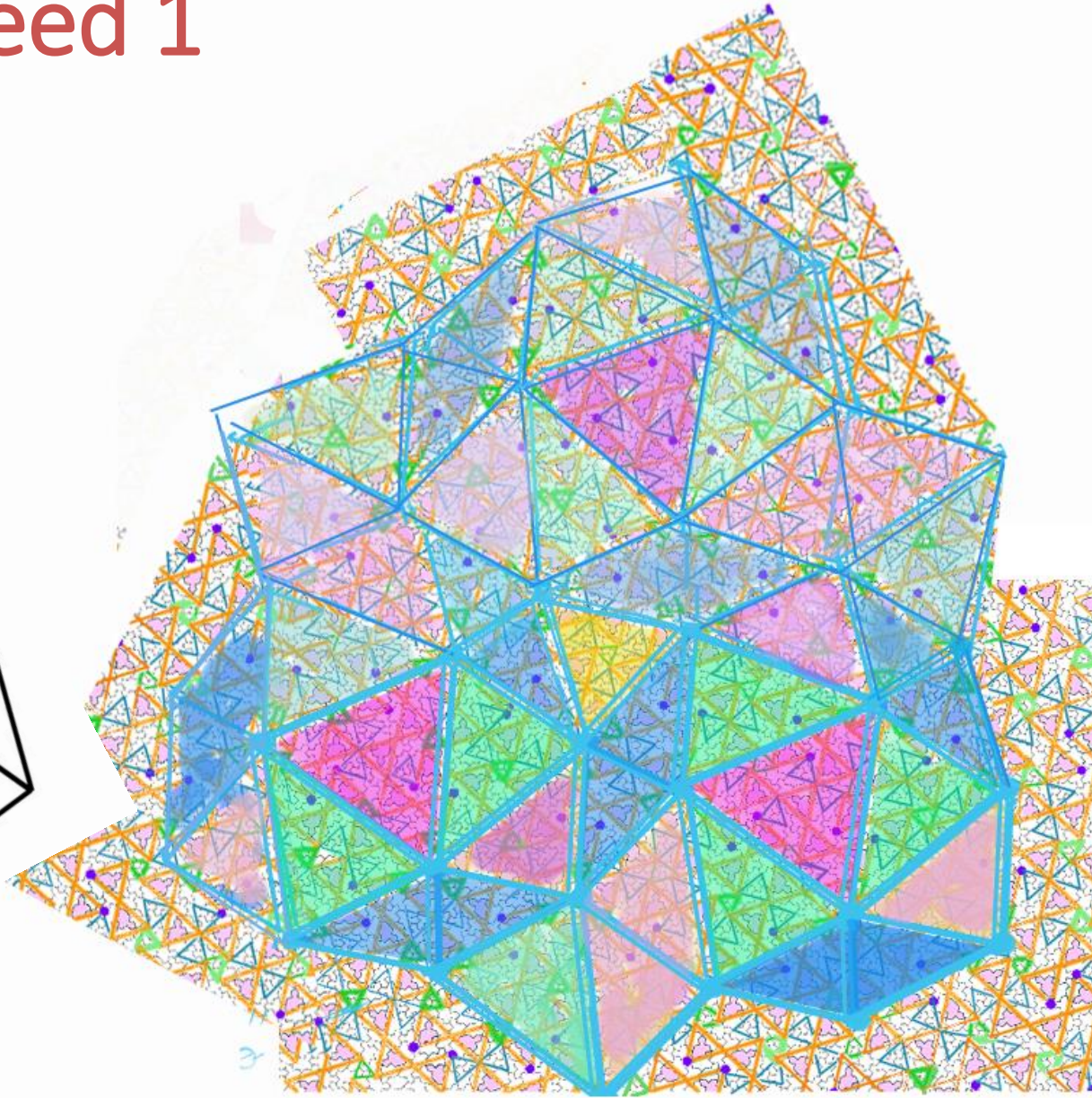
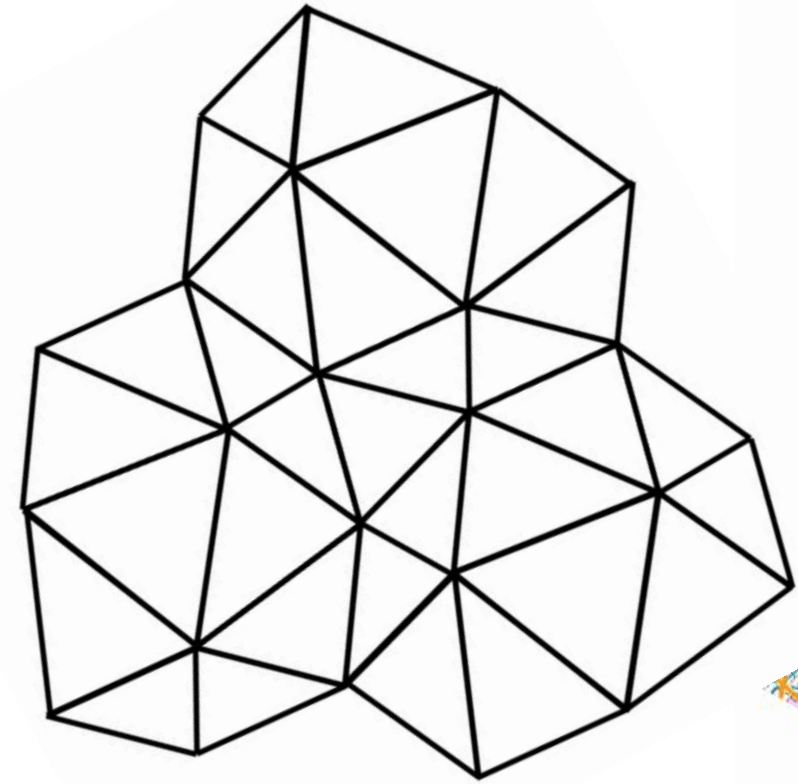


T5

Seed 1

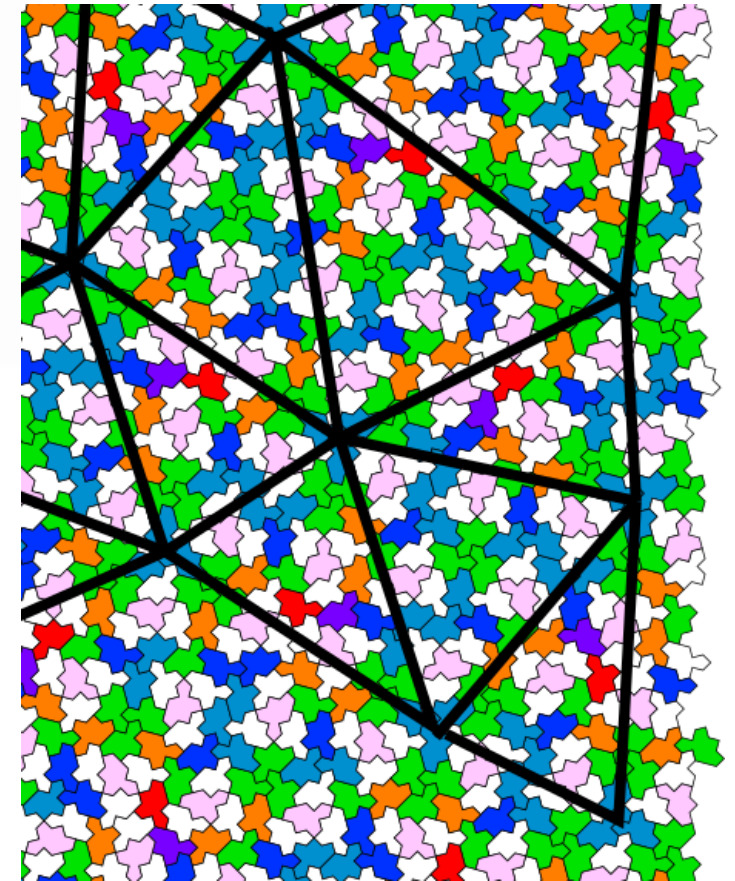
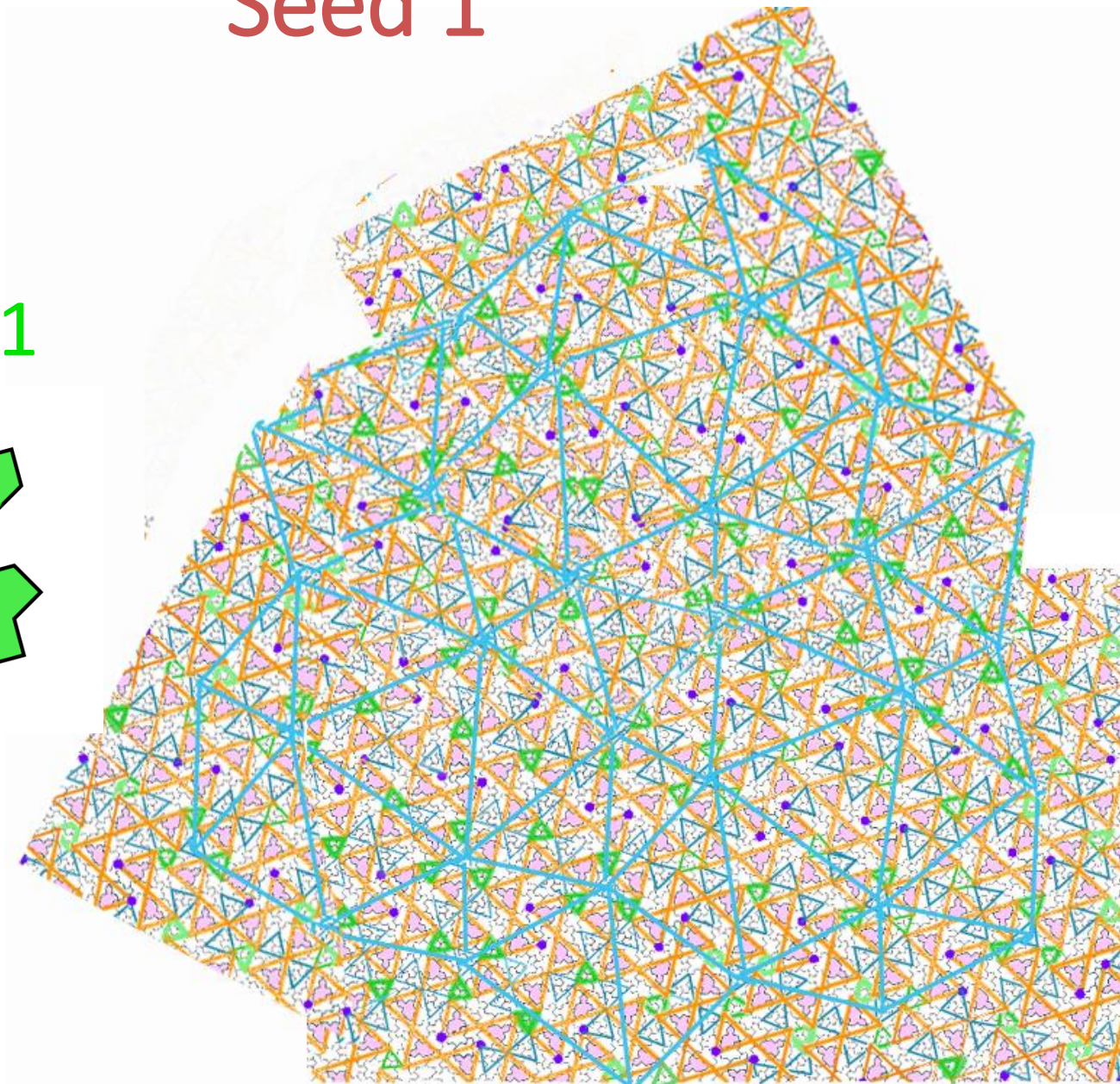
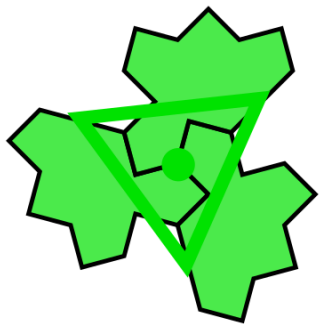


Seed 1

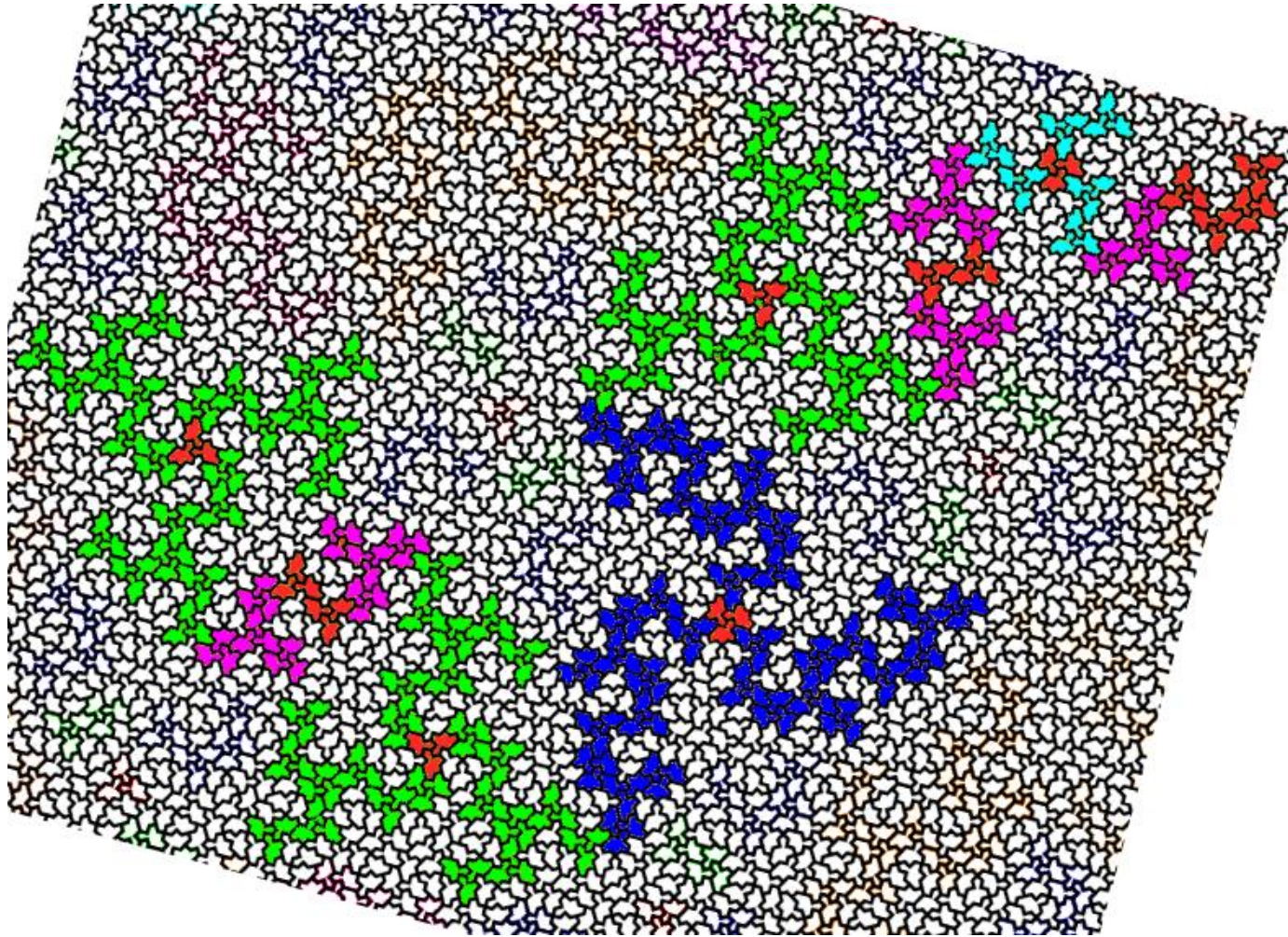


Seed 1

3-fold-S1

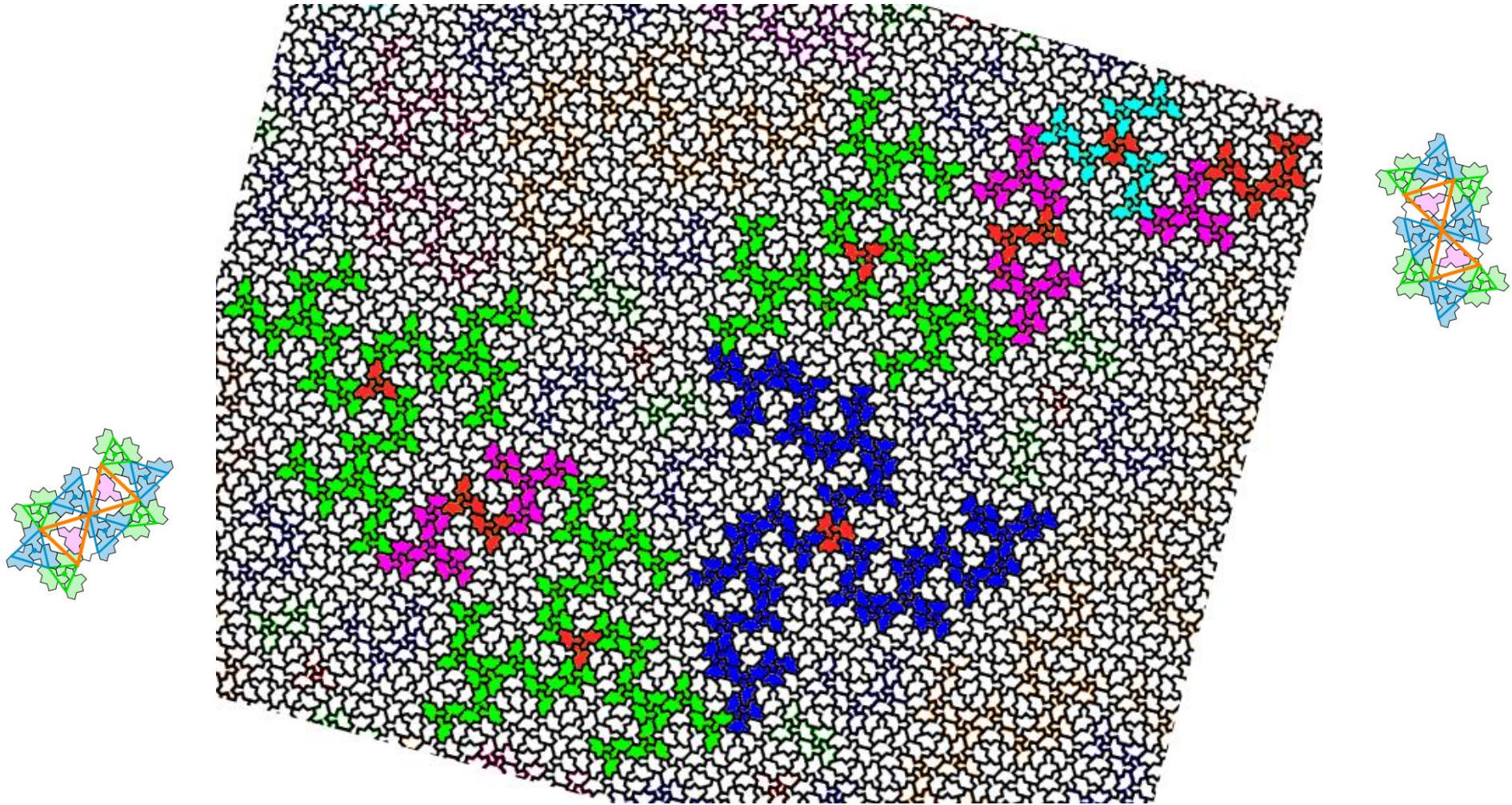


Star clusters by Boris Horvat: example of cluster 147



- Mirror + rotation of $\pi/12$ for comparison

cluster 147: two diabolos



- -> to finish ... Identify the type of the local 3-fold symmetry centers

Results so far

- Local 3-fold and 2-fold symmetry centers are identified
- Violet points -> different way to identify (chimera ...)
- Analysis of the size/amount of stars clusters depending on the level (Boris Horvat)
- Different patterns can be used to decompose the whole tiling:
 - 2-arms plus 3-arms+diabolo
 - full clusters
 - versions of 'anatomy' etc ...

Helpful to identify the global symmetry centers !

- Identification of 2-fold global symmetry center (center of diabolo figure)
- two possible 3-fold global symmetry centers: center of green triangle and center of blue triangle. They correspond to three 2-arms (P1/P3/P5) rotated by $2\pi/3$ around the center)
- triangular array of blue stars surrounded by three green stars

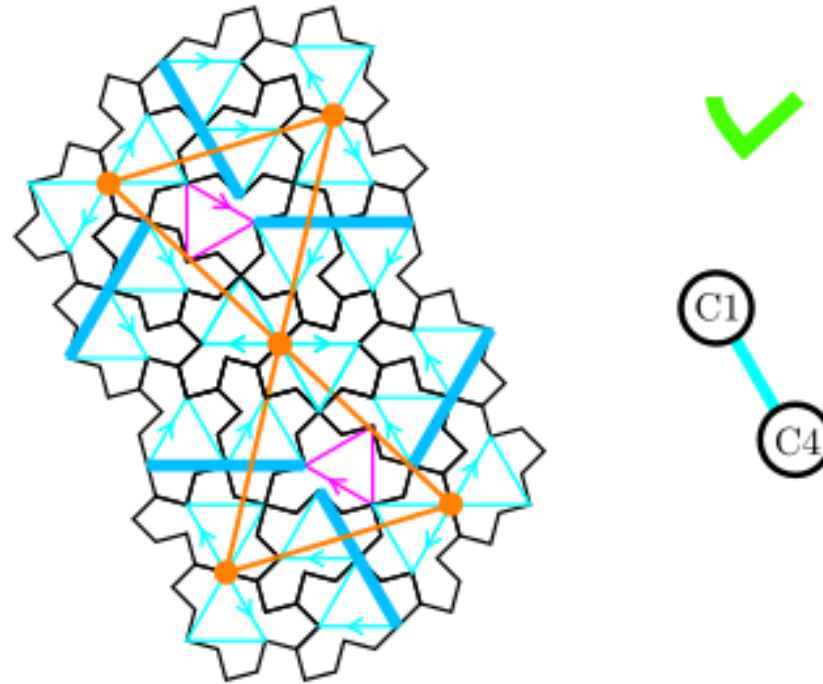
TO DO: inflation schemes around symmetry centers

Different possible methods should give the same result

- H7/H8 (by S. Tathan) applied to a symmetric subset of clusters
-> might be the most straightforward method for implementation as code is already running (but in RUST language) ?
- Full clusters with translations (Marianne ...)
-> will need more work to do a complete python implementation
- Substitution rules for large colored triangles (JFS)
-> will need more work to figure out the substitution rules
- generate only the star clusters based on 2-fold and 3-fold local rotations and some additional rules ???
-> to test the idea first ...

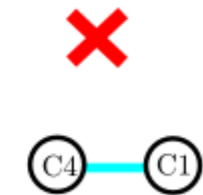
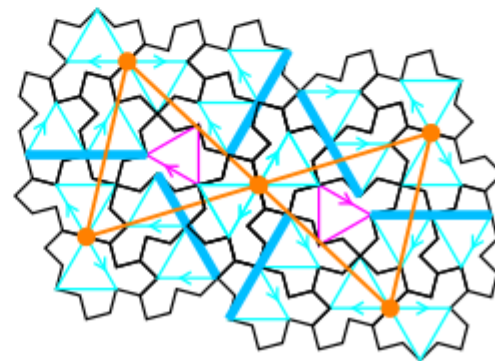
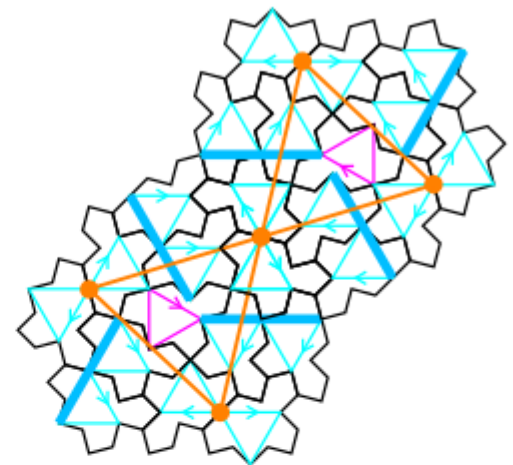
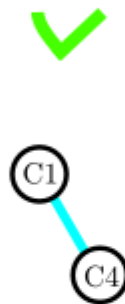
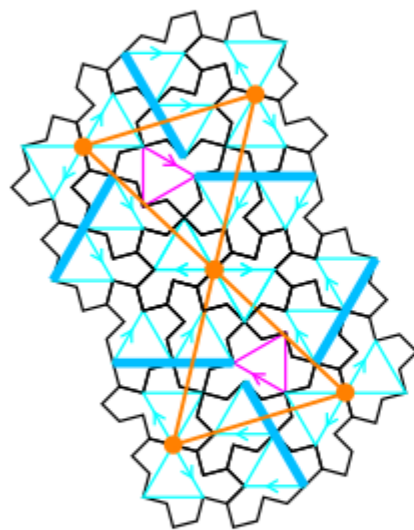
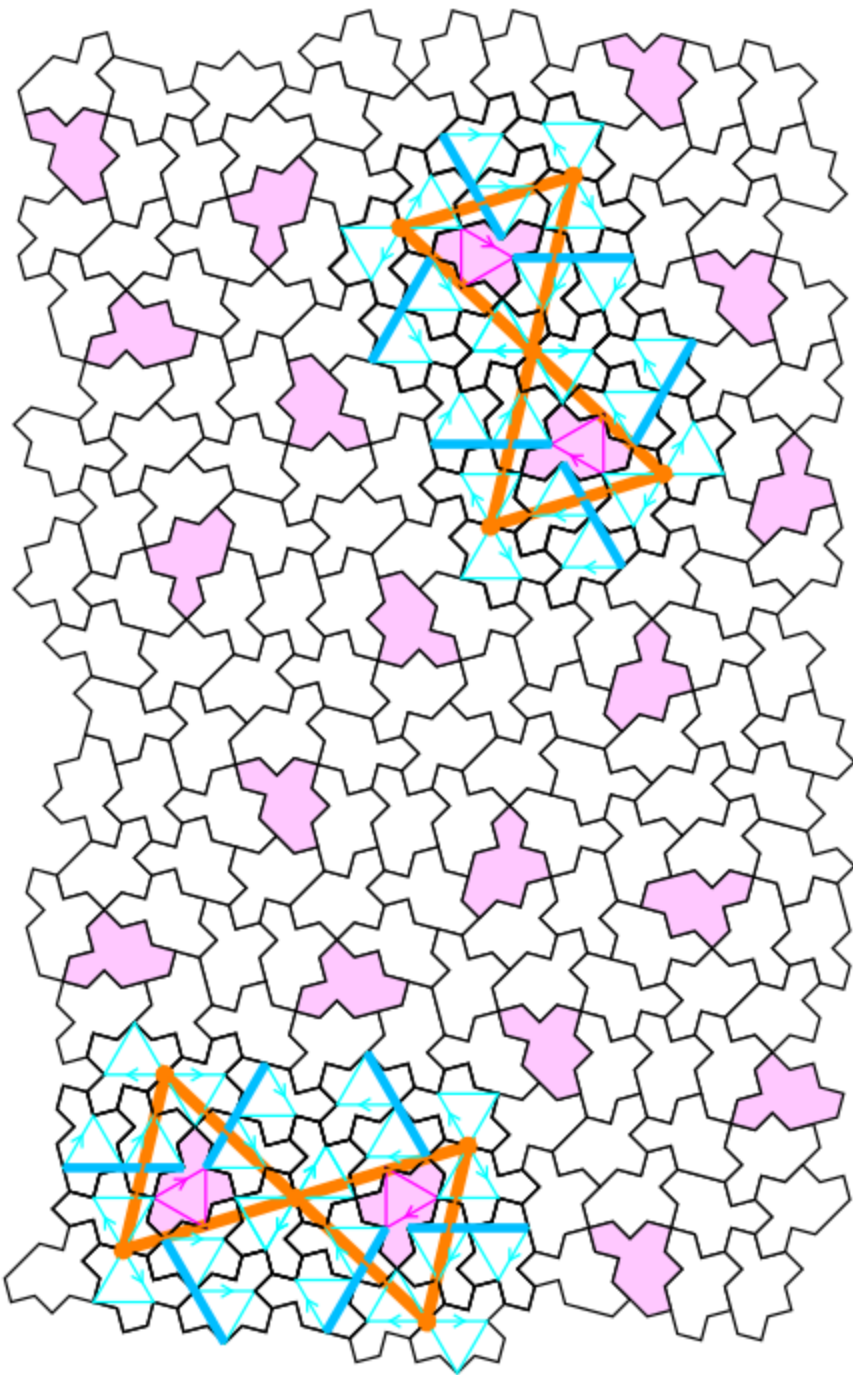
Homochiral inflation: use H7/H8 by Simon Tathan ?

- For 2-fold symmetry, apply inflation to two clusters in C1-C4 relative orientations :



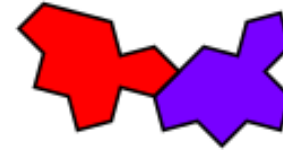
- Should keep the 2-folds symmetry at all inflation steps
- TO DO: test it manually and contact S. Tathan afterwards

Diabolo pattern



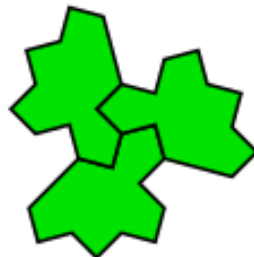
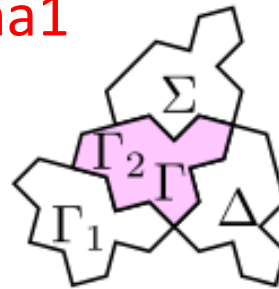
Identification of the greek letters for tiles

- ODD tiles
Pink color



- $\backslash$ theta and $\backslash$ lambda always in couple

- Small cluster $\backslash$ sigma, $\backslash$ delta, $\backslash$ gamma1
white color

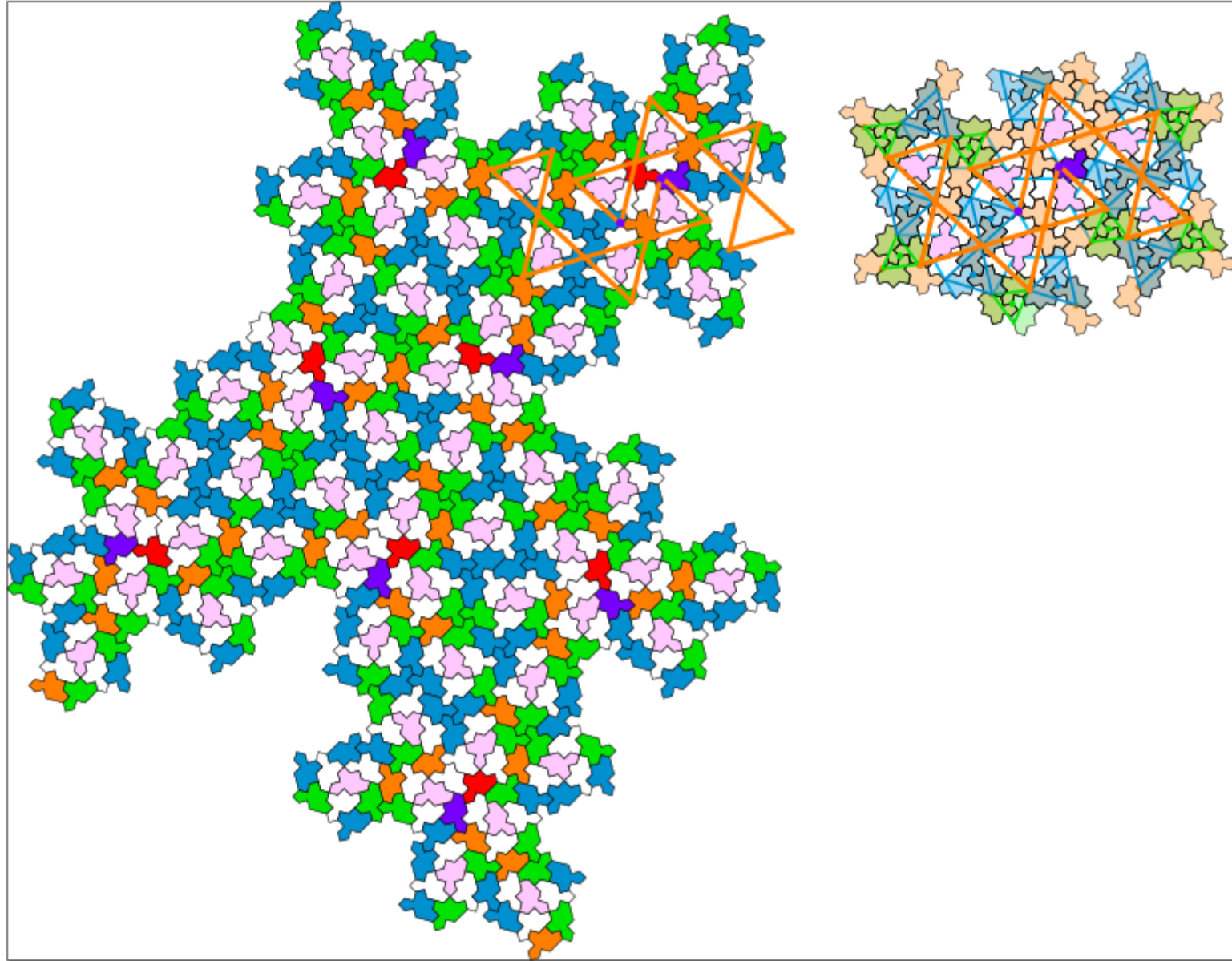


- All $\backslash$ chi tiles are linked to a $\backslash$ pi tile and consequently to a large star.

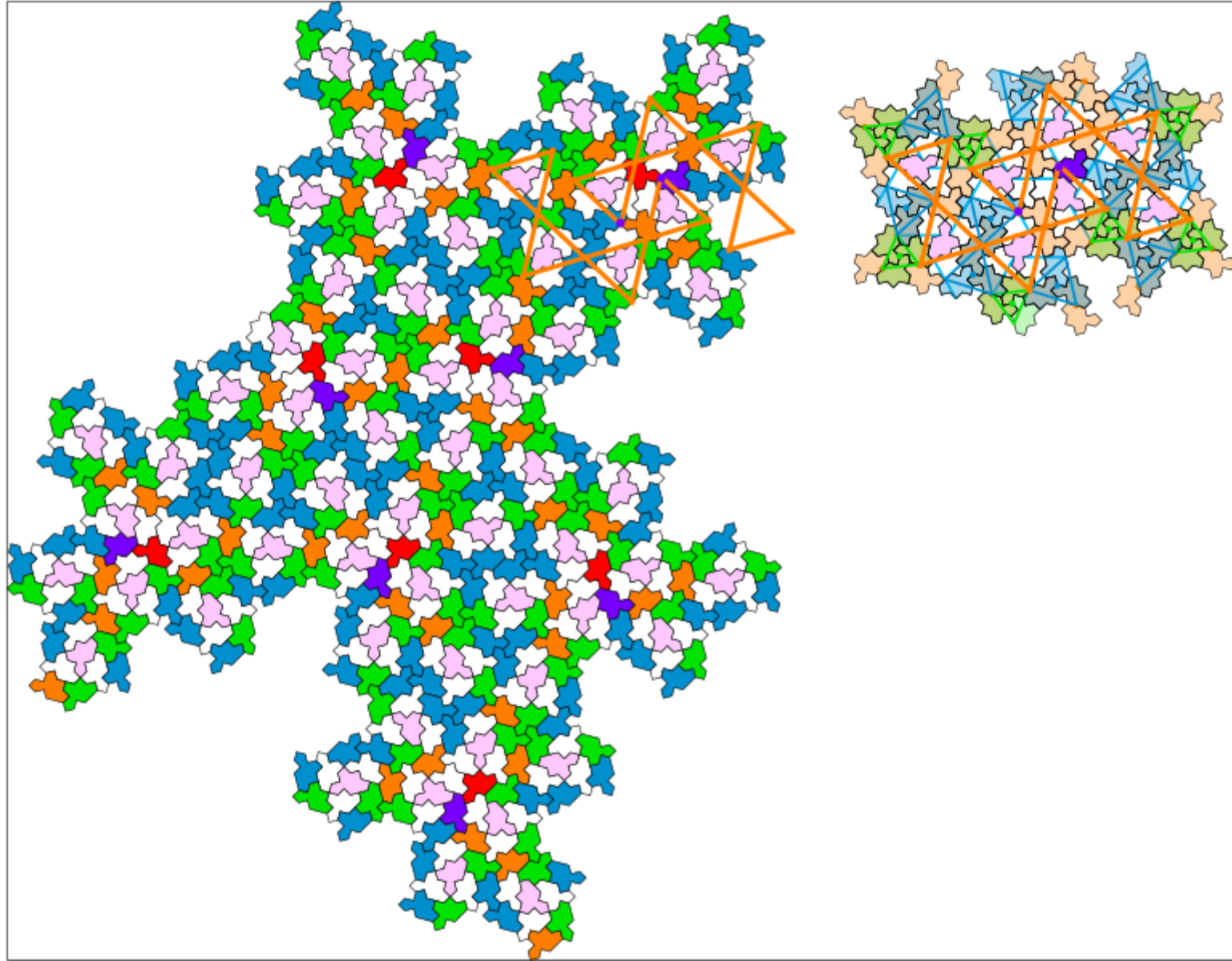
- $\backslash$ phi: Small green stars + some isolated EVEN tiles

- Blue large stars correspond to all $\backslash$ Pi and $\backslash$ Psi tiles with different combinations

Inflation using python code

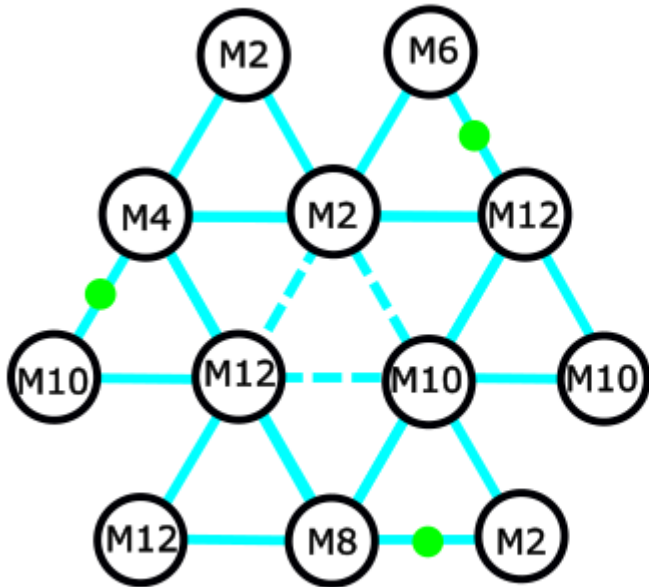


Inflation using python code

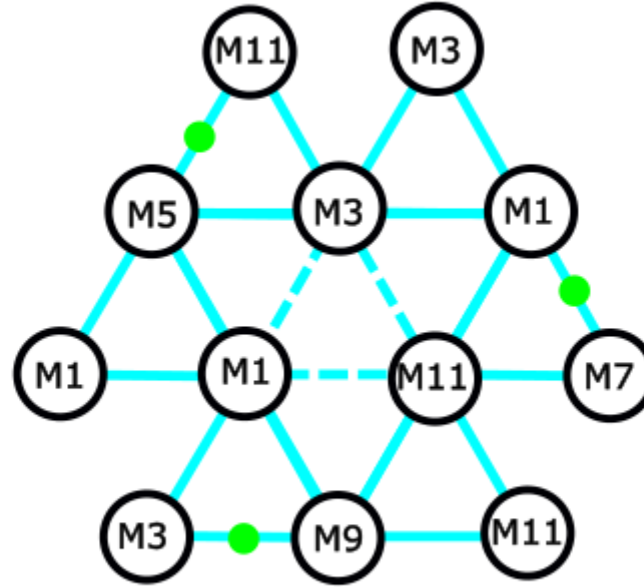


Inflation step 1: substitution rule for C1

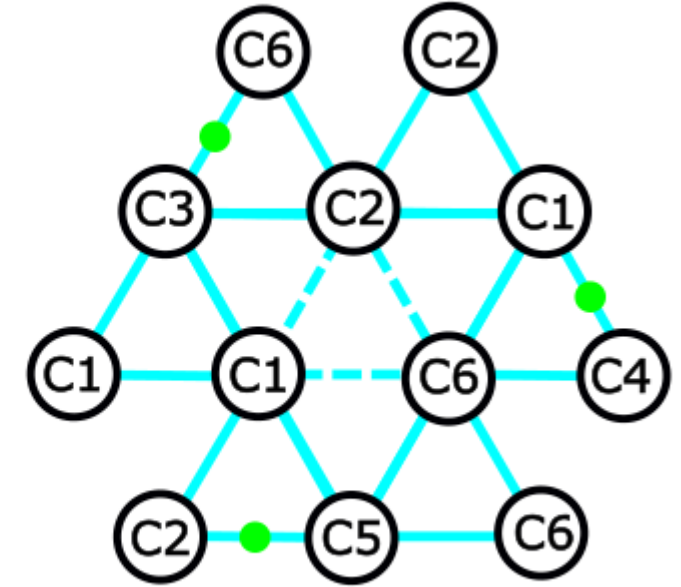
EVEN tiles in C1



ODD tiles for step 1

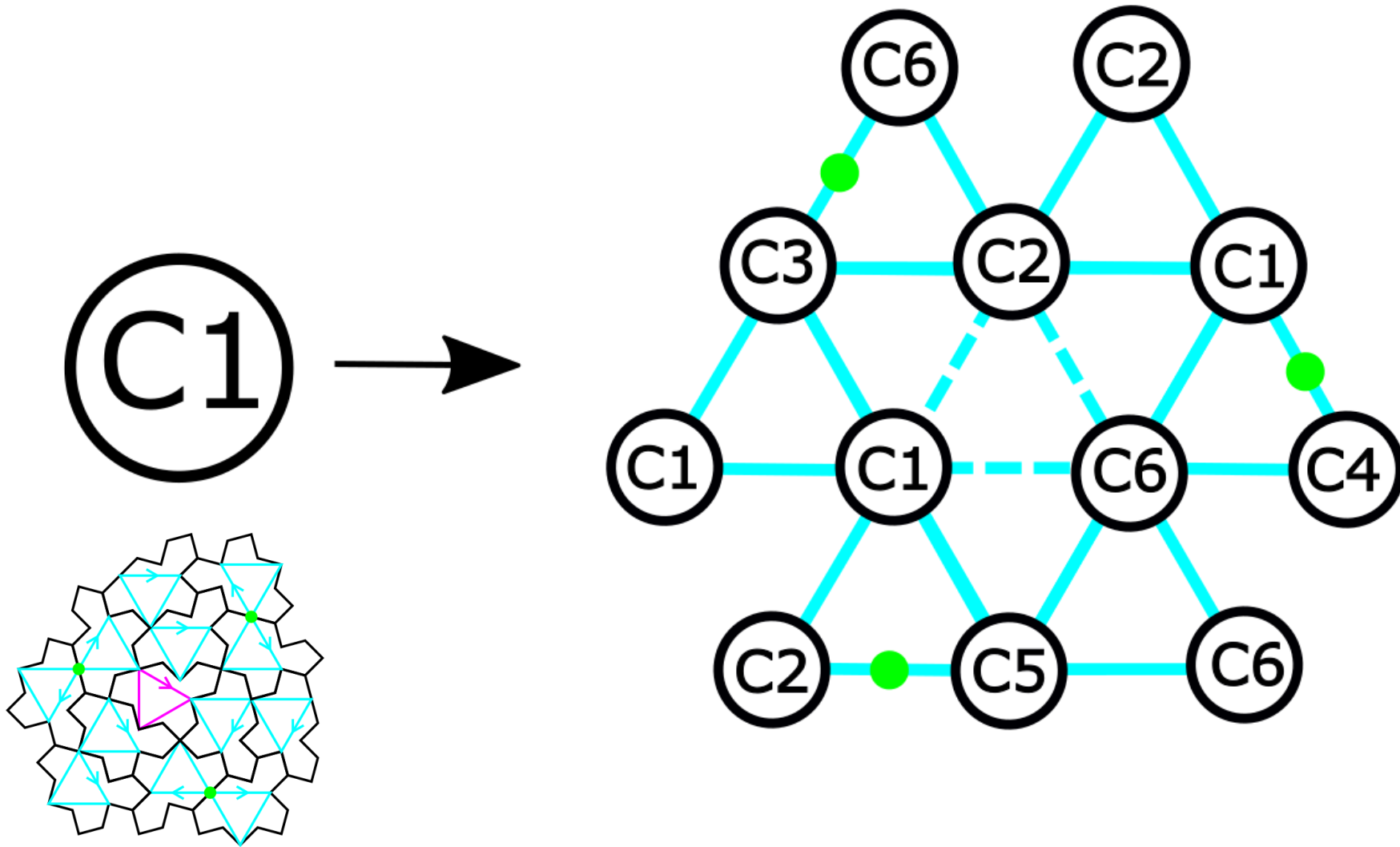


same as full clusters



- ODD tiles for step 1 are obtained by adding 1 (mod 12) to all EVEN tiles in the C cluster (6 tiles)
- Directions of translated and half-turned tiles are exchanged (**see the change of position for green points**). Then three tiles are translated (M1, M3, M11) and three are half-turned (M5→M11, M1→M7, M9→M3)
- ODD tile index is equivalent to the orientation of a full C cluster 😊

Inflation step 1



Homochiral python code on github

- <https://github.com/marimperorclerc/homochiral>

python code for drawing clusters with $\text{Tile}(a,b)$ in svg graphics

References

Code is based on the drawsvg package:

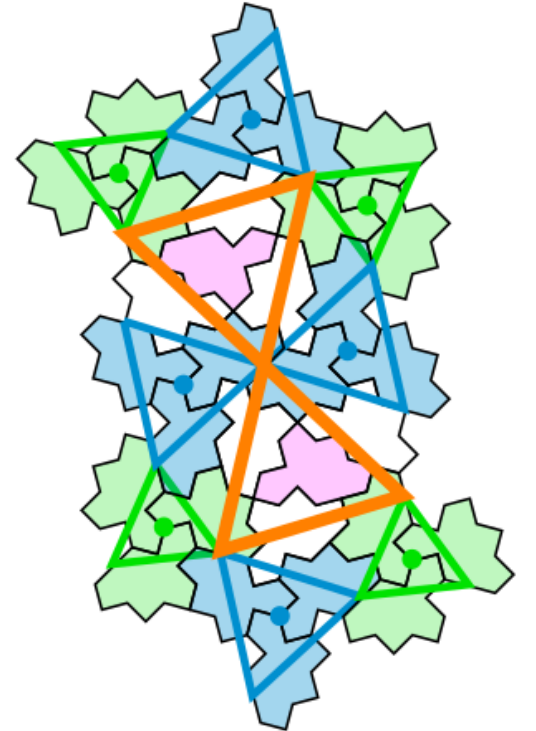
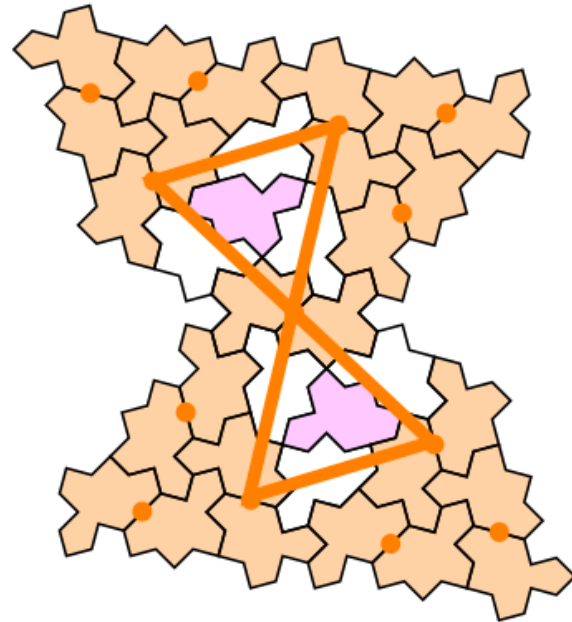
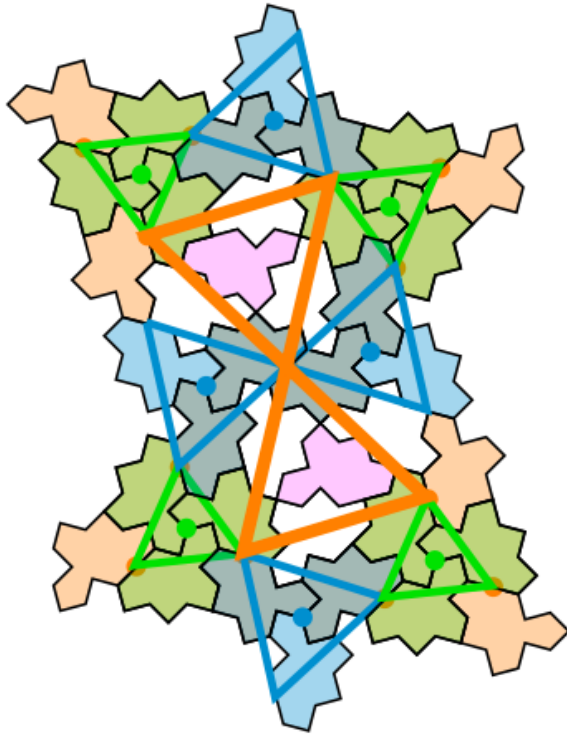
<https://pypi.org/project/drawsvg/>

And the python code with alternate chirality for inflation for spectre tiling:

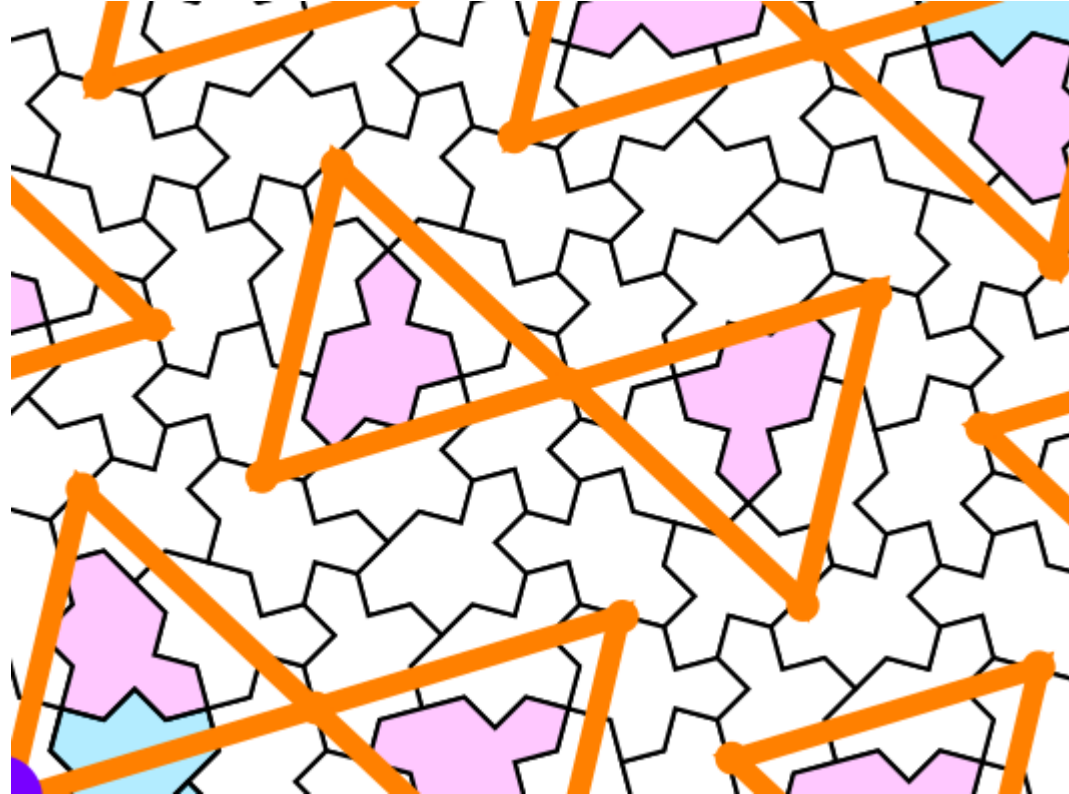
<https://github.com/shrx/spectre/blob/master/spectre.py>

- Adapted here (spectre-colored-tiles.ipynb) with different color choices
- See some .svg files in \svg-spectre folder

DIABOLO

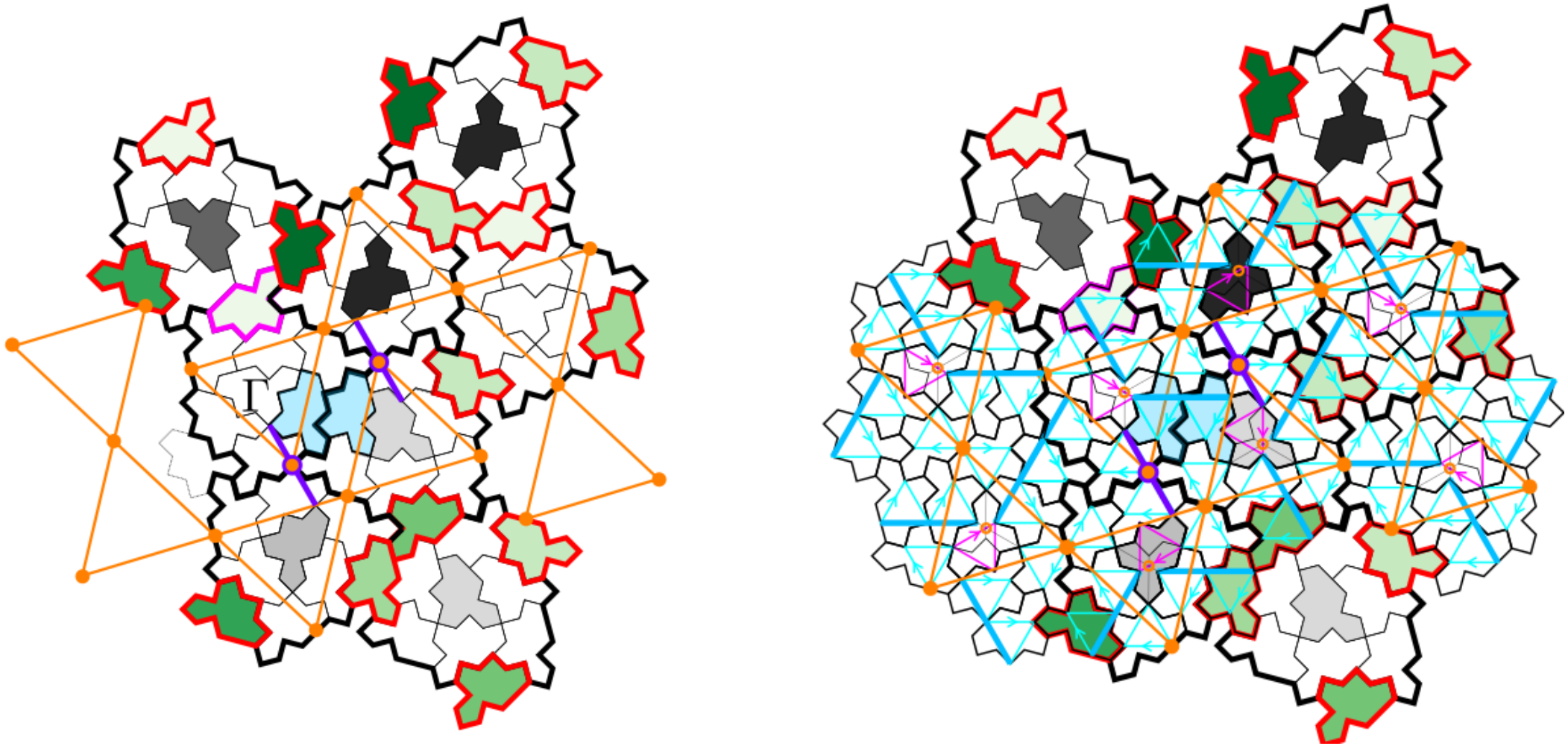


Diabolo pattern



- Local 2-fold axis

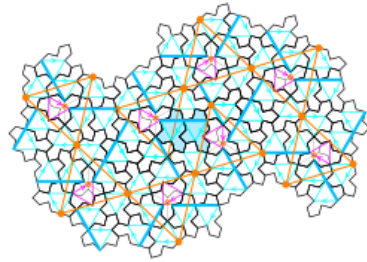
Link with triangular pattern



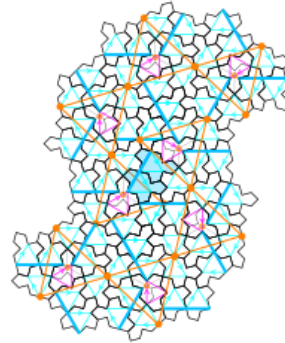
- The two other environnements with a violet lines are always coupled together

Six orientations of the 2-arms

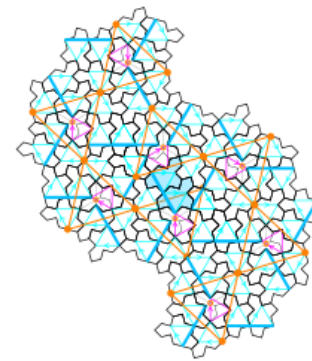
P1



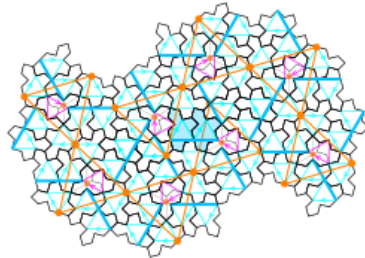
P2



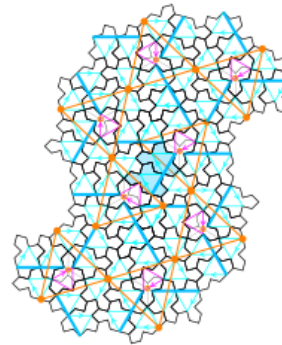
P3



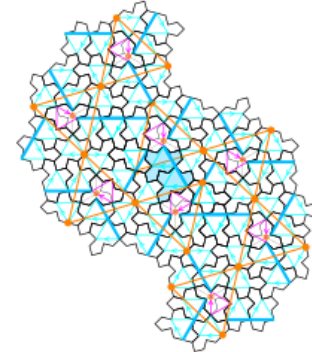
P4



P5

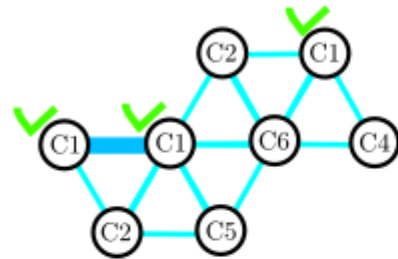


P6

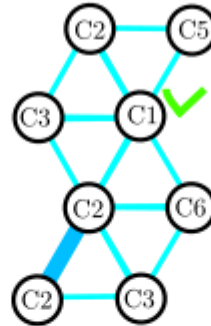


Six orientations of the 2-arms

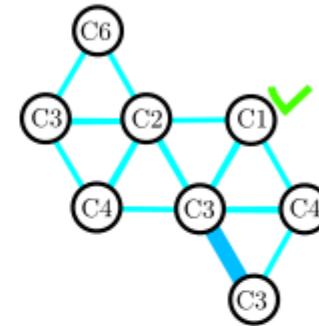
P1



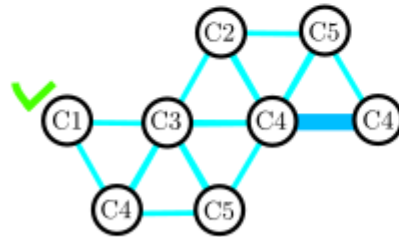
P2



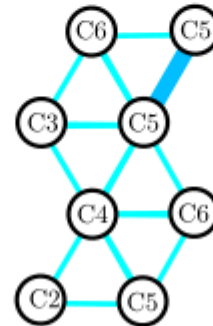
P3



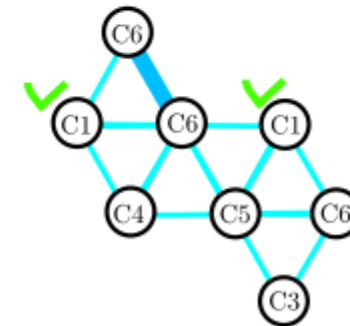
P4



P5



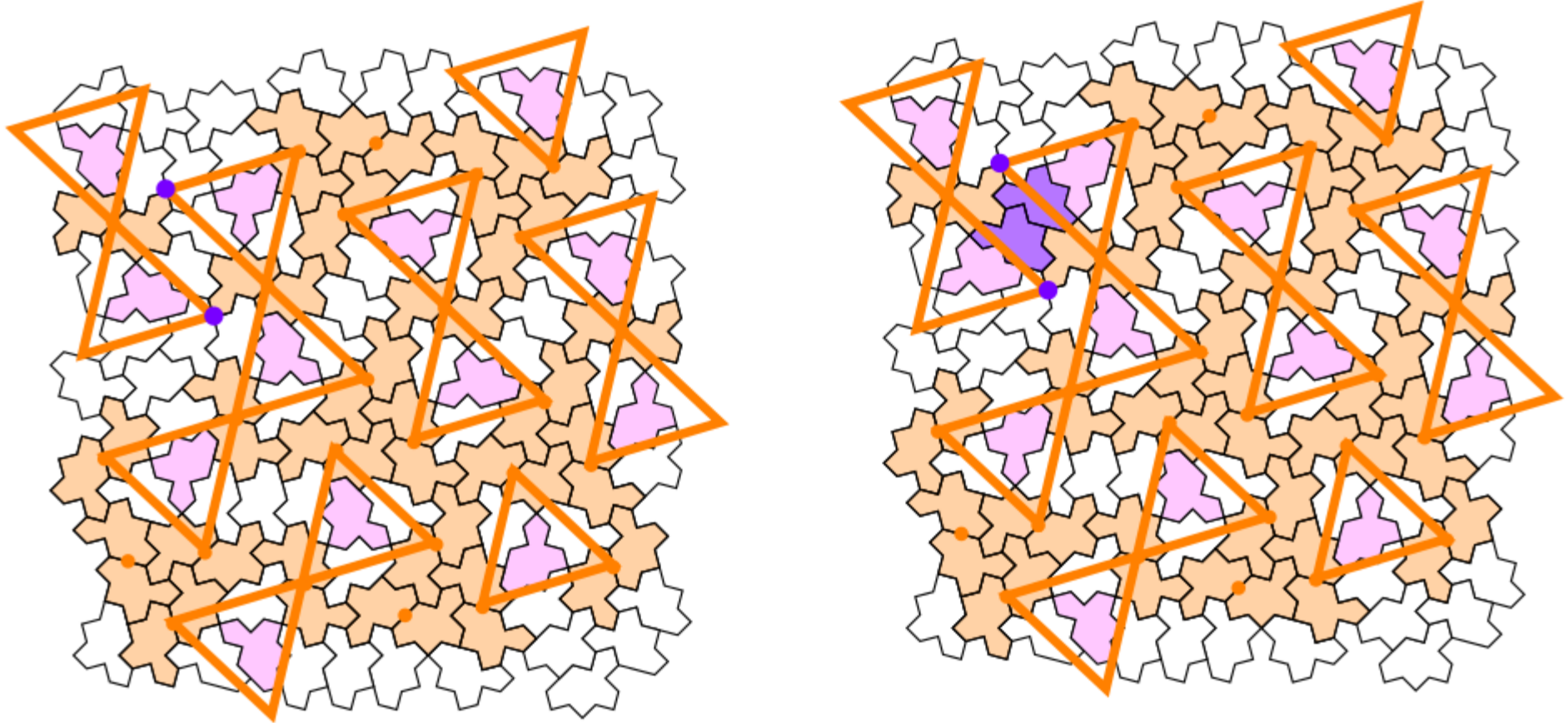
P6



- 8 combinations for a given cluster orientation (green crosses)

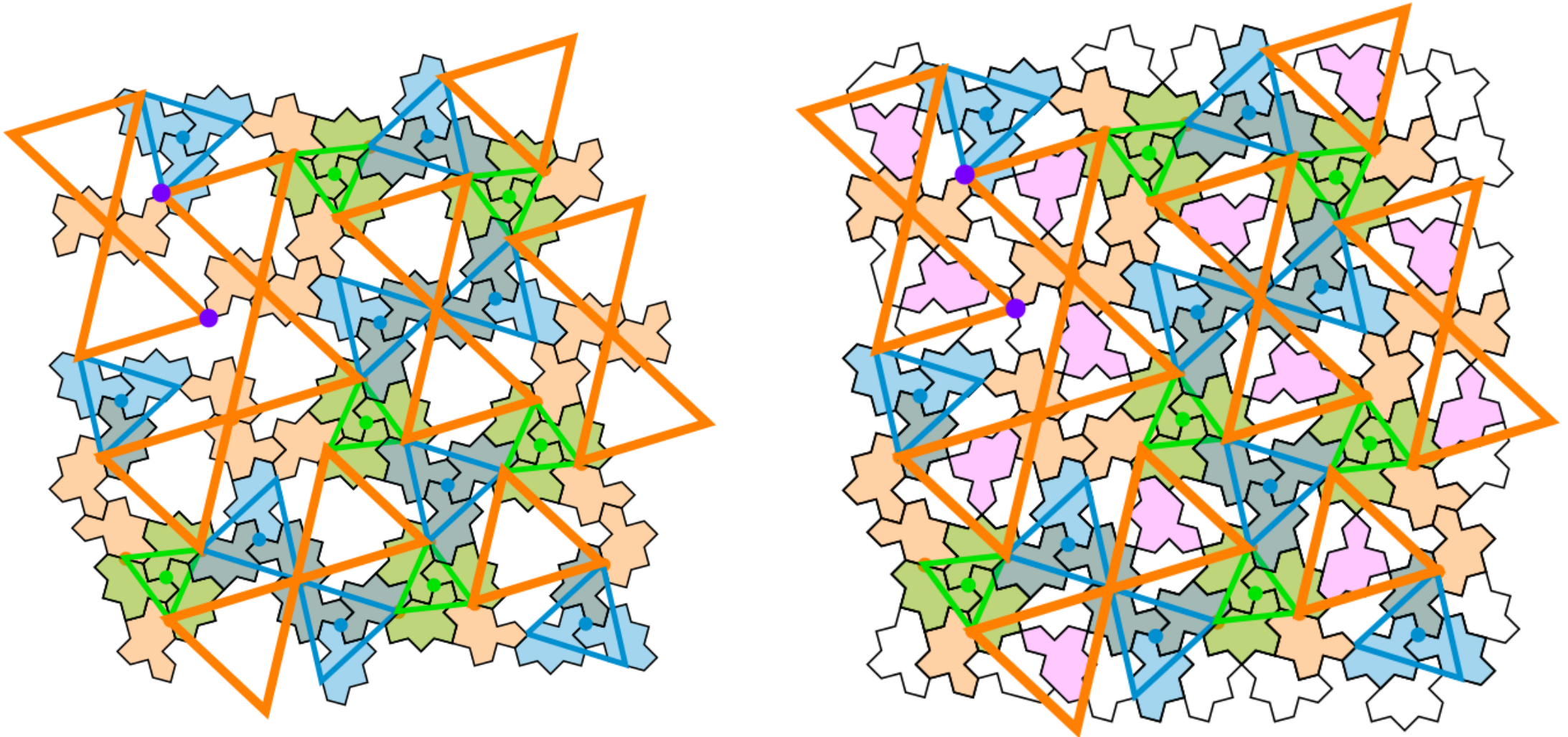
Example: Set of all local 2-fold symmetry centers

2-fold

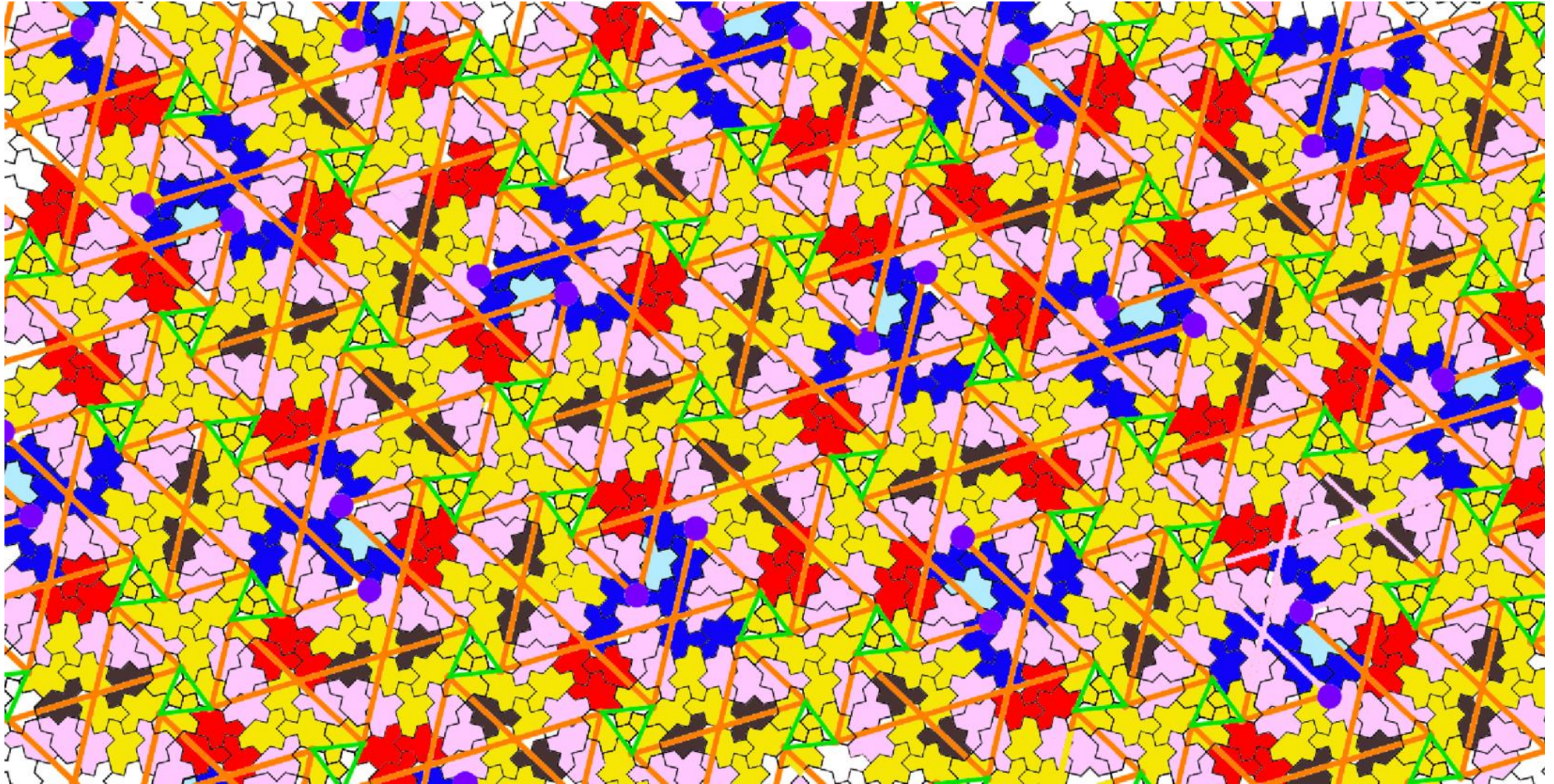


- around each ODD monotile (pink), three points (orange triangle)
- Violet points are not local 2-fold centers, but all other points are (orange)

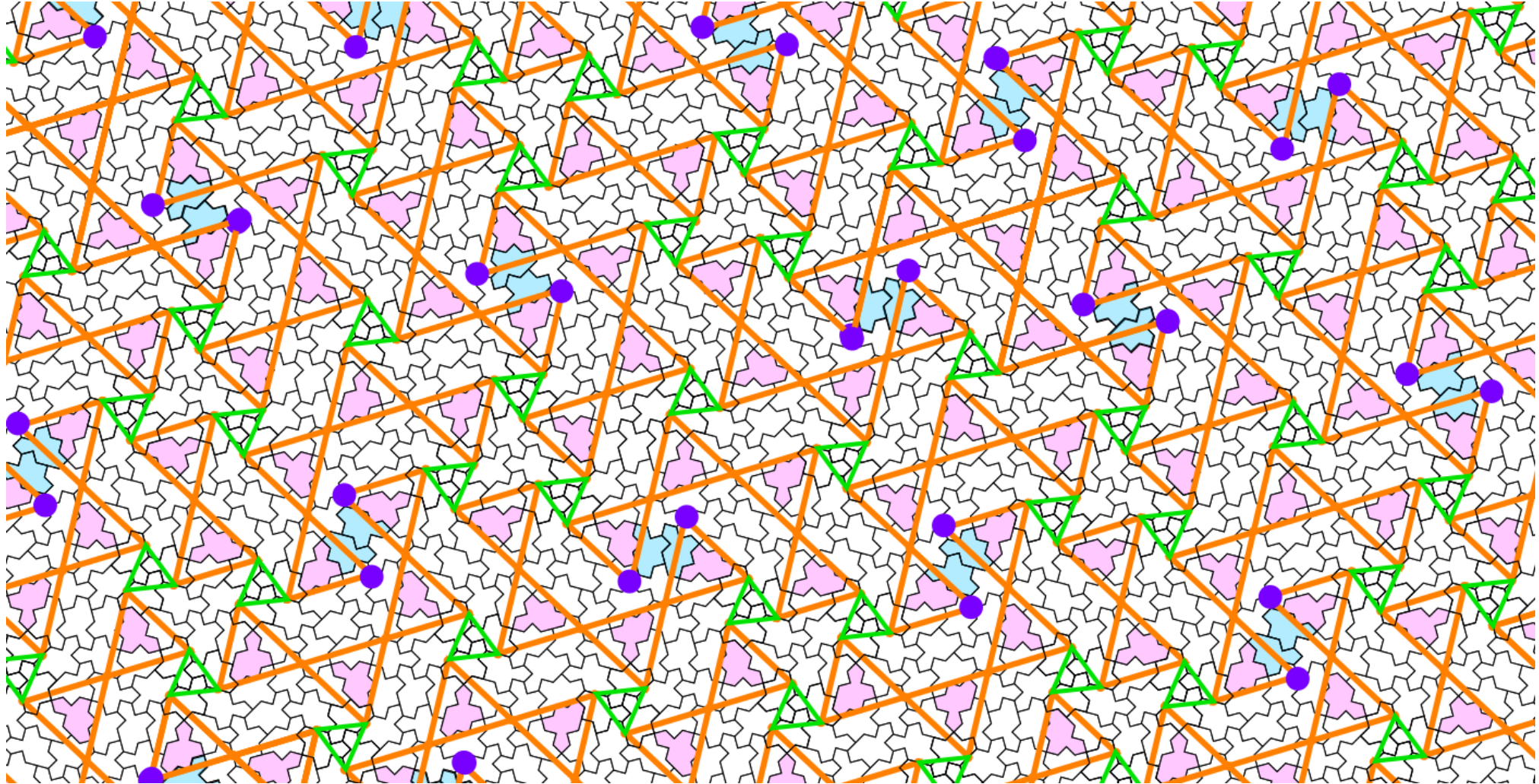
superposition



Comparision

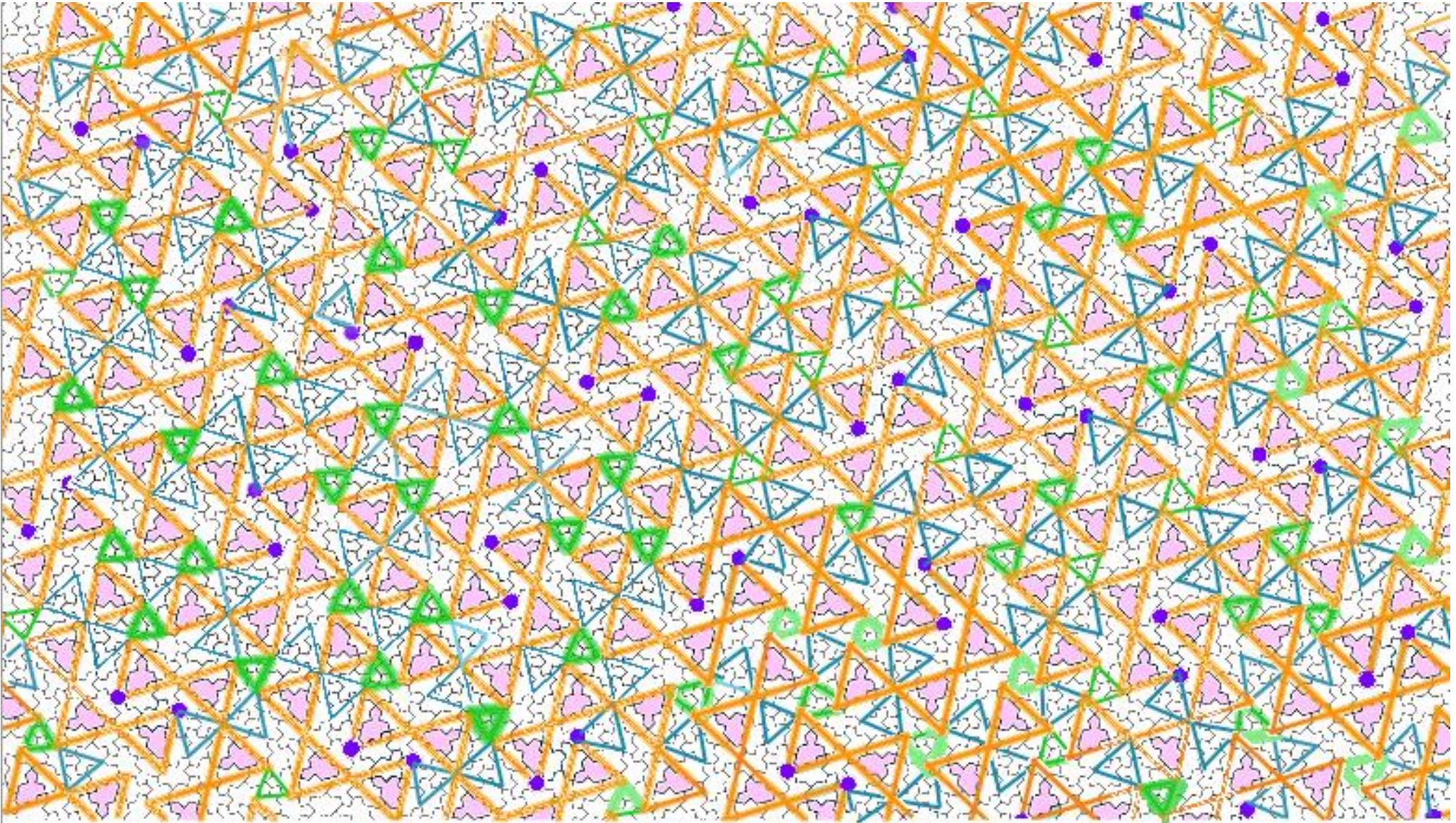


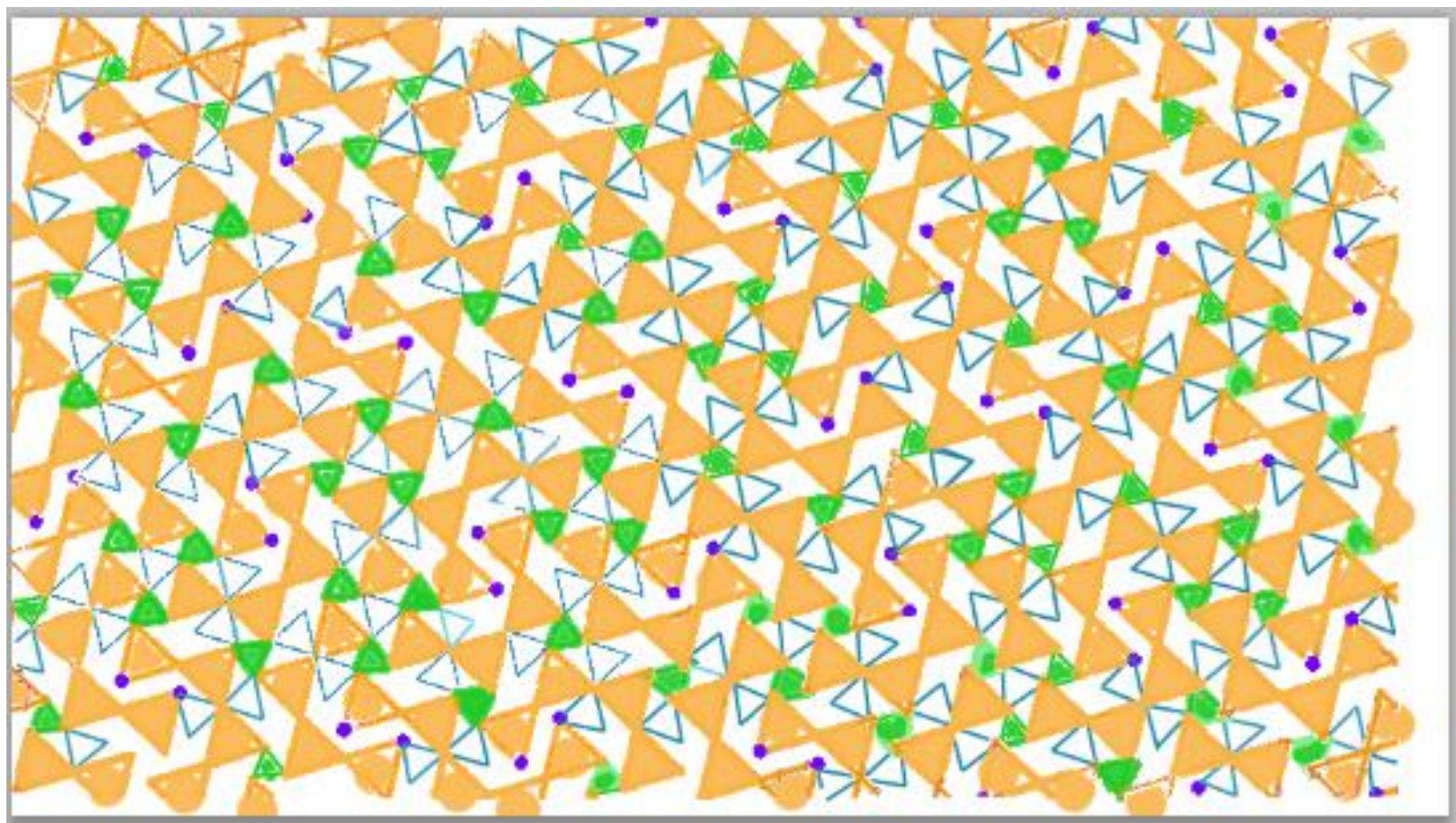
Small stars connect the orange triangles



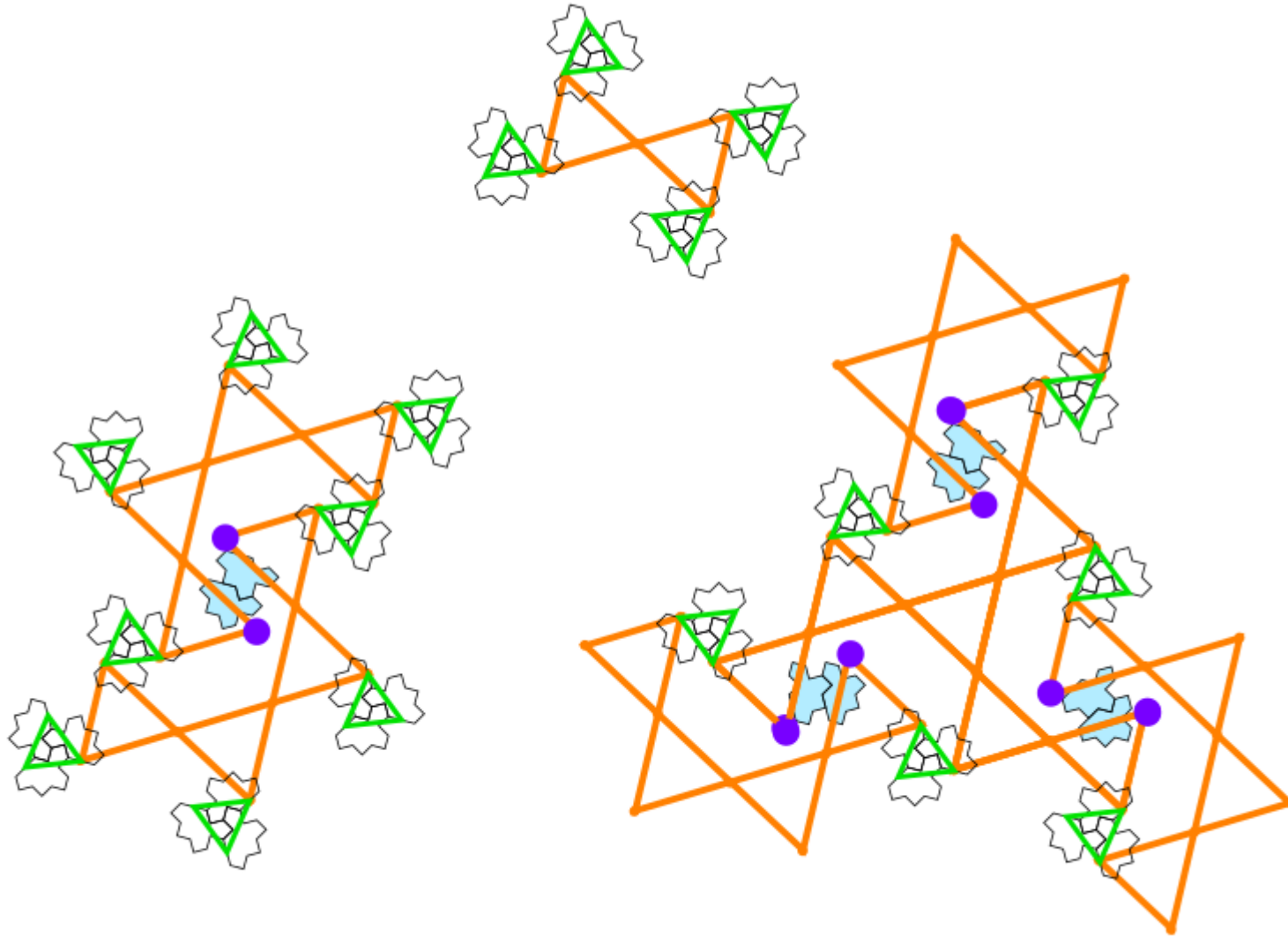
- Possibility to obtain the relative positions between diablo, 2-arms and 3-arms

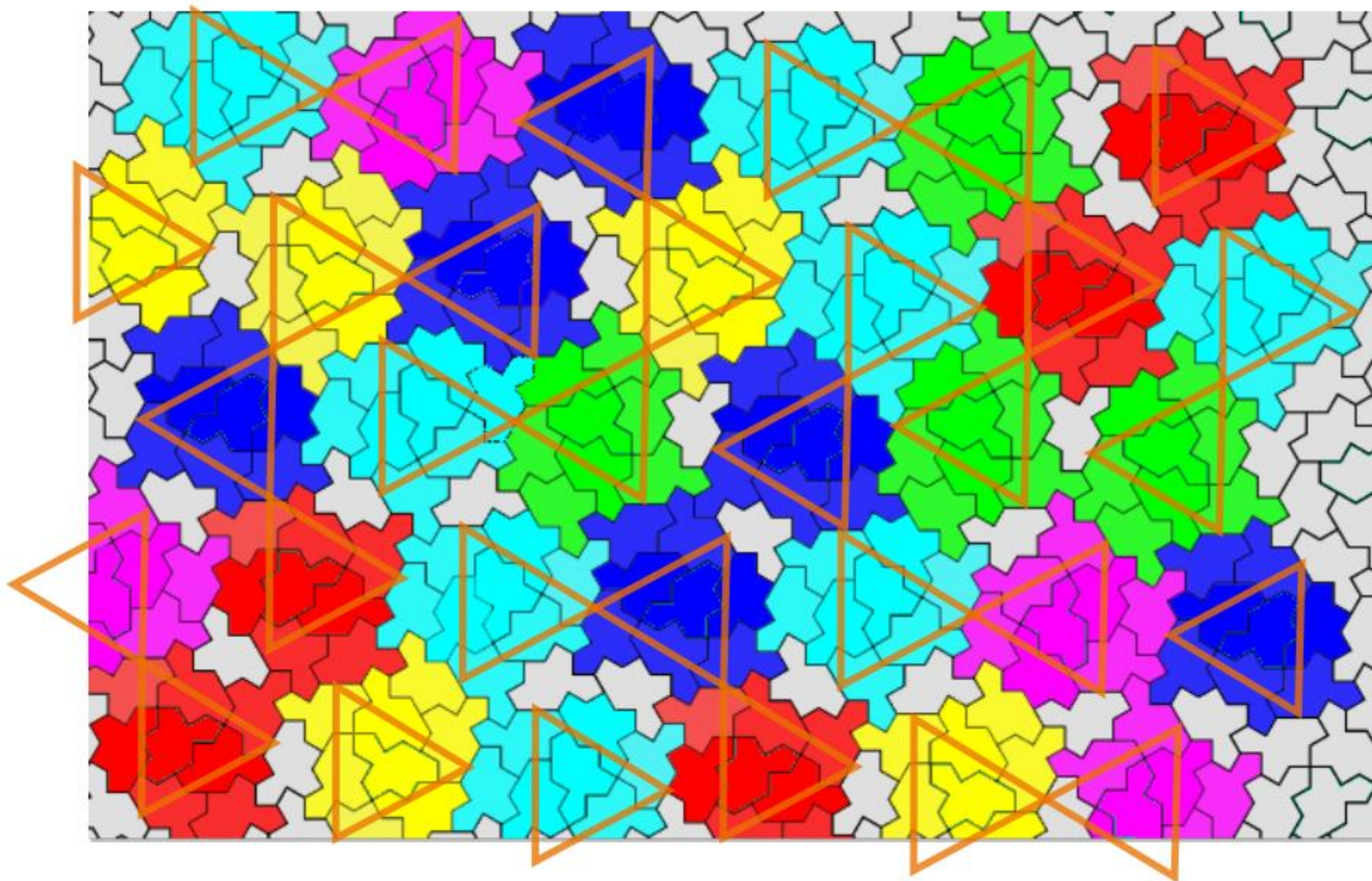
SMALL AND LARGE STARS



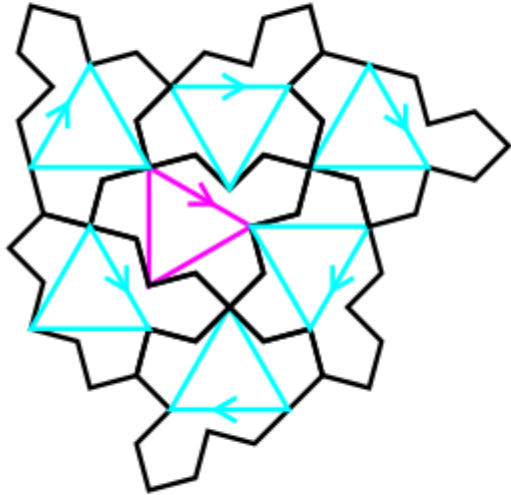


New puzzle

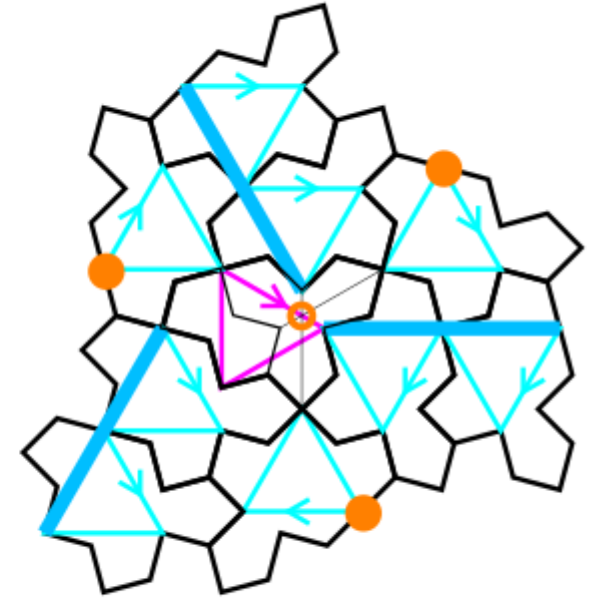
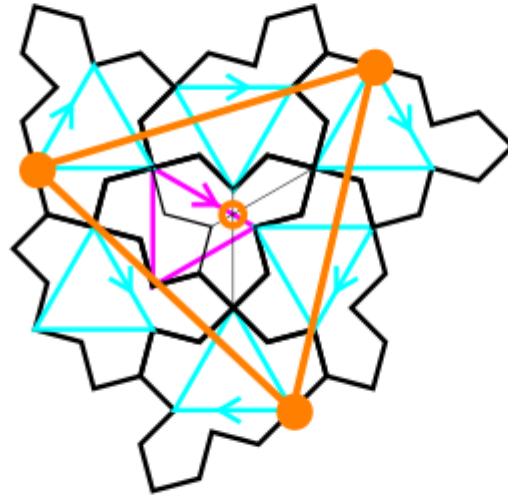




- Tile(1,1)



full cluster

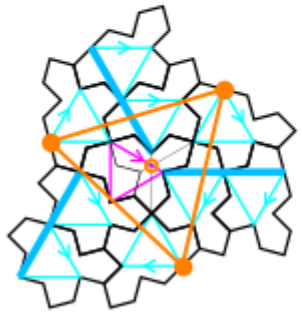


- C cluster: one ODD tile at the center + 6 EVEN tiles around
- Three glue points (orange color): regular triangle centered on the trifle shape of the ODD tile
- Full cluster: 3 more EVEN tiles obtained by translations (three dark blue lines)

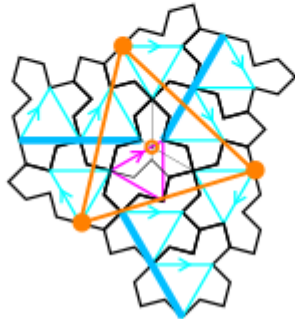
full cluster with its six rotations: $n(2\pi/6)$

- Tile(1,1)

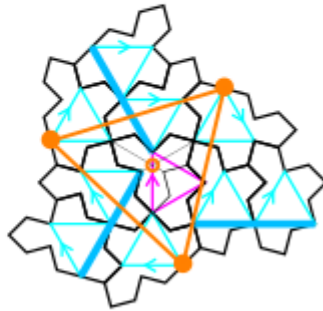
C1



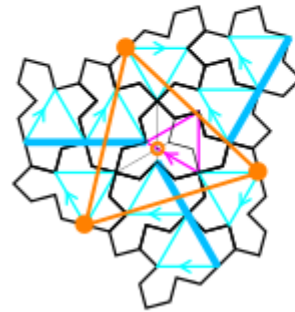
C2



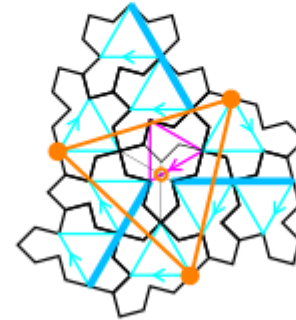
C3



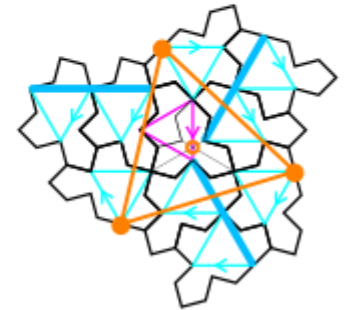
C4



C5

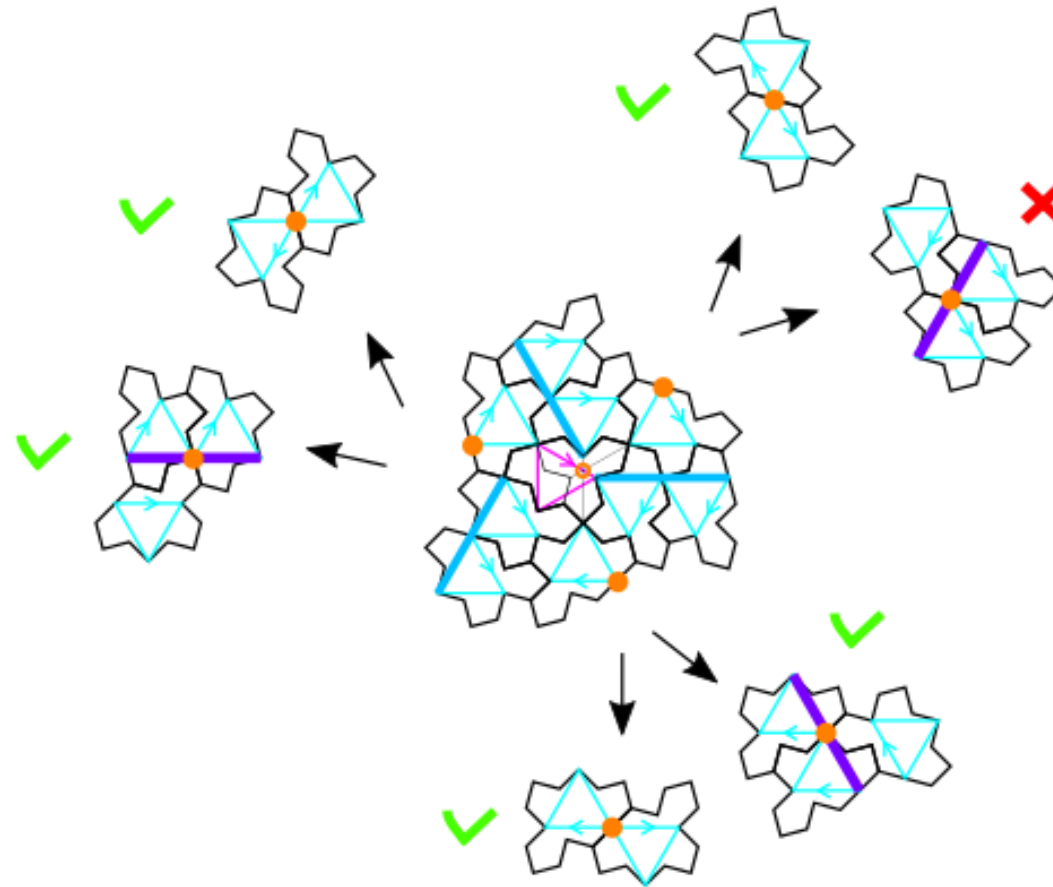


C6



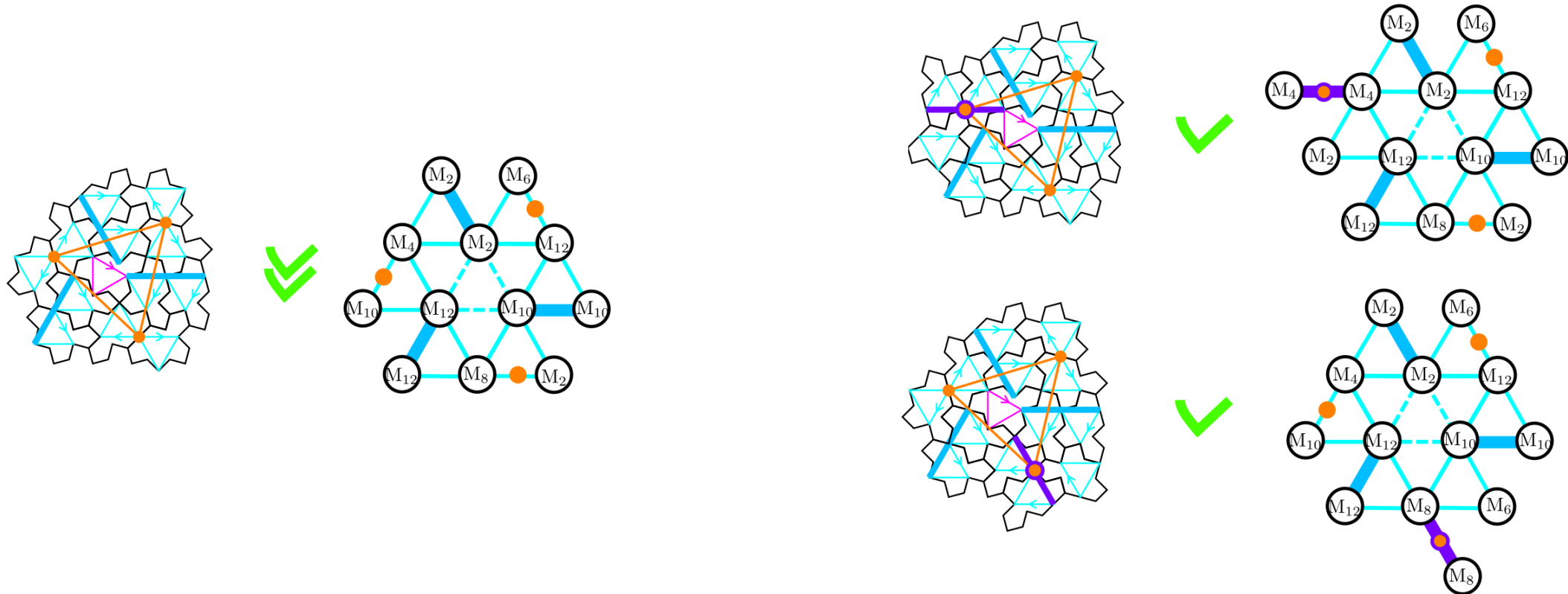
- 13 tiles in a full cluster: one ODD tile at the center + 9 EVEN tiles around
- Blue triangles (EVEN) have one horizontal edge when pink (ODD) have a vertical one

Possible configurations at the glue points ?



- Half-turned or translated monotile (violet line)

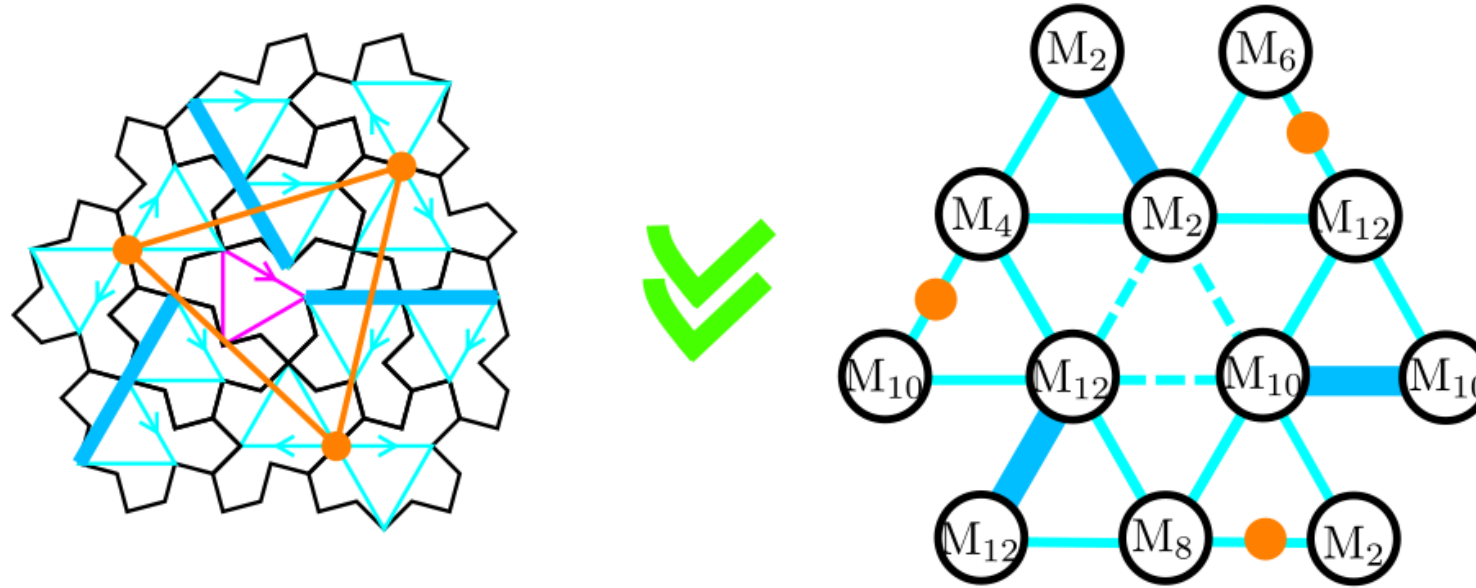
Three possible configurations at the glue points



- configuration with three half-turned tiles is the most common
- The other two are always coupled together in the tiling:
- One violet line in a 2-arms and three violet lines in a 3-arms

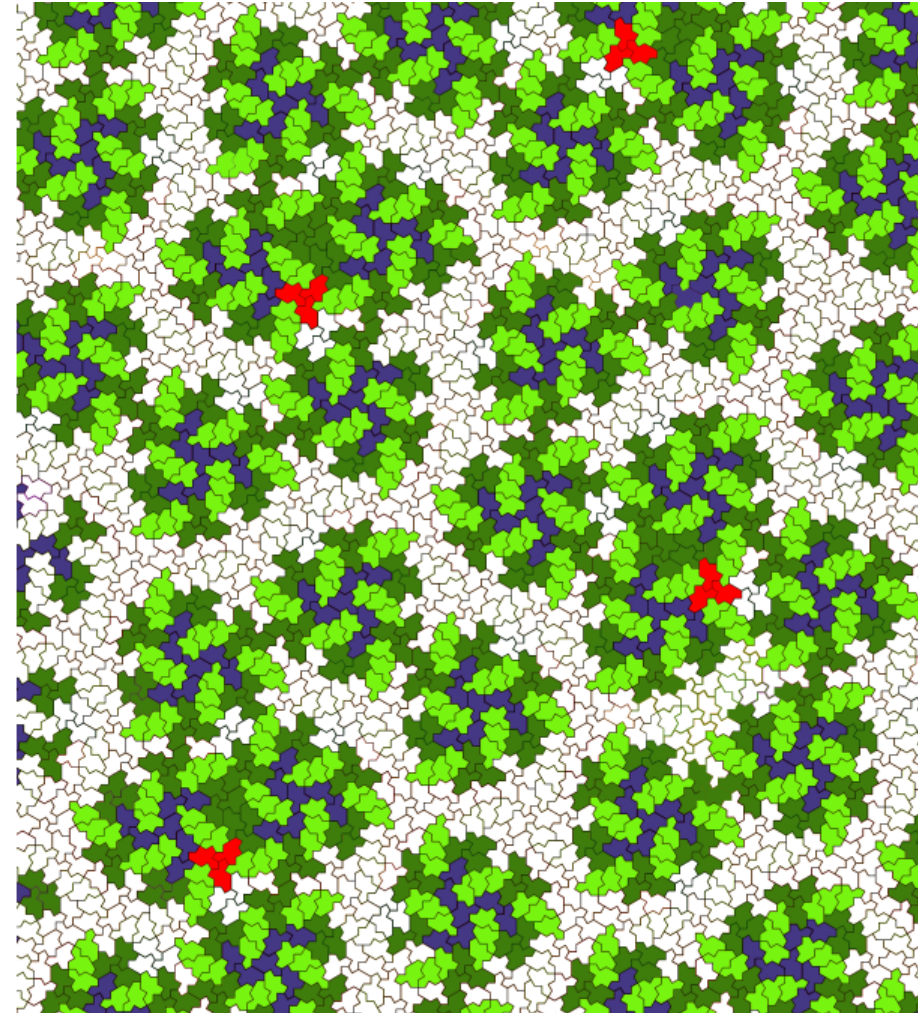
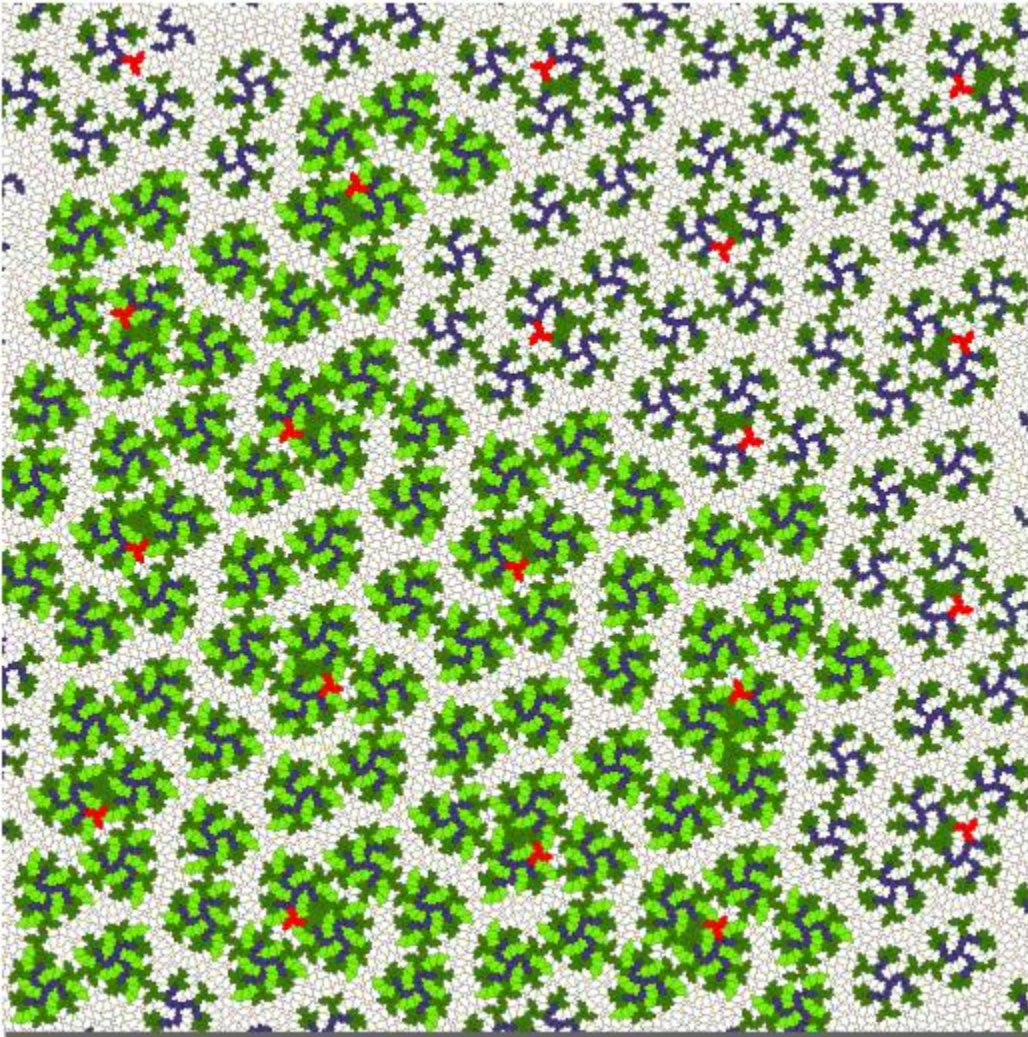
Orientation of the EVEN tiles in the most frequent configuration

- Tile(1,1)

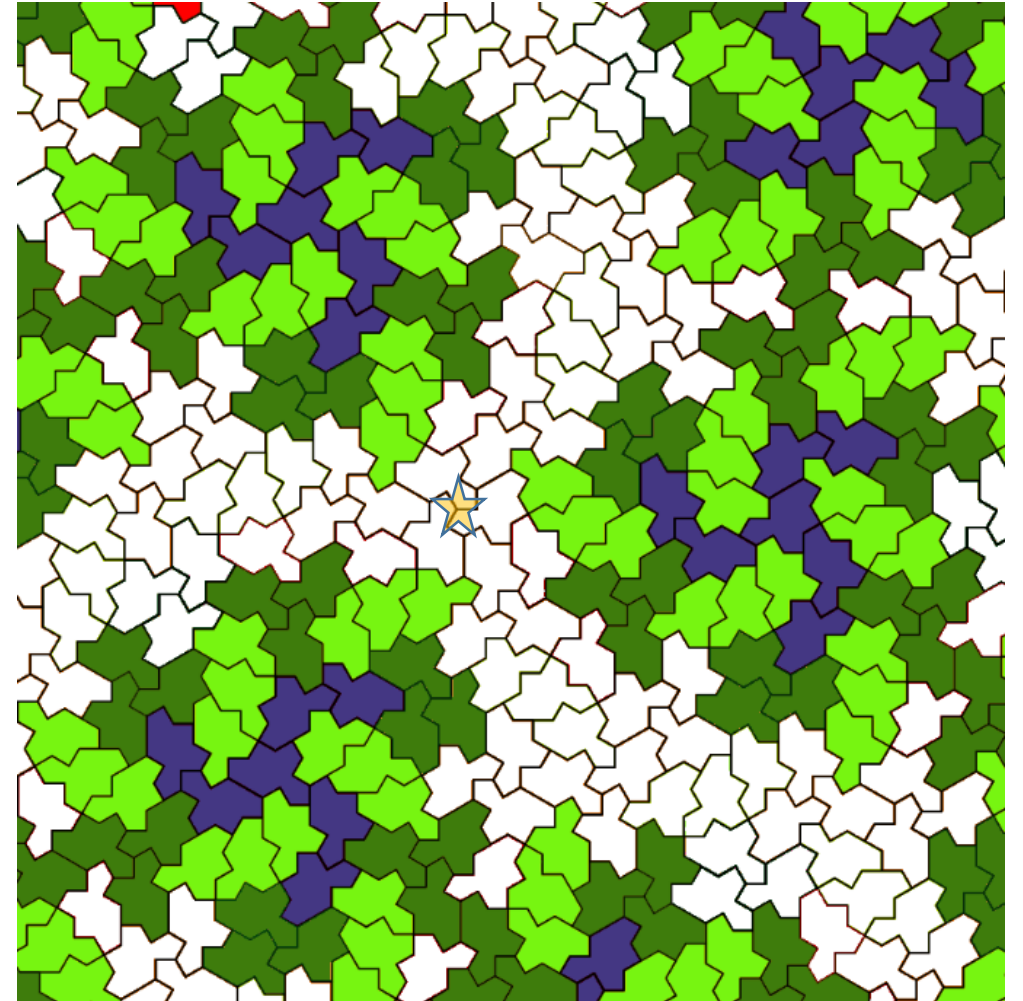
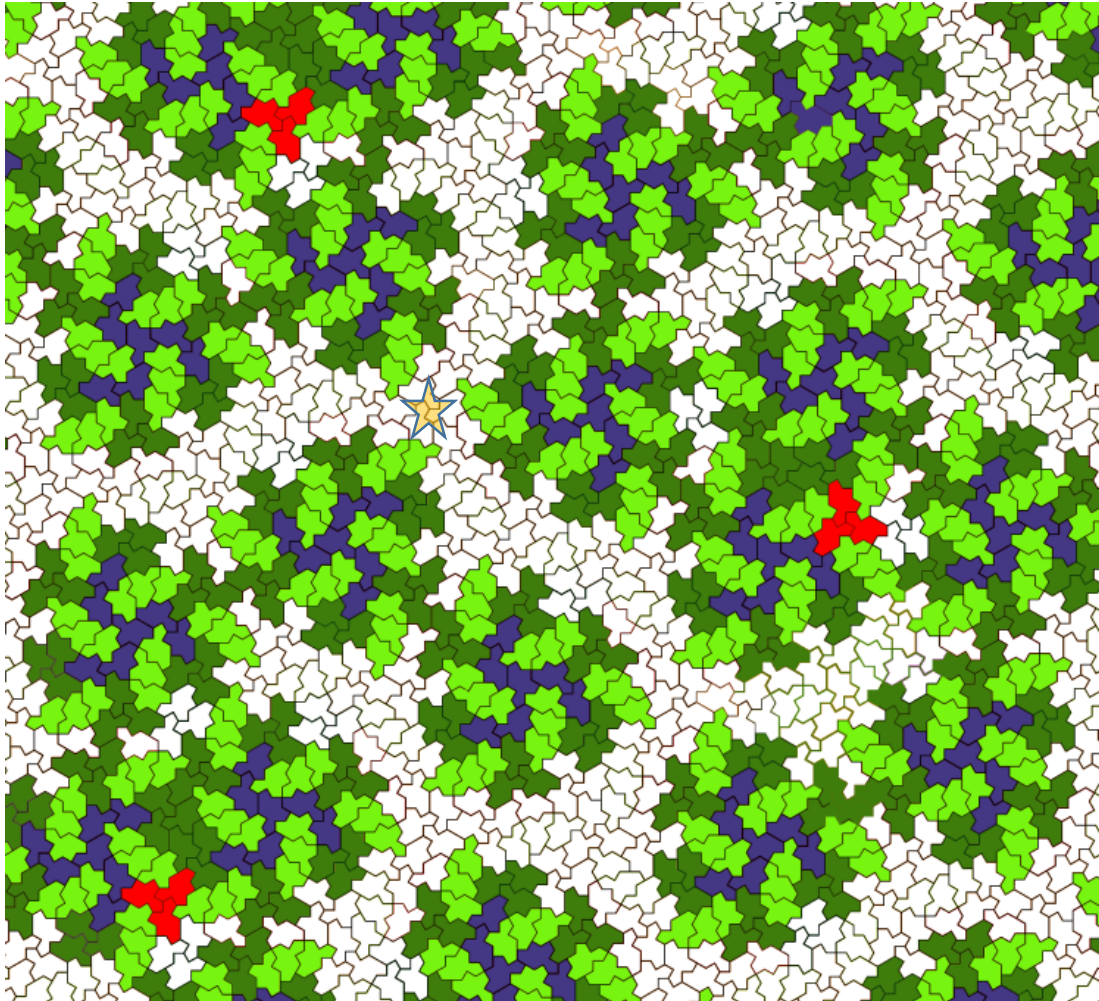


- 6 tiles in the C-cluster (corona made of M12, M4, M2, M12, M10, M8)
- 3 translated (M2, M10, M12) (along blue lines, the edges of blue triangles)
- 3 half-turned (M4 → M10, M12 → M6, M8 → M2)

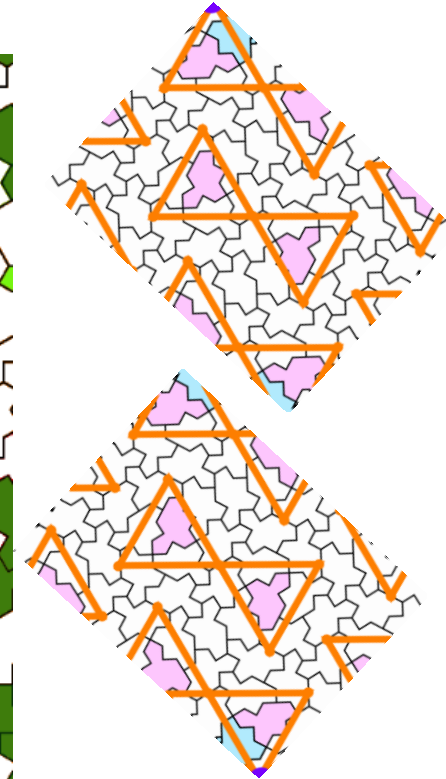
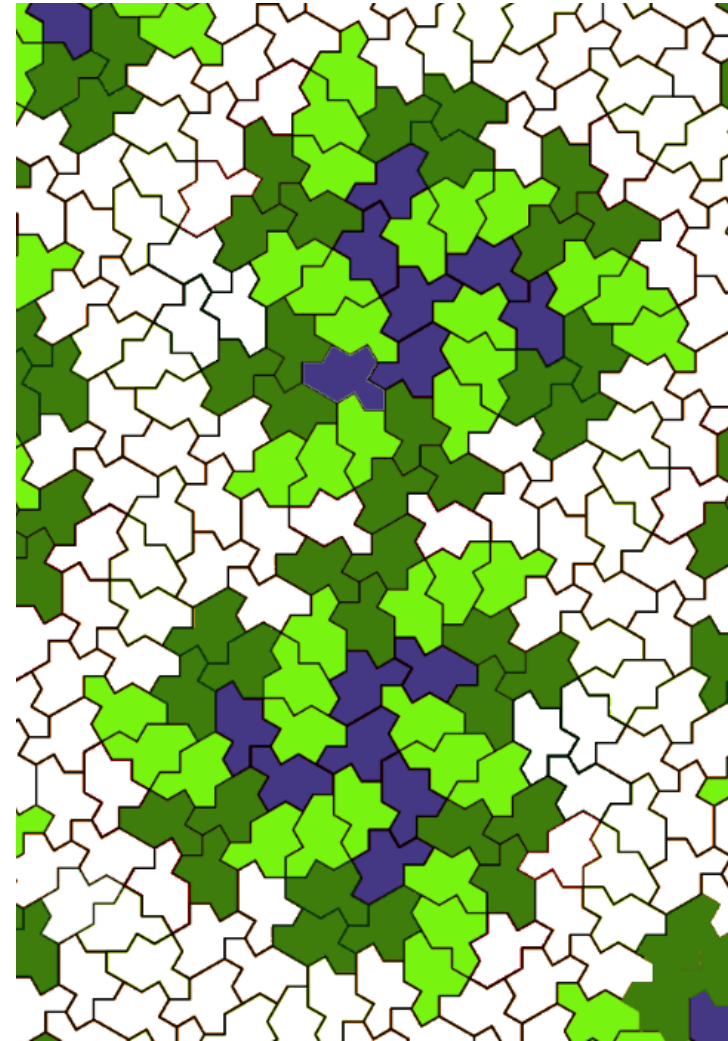
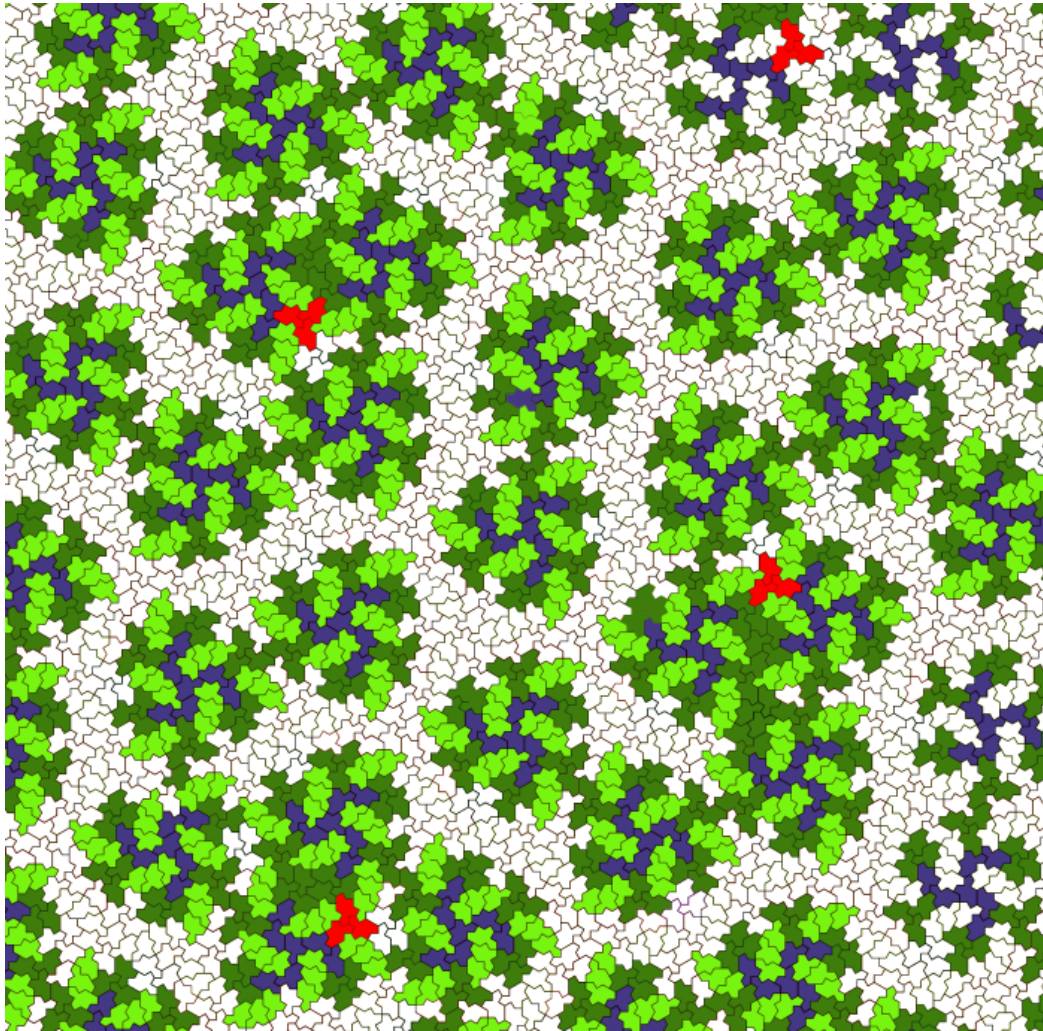
Diabolo/2-arms/3-arms at level 2 by Boris Horvat



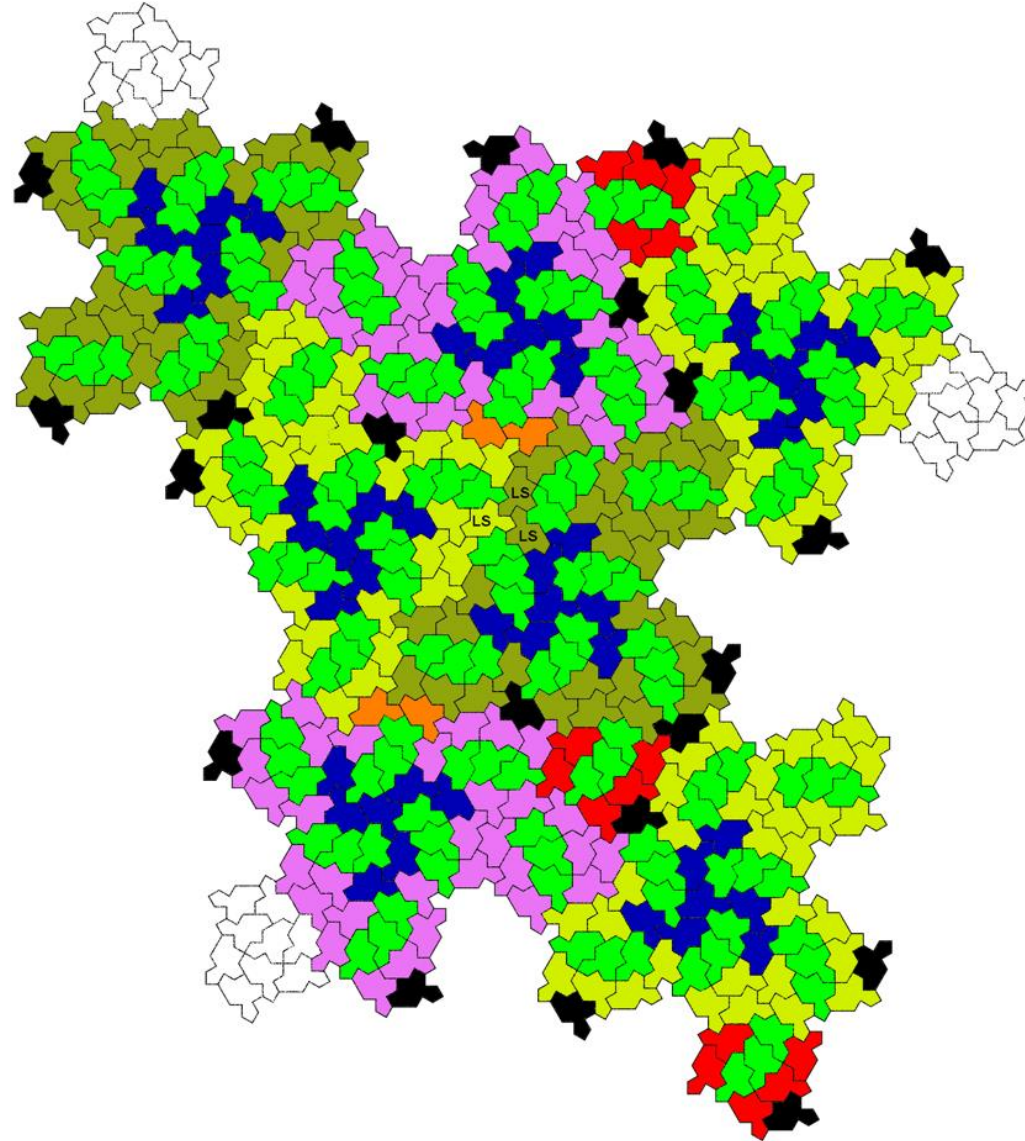
Diabolo/2-arms/3-arms at level 2 by Boris Horvat



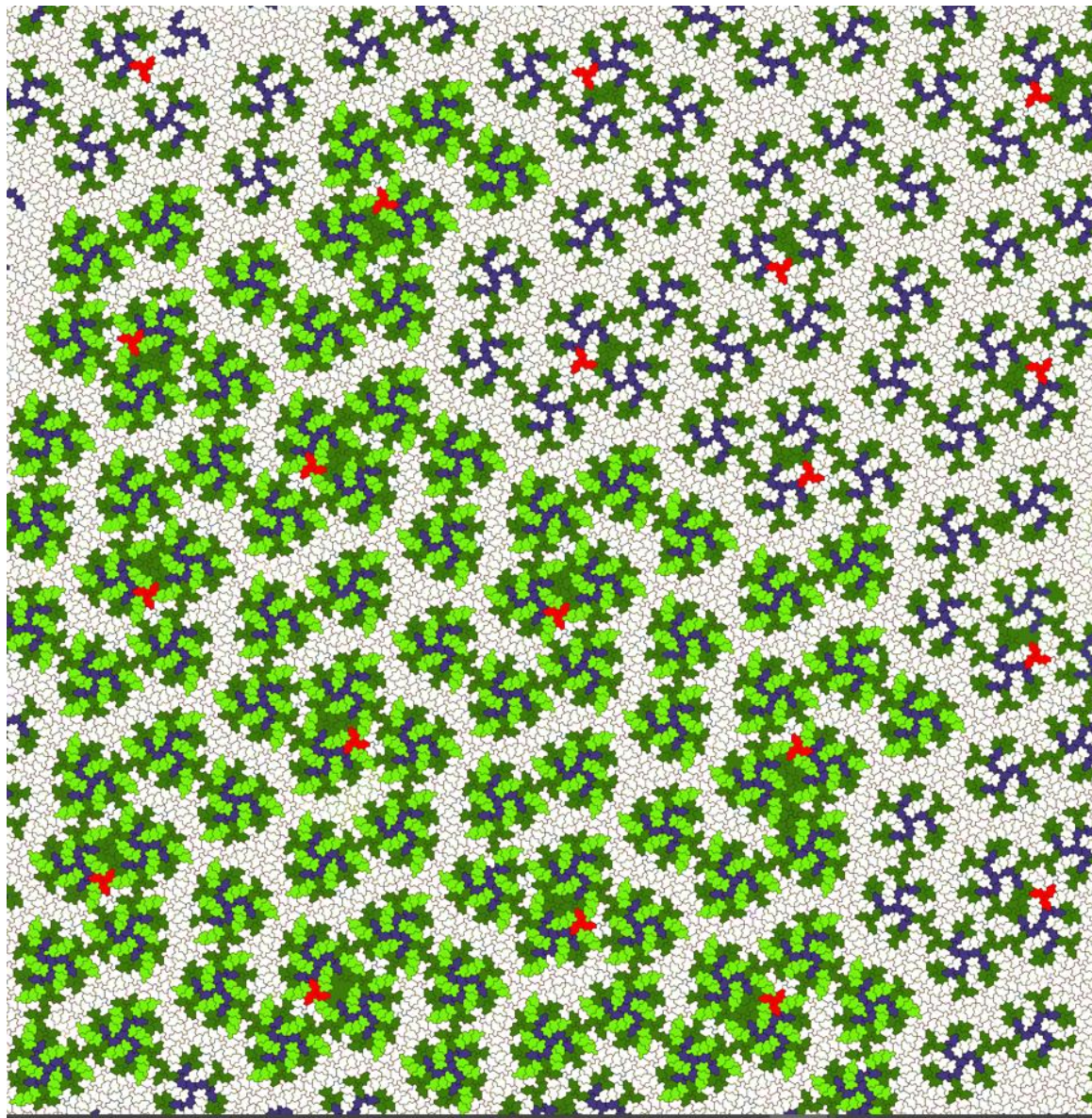
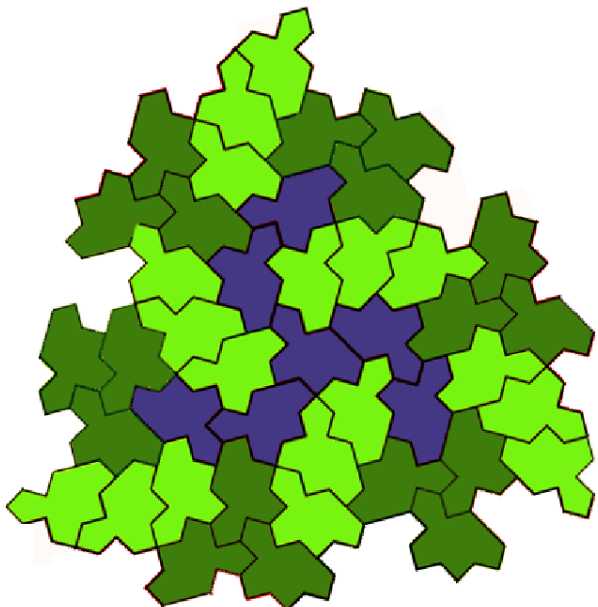
Diabolo/2-arms/3-arms at level 2 by Boris Horvat



Diabolo/2-arms/3-arms at level 2 by Boris Horvat

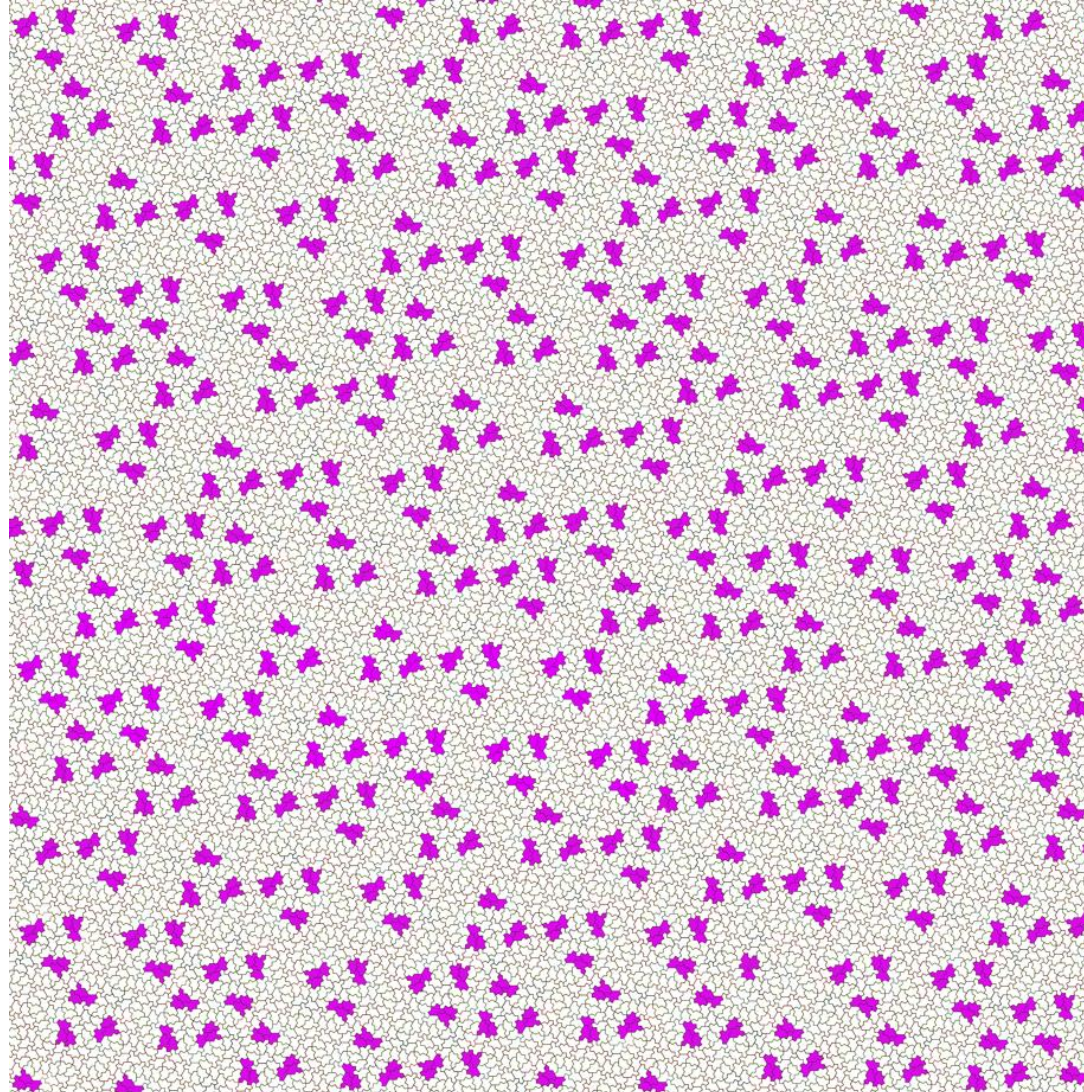


Chimera level 2 by Boris Horvat



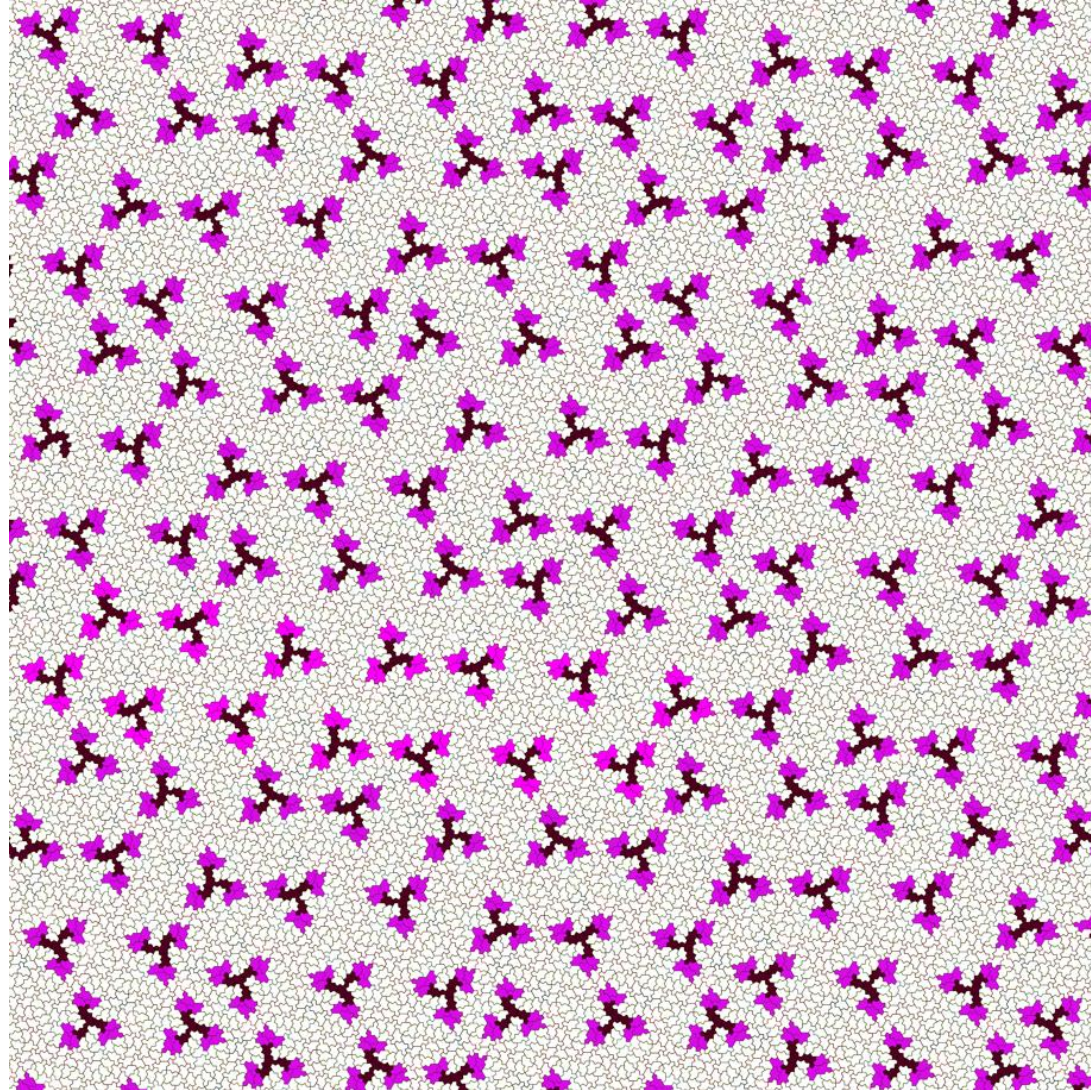
Triplets Marianne by Boris Horvat

**Diabolo
2-arms
3-arms
are visible !**

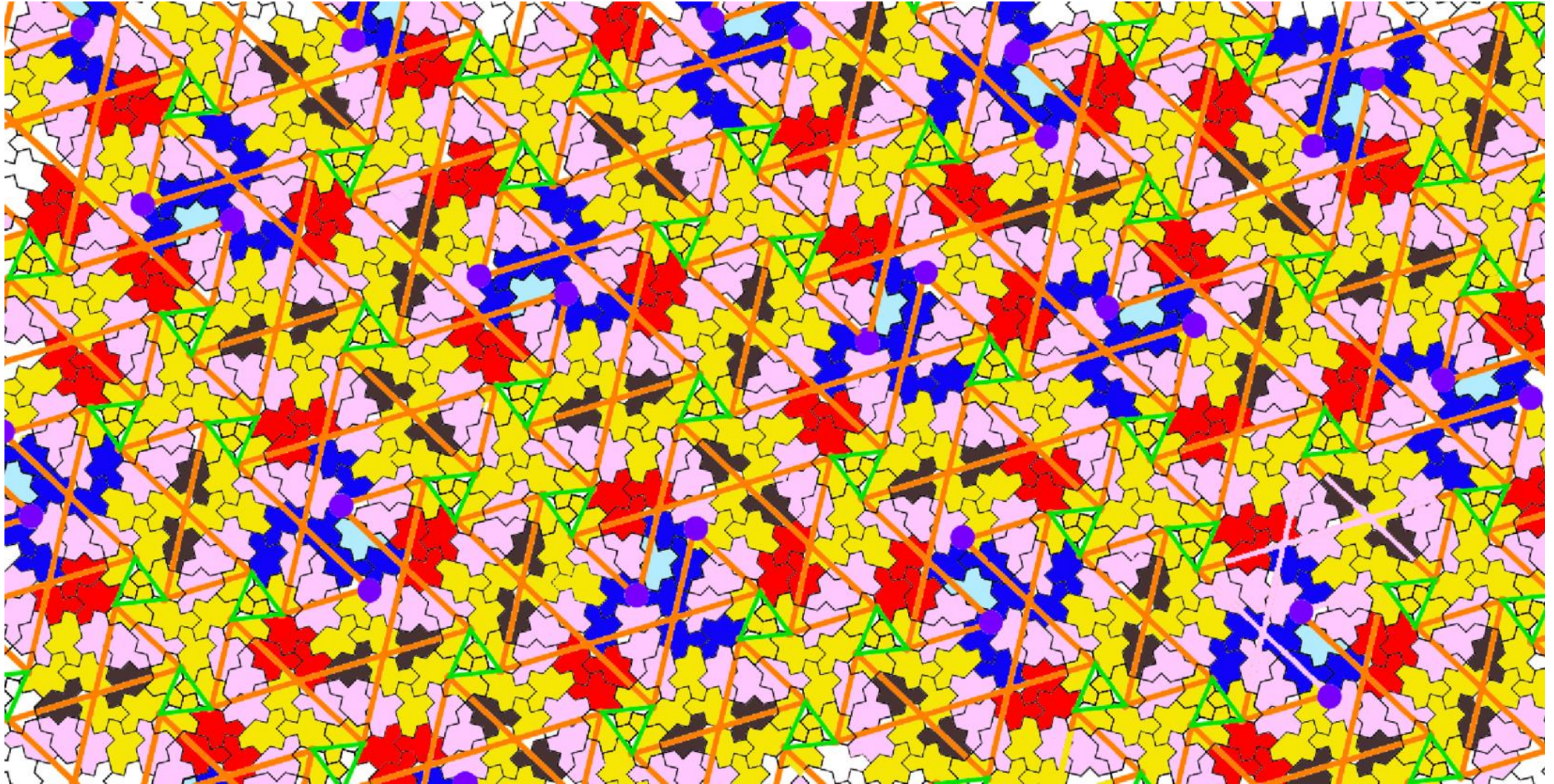


Triplets and propellers by Boris Horvat

Diabolo
2-arms
3-arms
are visible !



Comparision



Espaces entre triangles oranges

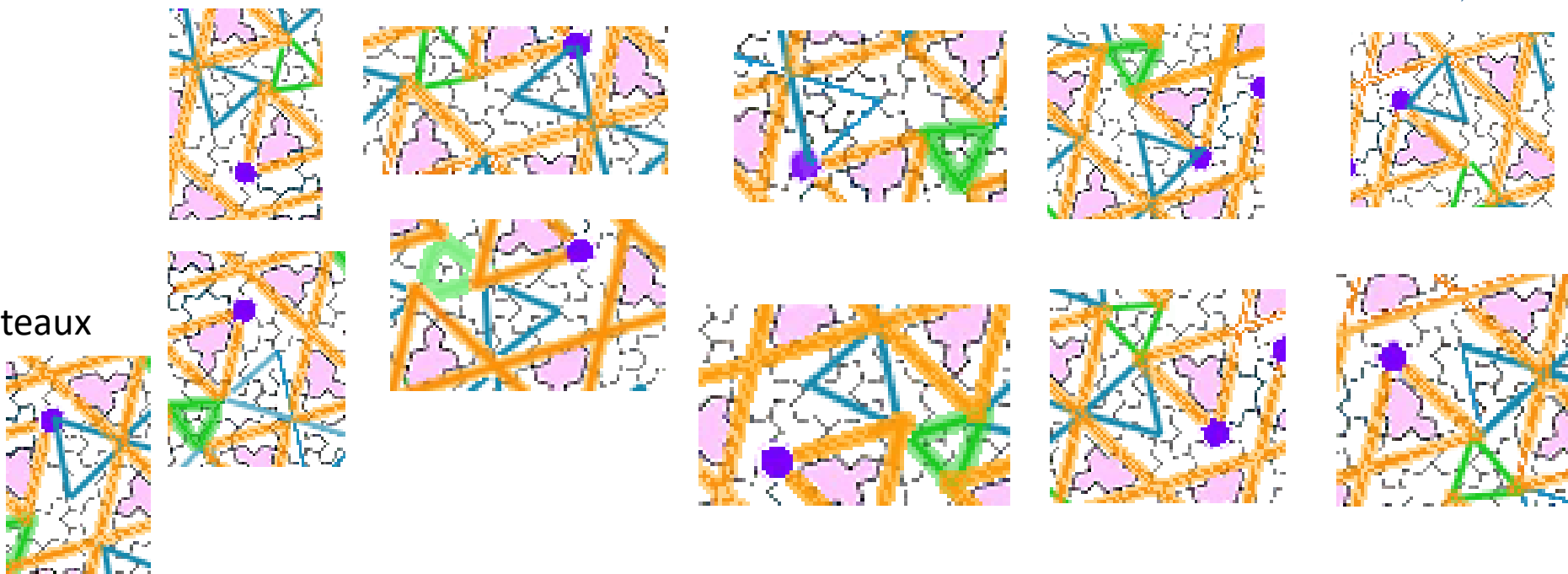
Parallélogrames



Triangles



Bateaux



Petits parallélogrames
2 orientations de tuiles

